

Optics and Modern Physics

CHAPTER 4

Diffraction and Polarization

4.1–4.20

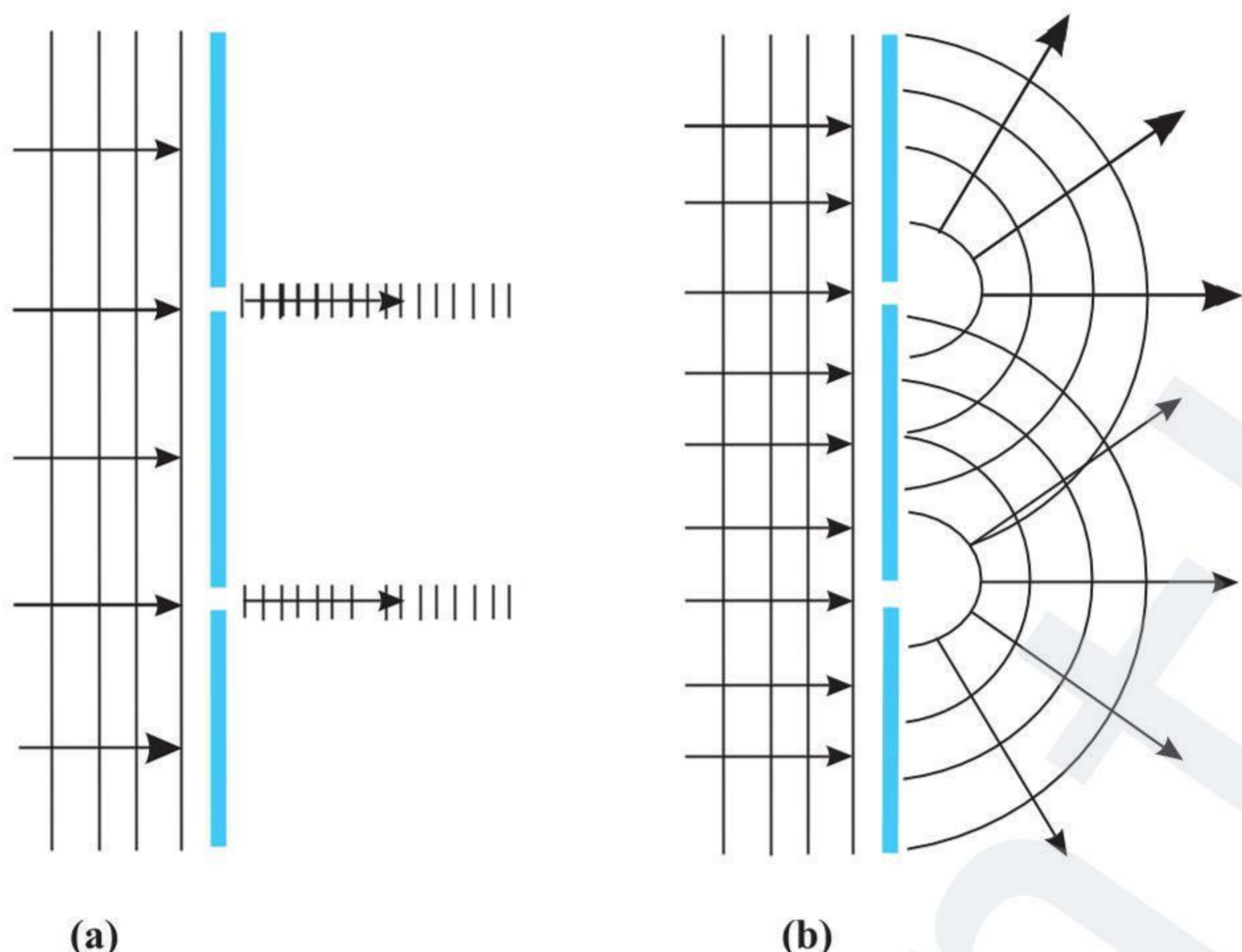
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4

Diffraction and Polarization

INTRODUCTION

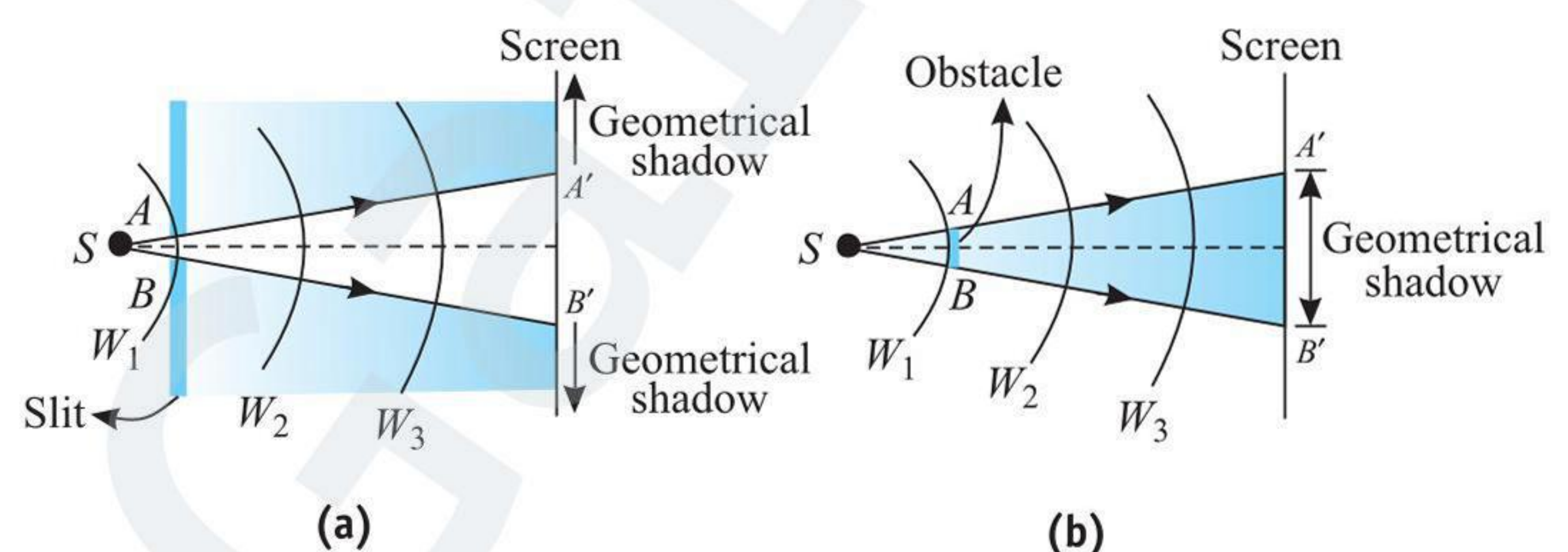
The basic property of light is its travelling in a straight line. Because of this, light casts the shadow of the object coming in its path. In previous chapter, we learned if light passes through a pair of narrow slits it shows interference phenomenon. In fact, the light waves from the two slits overlap as they spread out, filling what we expect to be shadowed regions with light and producing interference fringes. If light waves did not spread out after passing through the slits, no interference would occur.



Also close to a sharp edge of an opaque obstacle or through a narrow slit, we do observe some deviation of light from its straight line path.

- Let us consider a source of light S placed at a short distance from a narrow (circular) aperture AB . Let a screen be placed at some distance from the aperture [Fig. (a)], only the part $A'B'$ of the screen should get illuminated and no light should enter the regions $A'X$ and $B'Y$ of the screen as shown in Fig. (a). But we do not find complete darkness in these regions. $A'B'$ is the portion on the screen which is illuminated by light passing through the aperture. Beyond A' and B' on both the sides is the region of the geometrical shadow. If W_1 represents a primary wavefront falling on the aperture and if secondary wavefronts $W_2, W_3, \dots$ are drawn according to Huygens' principle, it is found that some portions of these wavefronts would go into the geometrical shadow beyond A' and B' . The shadow is, therefore, not sharp.
- Similarly, if an **opaque (circular) obstacle** AB is placed between the source S and the screen [Fig. (b)], some portions of the secondary wavefronts W_2 and W_3 etc. will enter the region of geometrical shadow $A'B'$ on the screen and as the shadow is again not sharp. But this illumination of the

geometrical shadow will be observed only if the aperture is very narrow or if the size of the obstacle is very small.



The phenomenon of bending of light round the sharp corners and spreading into the regions of the geometrical shadow is called **diffraction**.

FRESNEL AND FRAUNHOFER DIFFRACTIONS

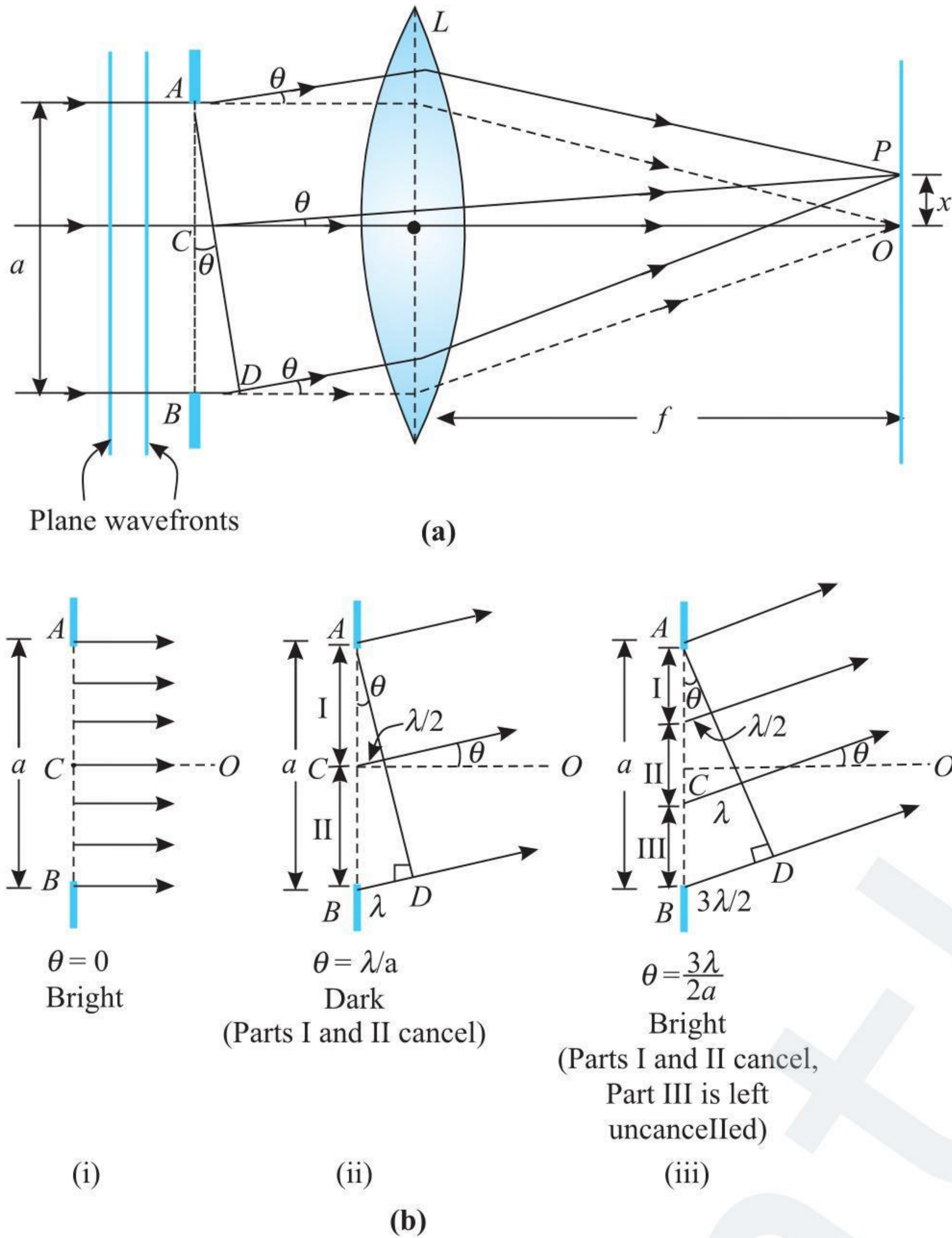
Before discussing how diffraction patterns are formed, we must briefly explain the distinction between Fresnel and Fraunhofer diffractions.

- Fresnel diffraction:** The source and the screen are placed at finite distances from the obstacle (or slit) and no lenses are used. Since the source of light is close to the slit, the wavefront is either spherical (in case of point source) or cylindrical (in case of a line source) in nature. The screen is also at a finite distance from the slit. This is also called **near-field diffraction**.
Fresnel gave the exact explanation of the phenomenon of diffraction. According to him, diffraction is due to the interference of secondary wavelets originating from the portion of the primary wavefront, which is not blocked by the slit or the obstacle.
- Fraunhofer diffraction:** In Fraunhofer diffraction, the incident waves as well as the diffracted waves can be approximated to have plane wavefronts. This can be achieved by placing the source and the screen far away from the obstacle or by using lenses so that the source and the screen are at focal planes. Fraunhofer diffraction is also called **far-field diffraction**. Owing to the very nature of the wavefronts, Fraunhofer diffraction (planar wavefronts) is much simpler to analyse than Fresnel diffraction (spherical wavefronts). Therefore, we shall limit ourselves to Fraunhofer diffraction.

FRAUNHOFER DIFFRACTION AT A SINGLE SLIT

Let a plane wavefront falling on a single narrow slit AB of width ' a ' [Fig. (a)]. According to our knowledge of that light travels in straight path, we should expect a sharp image of the slit on

the screen. But this is not the case because light travels only approximately in straight lines and is diffracted (i.e., spreads out) into the regions above A and below B where we do not expect it. The simple laws of geometrical optics do not apply and as such instead of a sharp image of the slit, we get the central image [Fig. (a)] flanked by a series of dark and bright fringes (or bands). The intensity of the bright bands goes on decreasing as we proceed outwards from the central band either way. This has been explained as follows.



According to Huygens' hypothesis, each point on the plane wavefront falling on AB gives rise to secondary wavelets which are initially in the same phase. These wavelets spread out in all directions.

(a) **Position of central maximum:** According to Huygens' principle, when light falls on the slit, it becomes a source of secondary wavelets. These wavelets are initially in phase and spread out in all directions. Let C be the centre of the slit AB . The wavelets which fall on the lens in a direction parallel to CO meet at point O in the same phase. This is because, the wavelets are initially in phase and their optical paths between the slit and the point O are also equal. Therefore, the wavelets reinforce each other and give rise to the central maximum at point O .

(b) **Positions of minima:** Consider a point P on the screen at which wavelets travelling in a direction making angle θ with CO are brought to focus by the lens. The wavelets from different parts of the slit do not reach point P in phase, although they are initially in phase. It is because, they cover unequal distances in reaching the point P . The wavelets from points A and B will have a path difference equal to BD .

$$BD = a \sin \theta \quad \left(\text{as } \sin \theta = \frac{BD}{AB} = \frac{BD}{a} \right)$$

- (i) Let the point P be so situated on the screen that $a \sin \theta = \lambda$, where λ is the wavelength of the incident monochromatic light. In this case, the path difference for the wavelets from A and C and similarly for those from C and B will be $\lambda/2$, where C is the midpoint of the slit.
- (ii) If we imagine the slit AB to be divided into two halves AC and CB (denoted by I and II) as shown in Fig. [(b) (ii)], then for every point in the upper half AC (i.e., Part I), there is a corresponding point in the lower half CB (i.e., Part II), for which the path difference is $\lambda/2$. The wavelets from each pair of corresponding points in the two halves will reach P in opposite phases and obviously will interfere destructively with each other and produce the **first minimum or a dark band** there. Thus, for the first minimum,

$$a \sin \theta = \lambda$$

$$\text{or } \sin \theta = \frac{\lambda}{a} \quad \dots(i)$$

Since θ is small, $\sin \theta \approx \theta$ and as such

$$\theta = \frac{\lambda}{a} \quad \dots(ii)$$

Proceeding on the same lines, it can be shown that the condition for the n^{th} minimum is

$$\theta_n = \frac{n\lambda}{a} \quad \dots(iii)$$

where $n = 1, 2, 3, \dots$, but not zero (as that is the condition for the central maximum)

Thus, the first, second third, ... minima occur when

$$\theta_1 = \frac{\lambda}{a}, \theta_2 = \frac{2\lambda}{a}, \theta_3 = \frac{3\lambda}{a} \dots \quad \dots(iv)$$

- (c) **Positions of maxima:** Let point P on the screen be situated such that the path difference

$$a \sin \theta = \frac{3\lambda}{2} \quad \dots(v)$$

Now consider the slit width to be divided into three equal parts as shown in Fig. [(b) (iii)] so that the path difference between the wavelets from the first two parts will again be $\lambda/2$. These wavelets will interfere destructively and cancel each other's effect. The wavelets from the third part will remain uncanceled. Since all of them are in the same phase, they reinforce each other and thus produce the first secondary maximum. Therefore, for the first secondary maximum, from Eq. (v),

$$\sin \theta = \frac{3\lambda}{2a}$$

$$\text{or } \theta = \frac{3\lambda}{2a} \quad (\text{as } \sin \approx \theta, \text{ where } \theta \text{ is small}) \quad \dots(vi)$$

Proceeding in the same way, it can be shown that the condition for the n^{th} secondary maximum is

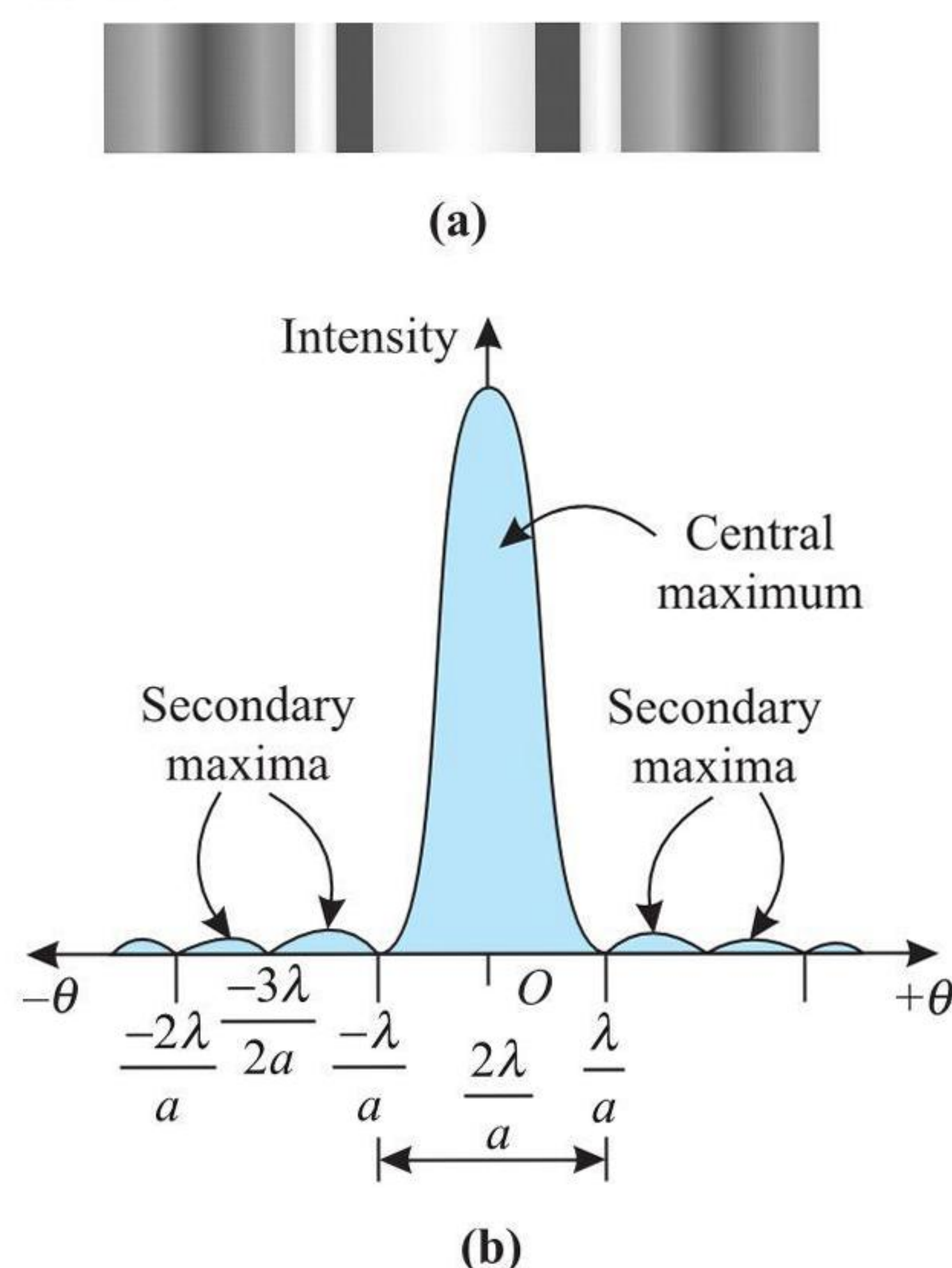
$$\theta'_n = (2n+1) \frac{\lambda}{2a} \quad \dots(vii)$$

where $n = 1, 2, 3, \dots$ and not zero.

Thus, the first, second, third...secondary maximum occur when

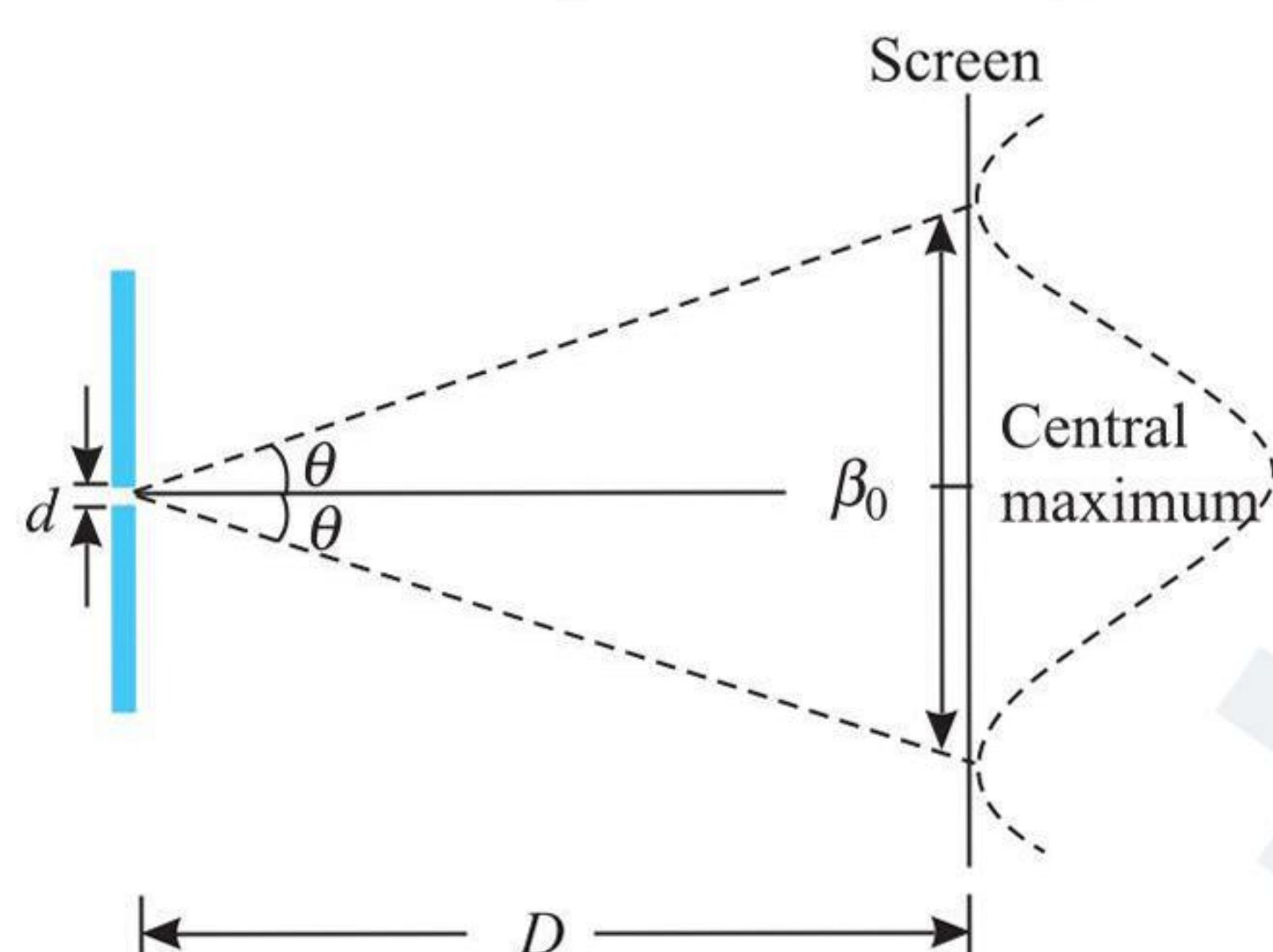
$$\theta'_1 = \frac{3\lambda}{2a}, \theta'_2 = \frac{5\lambda}{2a}, \theta'_3 = \frac{7\lambda}{2a} \dots$$

The relative intensities of the various maxima have been shown in Fig. (b).



Angular Width of Central Maximum

The angular width of the central maximum is the angular separation between the directions of the first minima on the two sides of the central maximum, as shown in figure.



The directions of first minima on either side of central maximum are given by $\theta = \lambda/a$. This angle is called half angular width of central maximum.

$$\therefore \text{Angular width of central maximum} = 2\theta = \frac{2\lambda}{a}$$

Linear Width of Central Maximum

If D is the distance of the screen from the single slit, then the linear width of central maximum will be

$$\beta_0 = D \times 2\theta = \frac{2D\lambda}{a}$$

$$\left[2\theta(\text{rad}) = \frac{\text{Arc}}{\text{Radius}} = \frac{\beta_0}{D} \right]$$

Linear Width of a Secondary Maximum

The angular width of n^{th} secondary maximum is the angular separation between the directions of n^{th} and $(n+1)^{\text{th}}$ minima.

$$\text{Direction of } n^{\text{th}} \text{ minimum, } \theta_n = n \frac{\lambda}{a}$$

$$\text{Direction of } (n+1)^{\text{th}} \text{ minimum, } \theta_{n+1} = (n+1) \frac{\lambda}{a}$$

$\therefore$ Angular width of n^{th} secondary maximum

$$= \theta_{n+1} - \theta_n = (n+1) \frac{\lambda}{a} - n \frac{\lambda}{a} = \frac{\lambda}{a}$$

Hence, the linear width of n^{th} secondary maximum = Angular width $\times D$

$$\text{or } \beta = \frac{D\lambda}{a}$$

Clearly, $\beta_0 = 2\beta$

Thus the central maximum of a diffraction pattern is twice as wide as any secondary maximum.

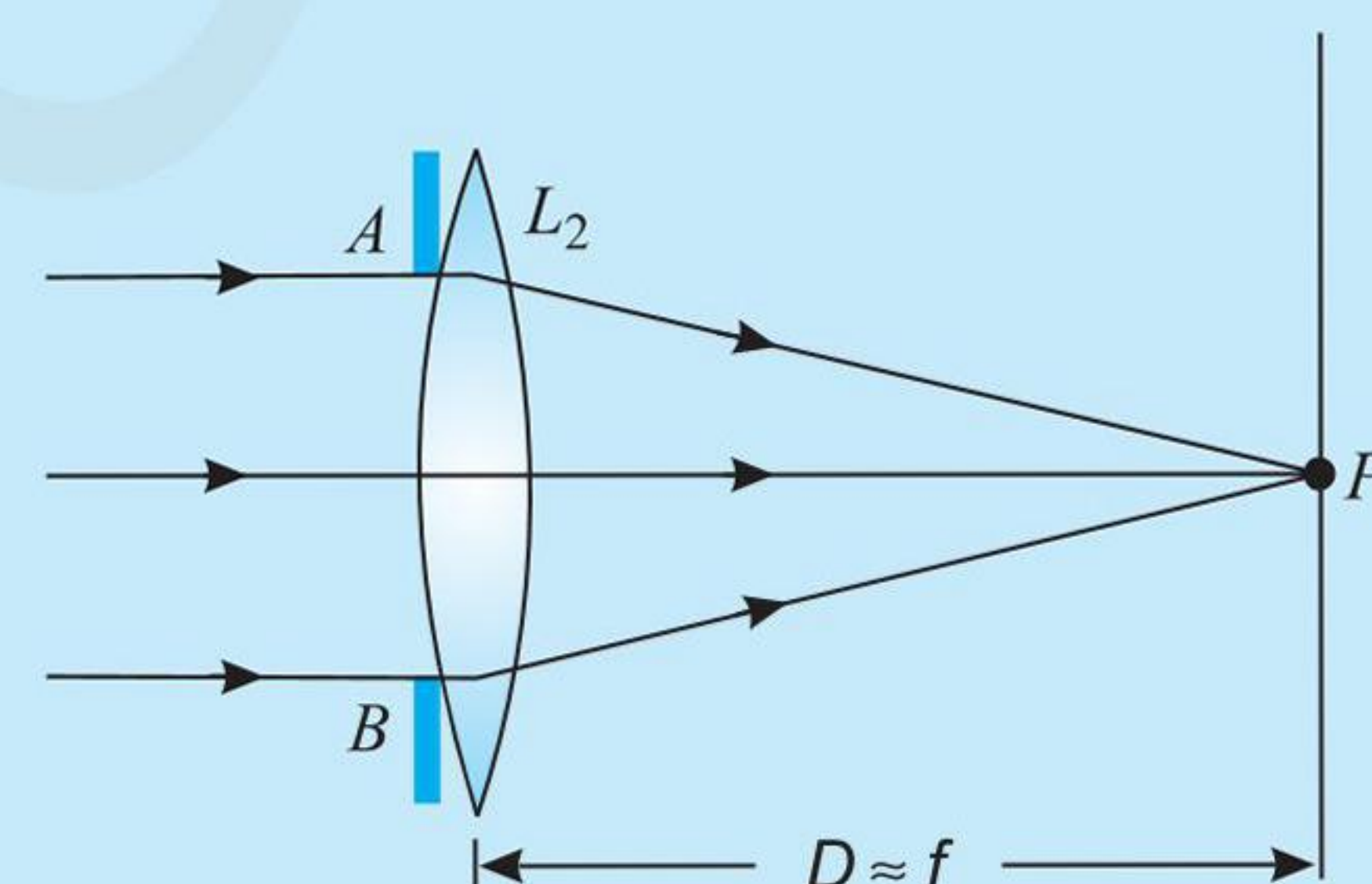
Clearly, width of a secondary maximum $\propto \frac{1}{\text{slit width}}$

As the slit width is increased, the secondary maxima get narrower. If the slit is sufficiently wide, the secondary maxima disappear and only the central maximum is obtained which is the sharp image of the slit. Thus a distinct diffraction pattern is possible only if the slit is very narrow.

Special Cases

Lens L_2 is shifted very near to slit AB . In this case distance between slit and screen will be nearly equal to the focal length

$$\text{of lens } L_2 \text{ (i.e. } D \approx f) \theta_n = \frac{x_n}{f} = \frac{n\lambda}{a} \Rightarrow x_n = \frac{n\lambda f}{a}$$



$$\text{Width of central maxima } w_x = 2x = \frac{2\lambda f}{a} \text{ and angular width of}$$

$$\text{central maxima } w_B = \frac{2x}{f} = \frac{2\lambda}{a}$$

Fringe width: Distance between two consecutive maxima (bright fringe) or minima (dark fringe) is known as fringe width. Fringe width of central maxima is doubled then the width of other maxima i.e.,

$$\beta = x_{n+1} - x_n = (n+1) \frac{\lambda D}{a} - n \frac{\lambda D}{a} = \frac{\lambda D}{a}$$

Important Points:

- Actually, there is no significant difference between an interference pattern and a diffraction pattern. Interference is usually referred to as the superposition of only a few (two, three or four) waves, whereas diffraction is concerned with the superposition of a very large number of waves.
- When $x = \frac{\lambda D}{d}$, a bright fringe is obtained in the interference pattern of two narrow slits distance d apart whereas when $x = \frac{\lambda D}{d}$, first minimum (dark band) is obtained in the diffraction pattern of a single slit of width a .

ILLUSTRATION 4.1

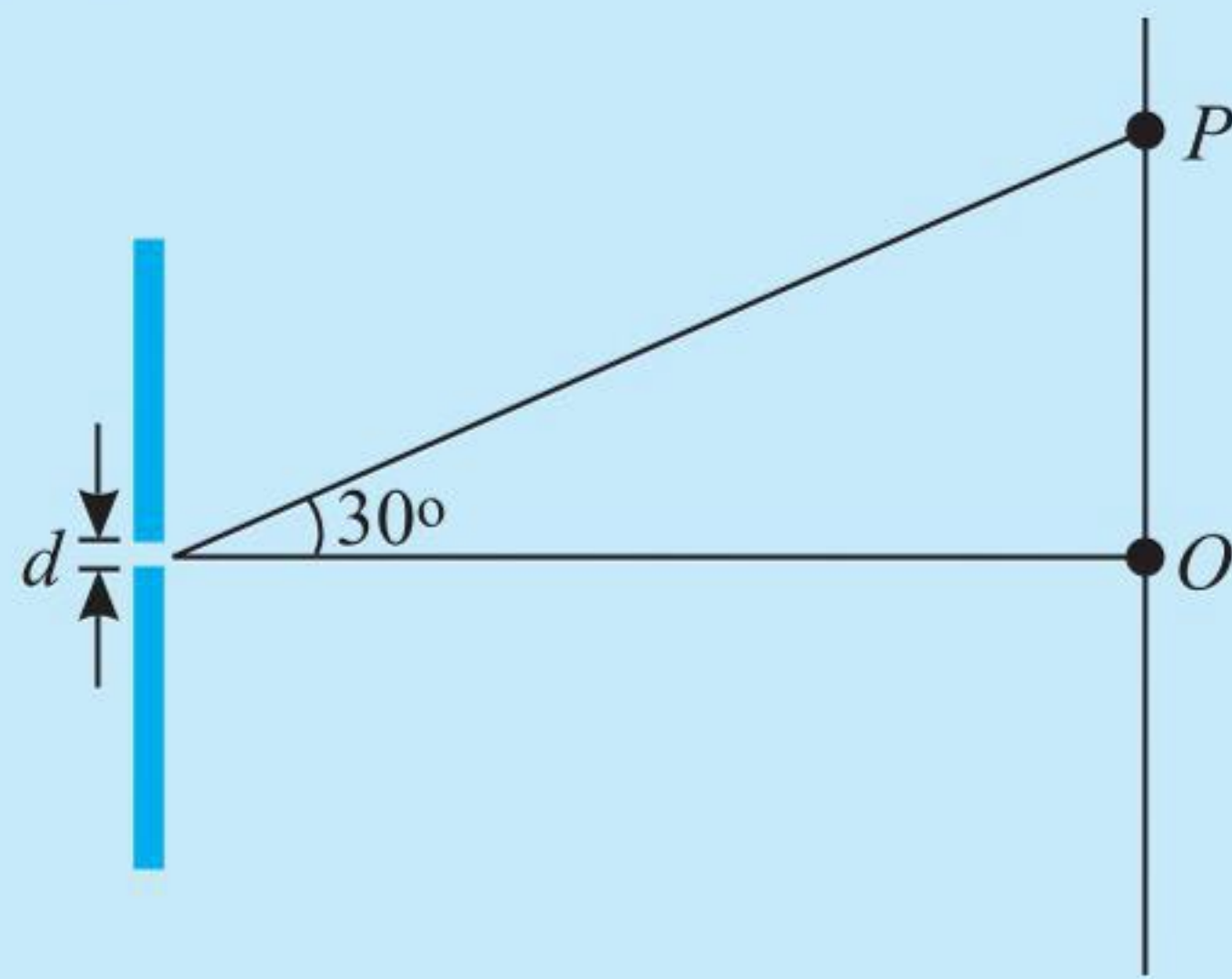
Why is the diffraction of sound waves more evident in daily experience than that of light wave?

Sol. The frequencies of sound waves lie between 20 Hz to 20 kHz, their wavelength ranges between 15 m to 15 mm. The diffraction occurs if the wavelength of waves is nearly equal to slit width.

The wavelength of light waves is 7000×10^{-10} m to 4000×10^{-10} m. For observing diffraction of light we need very narrow slit width. In daily life experience we observe the slit width very near to the wavelength of sound waves as compared to light waves. Thus, the diffraction of sound waves is more evident in daily life than that of light waves.

ILLUSTRATION 4.2

A slit of width ' d ' is illuminated by white light. For what value of ' d ' is the first minimum, for red light of $\lambda = 650$ nm, located at point P ? For what value of the wavelength of light will the first diffraction maxima also fall at P ?



Sol. For first minimum, $n = 1$

$$d \sin 30^\circ = \lambda$$

$$\text{or } d \times \frac{1}{2} = 650 \text{ nm} \quad \text{or } d = 1300 \text{ nm}$$

For first maximum to lie at P , $d \sin \theta = \frac{3}{2} \lambda'$

$$\text{or } \lambda' = \frac{2d \sin \theta}{3} = \frac{2 \times 1300 \times \sin 30^\circ}{3} = 433.3 \text{ nm}$$

ILLUSTRATION 4.3

A screen is placed 2 m away from the single narrow slit. Calculate the slit width if the first minimum lies 5 mm on either side of the central maximum. Incident plane waves have a wavelength of 5000 Å.

Sol. For the first minimum, $\theta = \frac{\lambda}{a}$... (i)

where a is the slit width. Since the first minimum is 5 mm on either side of the central maximum on the screen and it subtends an angle θ at the slit 2 m away,

$$\theta = \frac{5 \text{ mm}}{2 \text{ m}} = \frac{5 \times 10^{-3} \text{ m}}{2 \text{ m}} \quad \left[\text{as } \theta = \frac{l}{r} \right]$$

$$\text{or } \theta = 2.5 \times 10^{-3} \text{ rad} \quad \dots (ii)$$

From Eqs. (i) and (ii), $\frac{\lambda}{a} = 2.5 \times 10^{-3}$

$$\text{or } a = \frac{\lambda}{2.5 \times 10^{-3}} = \frac{5000 \times 10^{-8} \text{ m}}{2.5 \times 10^{-3}} = 2 \times 10^{-2} \text{ m} = 0.02 \text{ m}$$

ILLUSTRATION 4.4

A parallel beam of light of wavelength 600 nm is incident normally on a slit of width ' a '. If the distance between the slits and the screen is 0.8 m and the distance of the second order maximum from the centre of the screen is 15 mm, calculate the width of the slit.

Sol. Here, $\lambda = 600 \text{ nm} = 6 \times 10^{-7} \text{ m}$, $D = 0.8 \text{ m}$,
 $n = 2$, $x_2 = 15 \text{ mm} = 15 \times 10^{-3} \text{ m}$

Distance of n^{th} order secondary maximum from the centre of the screen, $x_n = \frac{(2n+1)\lambda D}{2a}$

$$\text{Thus, } x_2 = \frac{5\lambda D}{2a}$$

$$\text{or } a = \frac{5\lambda D}{2x_2} = \frac{5(6 \times 10^{-7} \text{ m})(0.8 \text{ m})}{2(15 \times 10^{-3} \text{ m})} = 8 \times 10^{-5} \text{ m}$$

ILLUSTRATION 4.5

Light of wavelength 550 nm is incident as parallel beam on a slit of width 0.1 mm. Find the angular width and the linear width of the principal maxima in the resulting diffraction pattern on a screen kept at a distance of 1.1 m from the slit. Which of these widths would not change if the screen were moved to a distance of 2.2 m from the slit?

Sol. Here $\lambda = 550 \text{ nm} = 550 \times 10^{-9} \text{ m}$

$$a = 0.1 \text{ mm} = 0.1 \times 10^{-3} \text{ m}, D = 1.1 \text{ m}$$

Angular width of principal maximum,

$$2\theta = \frac{2\lambda}{a} = \frac{2 \times 550 \times 10^{-9}}{0.1 \times 10^{-3}} = 1.1 \times 10^{-2} \text{ rad}$$

Linear width of principal maximum,

$$\beta_0 = \frac{2D\lambda}{a} = 1.1 \times 0.011 = 0.0121 \text{ m} = 12.1 \text{ mm}$$

When the screen is moved to a distance of 2.2 m, the angular width would not change because it is independent of this distance D .

ILLUSTRATION 4.6

Two wavelengths of sodium light 590 nm and 596 nm are used in turn to study the diffraction taking place at a single slit of aperture 2×10^{-6} m. The distance between the slit and the screen is 1.5 m. Calculate the separation between the position of first maximum of the diffraction pattern obtained in the two cases.

Sol. For the first secondary maximum, $\theta = \frac{3\lambda}{2a}$

Denoting the wavelengths of the spectral lines D_1 and D_2 by λ_1 and λ_2 ,

$$\theta_1 = \frac{3\lambda_1}{2a} \quad \text{and} \quad \theta_2 = \frac{3\lambda_2}{2a}$$

$$\text{Clearly, } \Delta\theta = (\theta_2 - \theta_1) = \frac{3\lambda_2}{2a} - \frac{3\lambda_1}{2a} = \frac{3(\lambda_2 - \lambda_1)}{2a}$$

$$\text{or } \Delta\theta = \frac{3(596 - 590) \times 10^{-9}}{2(2 \times 10^{-6})} = 4.5 \times 10^{-3} \text{ rad}$$

Thus, spacing between the first maxima of two sodium lines
 $= \Delta\theta \times (1.5) = (4.5 \times 10^{-3})(1.5) = 6.75 \text{ mm}$

VALIDITY OF RAY OPTICS: FRESNEL'S DISTANCE

Geometrical optics is based upon the rectilinear propagation of light. Diffraction effects show that light does not travel in straight lines.

The distance of the screen from the slit, so that spreading of light due to diffraction from the centre of screen is just equal to size of the slit, is called **Fresnel distance**. It is denoted by Z_F .

The diffraction pattern of a slit consists of secondary maxima and minima on the two sides of the central maximum. Therefore, one can say that on diffraction from a slit, light spreads on the screen in the form of central maximum. The angular position of first secondary minimum is called **half angular width** of the central maximum and it is given by

$$\theta = \frac{\lambda}{d} \quad (\text{provided } \theta \text{ is small})$$

If the screen is placed at a distance D from the slit, then the linear spread of the central maximum is given by

$$y_1 = D\theta_1 = \frac{D\lambda}{d}$$

It is, in fact, the distance of first secondary minimum from the centre of the screen. It follows that as the screen is moved away (D is increased), the linear size of the central maximum i.e. spread of light on the screen goes on increasing.

From the definition of Fresnel distance,

when $D = D_F$, $y_1 = d$ (size of the slit)

Setting this condition in the above equation, we have

$$d = \frac{D_F \lambda}{d} \quad \text{or} \quad D_F = \frac{d^2}{\lambda}$$

It follows that if screen is placed at a distance beyond D_F , the spreading of light due to diffraction will be quite large as compared to the size of the slit.

If $D < D_F$, then there will not be too much broadening by diffraction i.e., the light will travel along straight lines and the concepts of ray optics will be valid.

$$\text{As } D < D_F \quad \text{or} \quad D < \frac{d^2}{\lambda} \quad \text{or} \quad d > \sqrt{\lambda D}$$

For a given value of D , the quantity $\sqrt{\lambda D}$ is called the size of Fresnel zone and is denoted by d_F .

$$\text{i.e., } d_F = \sqrt{\lambda D}$$

Hence the concepts of ray optics can be conveniently used without introducing any appreciable error if the size of the aperture is greater than the size of the Fresnel zone,

$$\text{i.e., } d > d_F$$

Here, $a = 4 \text{ mm} = 4 \times 10^{-3} \text{ m}$

$$\lambda = 400 \text{ nm} = 400 \times 10^{-9} \text{ m}$$

$$\text{Thus, } D_F = \frac{(4 \times 10^{-3})^2}{400 \times 10^{-9}} = 40 \text{ m}$$

Difference between interference and diffraction

Interference	Diffraction
(1) It is the phenomenon of superposition of two waves coming from two different coherent sources	(1) It is the phenomenon of superposition of two waves coming from two different parts of the same wavefront
(2) In interference pattern, all bright lines are equally bright and equally spaced	(2) All bright lines are not equally bright and equally wide. Brightness and width go on decreasing with the angle of diffraction
(3) All dark lines are totally dark	(3) Dark lines are not perfectly dark. Their contrast with bright lines and width goes on decreasing with angle of diffraction
(4) In interference bands are large in number	(4) In diffraction bands are a few in number

CONCEPT APPLICATION EXERCISE 4.1

1. A slit of width 0.15 cm is illuminated by light of wavelength $5 \times 10^{-7} \text{ m}$ and a diffraction pattern is obtained on a screen 21 m away. Calculate the width of the central maxima.
2. Light of wavelength 5900 Å falls normally on a slit of width $11.8 \times 10^{-5} \text{ cm}$ and the resulting Fraunhofer diffraction is received on a screen. Calculate the angular position of the first minimum. Also, find the angular width of the central maximum.
3. Microwaves of frequency 12,000 MHz are incident normally on a rectangular slit of width 10 cm. Calculate angular spread of the central maximum of the diffraction pattern of the slit.
4. Monochromatic light passes through a slit whose width is 0.050 mm. At what angle will the third minimum be seen and what is the width of the central maximum on a screen located 1.0 m from the slit for $\lambda = 400 \text{ nm}$?
5. Parallel light of wavelength 5000 Å falls normally on a single slit. The central maximum spreads out to 30° on either side of the incident light. Find the width of the slit. For what width of the slit the central maximum would spread out to 90° from the direction of the incident light?
6. Calculate the distance that a beam of light of wavelength 500 nm can travel without significant broadening, if the diffracting aperture is 3 mm wide.

ANSWERS

1. 1.4 mm 2. $30^\circ, 60^\circ$ 3. 28.7° 4. $1.4^\circ, 1.6 \text{ cm}$
5. $10^{-6} \text{ m}, 5 \times 10^{-7} \text{ m}$ 6. 18 m

ILLUSTRATION 4.7

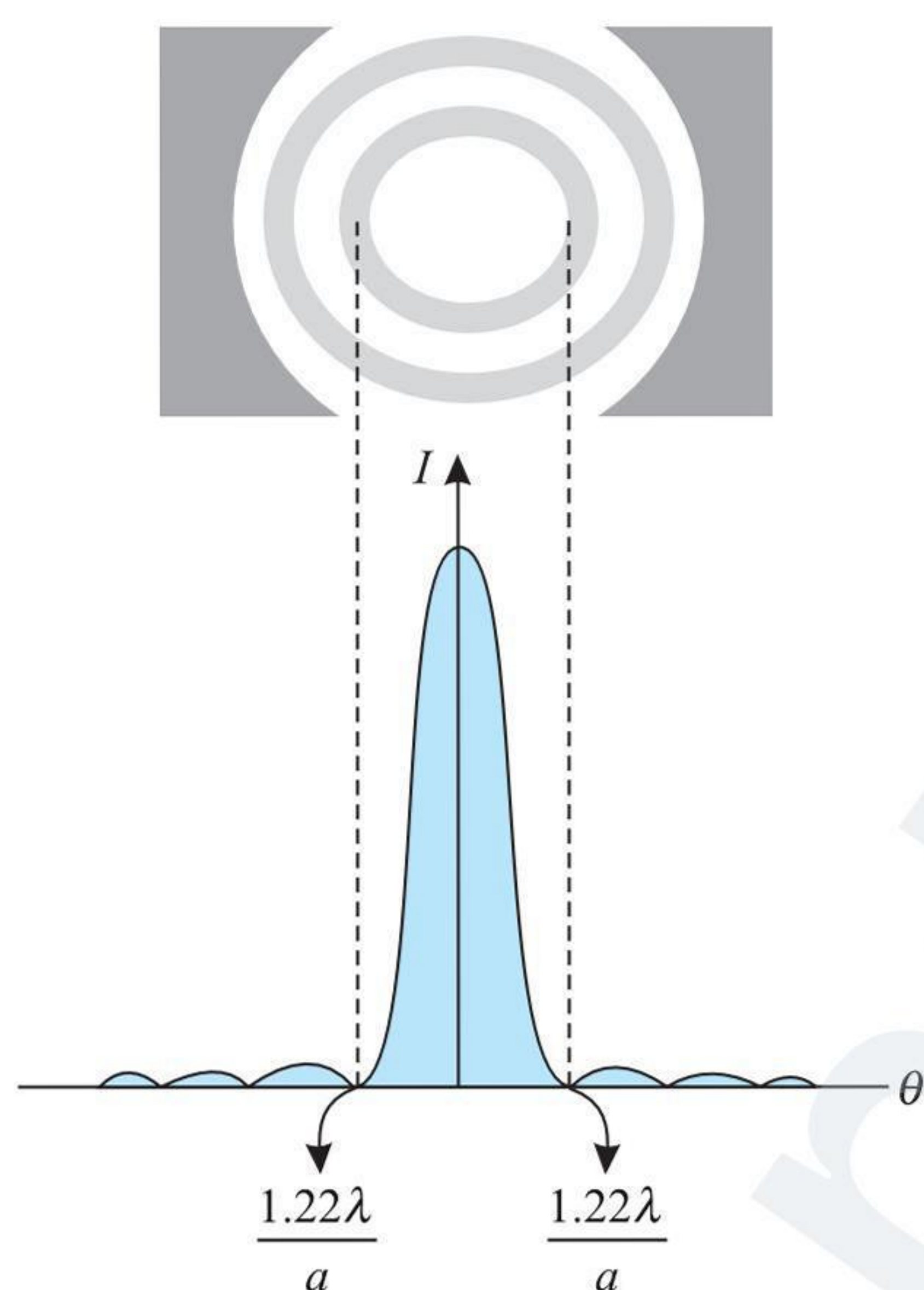
Estimate the distance for which ray optics a good approximation for an aperture of 4 mm and the wavelength is 400 nm?

Sol. Fresnel distance (distance for which ray optics is a good approximation), $D_F = \frac{a^2}{\lambda}$

DIFFRACTION BY A CIRCULAR APERTURE

Here we consider diffraction by a circular aperture—that is, a circular opening, such as a circular lens, through which light can pass. In all the optical instruments, we use lenses or mirrors. The optical instruments are based on the laws of geometrical optics for defining magnification. Therefore, the image of a point object produced by an optical instrument must be a geometric point. In practice, the image of a bright geometric point produced by a lens is never a geometric point. It is always a circular patch of light with a bright disc at the centre surrounded by alternate dark and bright rings. It arises because of diffraction of light.

Suppose that light of wavelength λ is incident on a lens of diameter a (or a mirror with a circular aperture of diameter a). The light diffracted from this circular aperture will show a diffraction pattern consisting of circular rings as shown in figure.



If θ is the angle between the central maximum (called Airy disc) and the first minimum (the first dark fringe). The (complex) analysis of such patterns shows that the first minimum for the diffraction pattern of a circular aperture of diameter a is located by

$$\sin \theta = 1.22 \frac{\lambda}{a} \quad (\text{first minimum – circular aperture}) \quad \dots(i)$$

Since θ is small, we may also write $\sin \theta \approx \theta$

$$\text{Thus, } \theta = 1.22 \frac{\lambda}{a}$$

We write a similar equation for a narrow slit

$$\sin \theta = \frac{\lambda}{a} \quad (\text{first minimum – single slit}) \quad \dots(ii)$$

which locates the first minimum for a long narrow slit of width a . The main difference is the factor 1.22, which enters because of the circular shape of the aperture.

RESOLVING POWER OF OPTICAL INSTRUMENTS

RAYLEIGH'S CRITERION

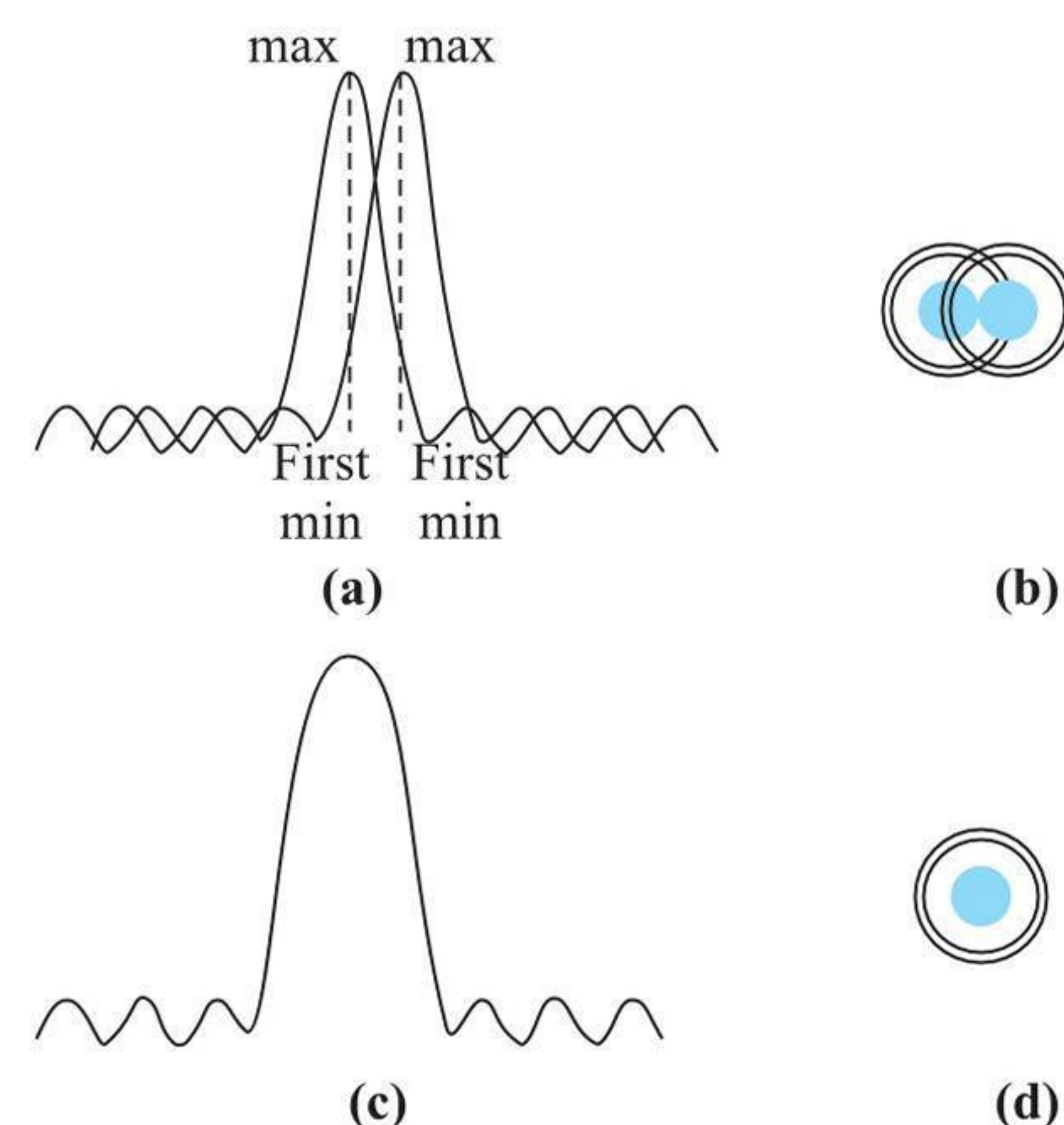
Two bright point objects lying very close to each other are seen with a naked eye or with the help of an optical instrument (a microscope or a telescope), then they may or may not be seen as two separate distinct objects due to the overlapping of their diffraction patterns. The ability to separate distinct objects depends upon limit of resolution and the resolving power of an optical instrument.

The smallest linear or angular separation between two point objects at which they can be just separately seen or resolved by an optical instrument is called the **limit of resolution of the instrument**.

The resolving power of an optical instrument is its ability to resolve or separate the images of two nearby point objects so that they can be distinctly seen. It is equal to the reciprocal of the limit of resolution of the optical instrument.

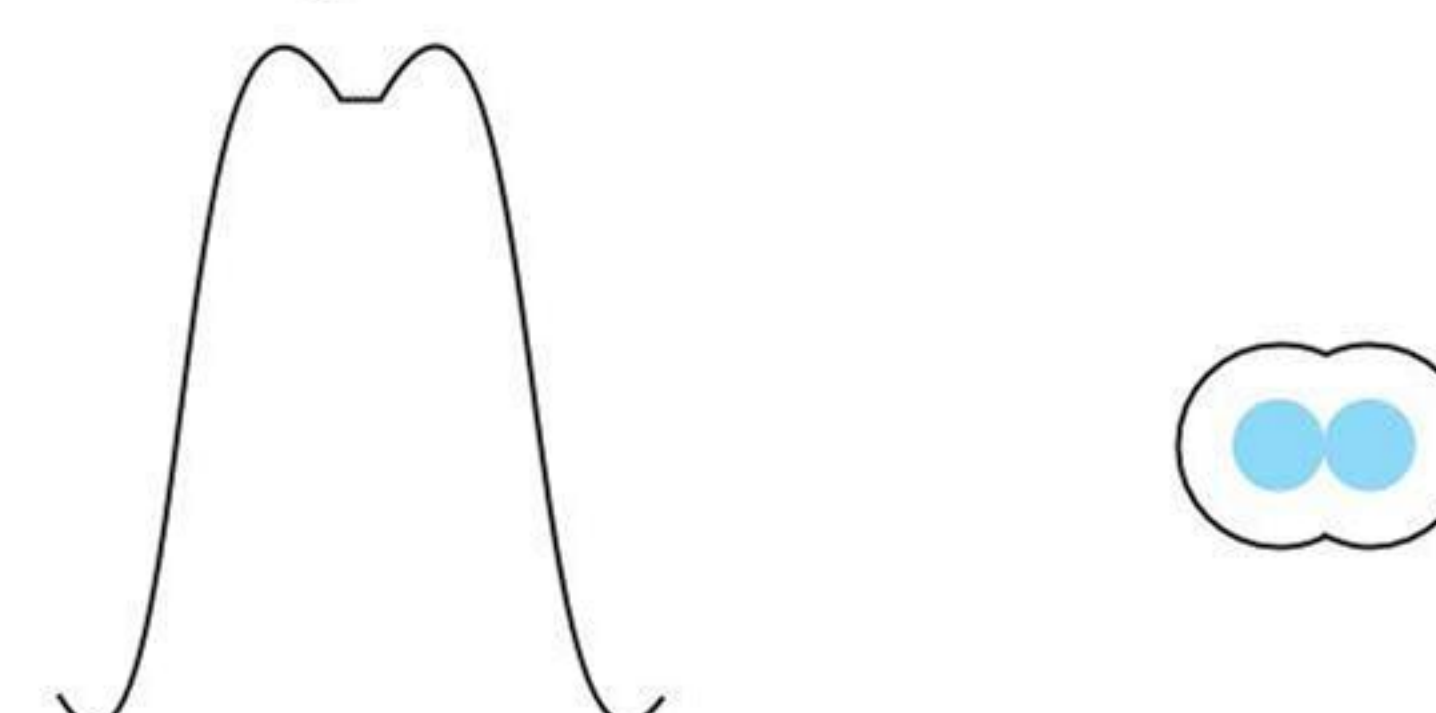
The smaller the limit of resolution of an optical instrument, greater is its resolving power.

Rayleigh's criterion for resolution: If we look through a telescope at two stars lying close to each other, their diffraction patterns overlap and the resultant pattern is little broader but otherwise similar to that of a single star, as shown in figure. So the two stars cannot be resolved or separately seen.



Figures (a) and (b) show overlapping of diffraction patterns of two close point objects and Figs. (c) and (d) show their resultant diffraction patterns.

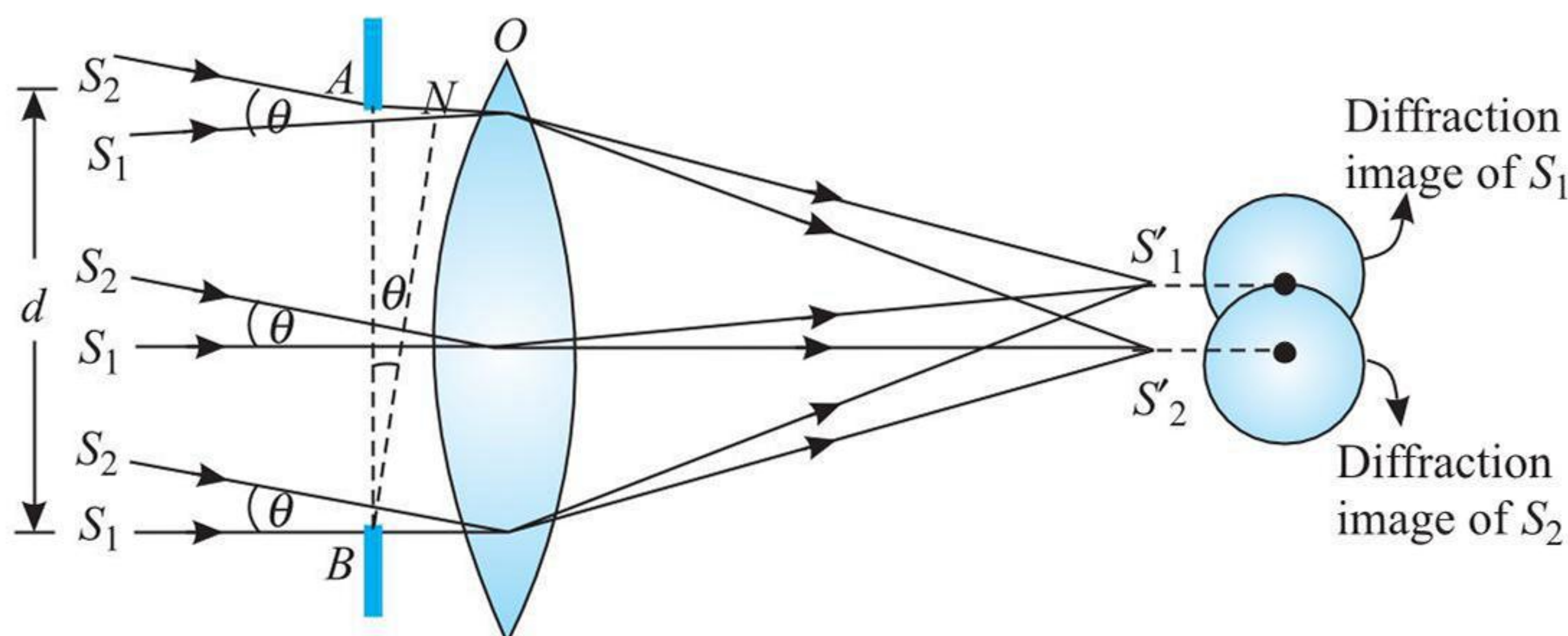
According to Rayleigh's criterion, the images of two point objects are just resolved when the central maximum of the diffraction pattern of one falls over the first minimum of the diffraction pattern of the other, as shown in figure. When seen through the telescope, the resultant diffraction has a well-marked depression at the top, showing that these are really two stars and not one. Thus the images of two stars have been just resolved.



Observation from telescope at Rayleigh's criterion

RESOLVING POWER OF A TELESCOPE

Let us consider that two distant stars S_1 and S_2 are observed through a telescope, whose objective (O) is of diameter ' a '. Let θ be the angular separation between the two stars and λ , the wavelength of the light, in which the two stars are observed. The objective lens forms the diffraction images S_1' and S_2' of the two stars as shown in figure.



The path difference between the rays of light from the stars S_1 and S_2 is equal to AN . The two images S_1' and S_2' will be just resolved, when the path difference AN is equal to the wavelength (λ) of the light. Since the aperture of the objective is circular, it corresponds to the smallest angular separation θ between the two stars, so that their images S_1' and S_2' are just resolved and is given by

$$\theta = \frac{1.22 \lambda}{a} \quad \dots(i)$$

Here, θ is called **limit of resolution of a telescope**

Resolving power of a telescope is defined as the reciprocal of the smallest angular separation between two distant objects, so that they appear just separated, when seen through the telescope. Therefore, the resolving power of the telescope,

$$R = \frac{1}{\theta} = \frac{a}{1.22 \lambda} \quad \dots(ii)$$

From the above expression, $R \propto a$, i.e., the larger the aperture of a telescope, higher is its resolving power. Therefore, telescopes with larger objectives form sharper images and are capable of resolving objects of smaller angular separation. The images are also brighter, not only because the light is concentrated in a **smaller Airy's disc**, but also due to the reason that a larger lens has more light gathering capacity.

Also, $R \propto \frac{1}{\lambda}$, it means the light of smaller wavelength is used to see the objects.

ILLUSTRATION 4.8

Calculate the separation of two points on the moon that can be resolved using 600 cm telescope. Given the distance of the moon from the earth is 3.8×10^{10} cm. The wavelength most sensitive to the eye is 5.5×10^{-5} cm.

Sol. Here $D = 600$ cm, $\lambda = 5.5 \times 10^{-5}$ cm

$$\text{Limit of resolution, } \theta = \frac{1.22 \lambda}{D} = \frac{1.22 \times 5.5 \times 10^{-5}}{600} = 1.1 \times 10^{-5} \text{ rad}$$

Let x be the distance between two points on the moon and d be the distance between the moon and the objective of the telescope.

$$\text{Then } \theta = \frac{x}{d}$$

$$\text{or } x = d \times \theta = 3.8 \times 10^{10} \times 1.1 \times 10^{-5} = 4180 \text{ cm}$$

ILLUSTRATION 4.9

Two stars at a distance of 10 light years are viewed through a telescope having a lens of 20 cm diameter. What is the minimum separation of these stars for which they would be distinguishable as separate objects? $\lambda = 6000 \text{ \AA}$

Sol. Here, $a = 20$ cm, $\lambda = 6000 \text{ \AA} = 6 \times 10^{-5}$ cm

$$D \text{ (distance of the star)} = 10 \text{ ly}$$

$$\text{Thus, } \Delta\theta = \frac{1.22 \lambda}{a} = \frac{1.22(6 \times 10^{-5} \text{ cm})}{20 \text{ cm}} = 0.366 \times 10^{-5} \text{ rad}$$

Minimum separation between the stars

$$= D\Delta\theta = (10 \text{ ly})(0.366 \times 10^{-5} \text{ rad})$$

$$= 3.66 \times 10^{-5} \text{ ly}$$

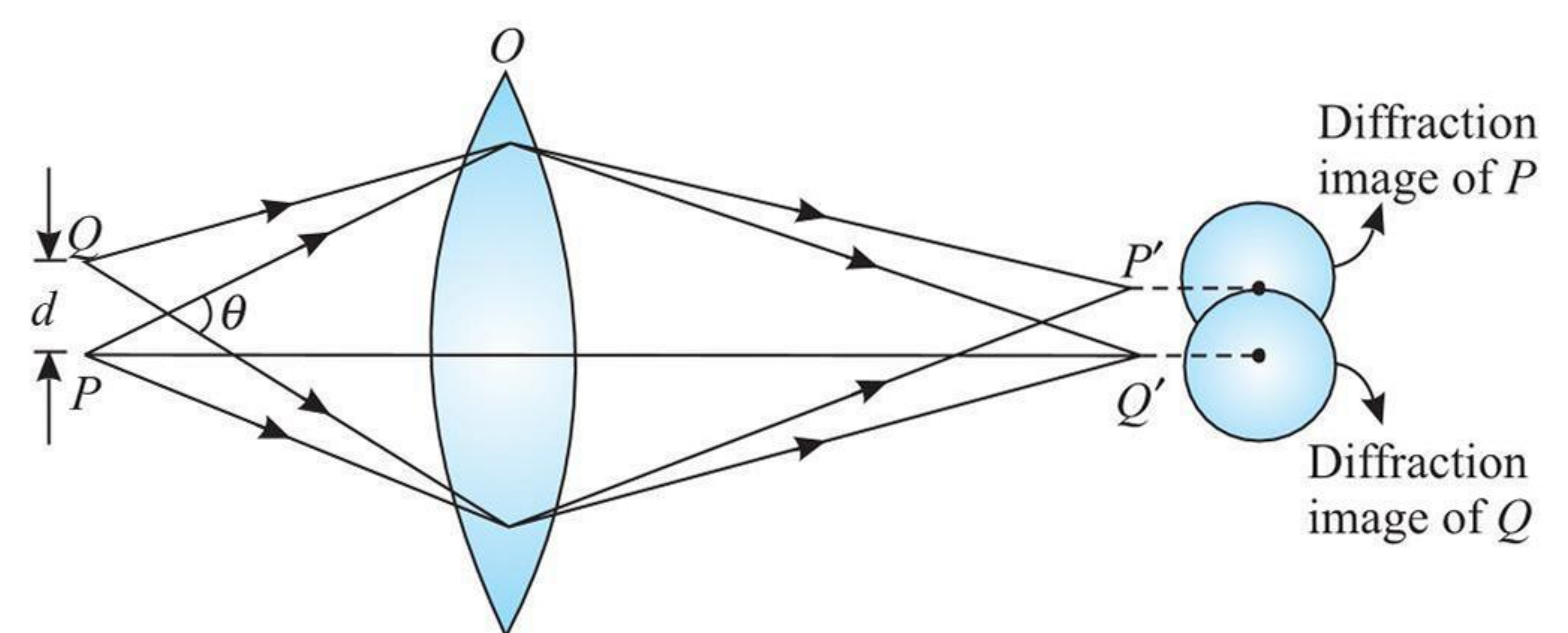
$$= 3.66 \times 10^{-5} (9.461 \times 10^{12} \text{ km})$$

$$= 3.46 \times 10^8 \text{ km}$$

RESOLVING POWER OF A MICROSCOPE

The resolving power of a microscope represents its ability to form distinctly separate images of two objects lying close to each other.

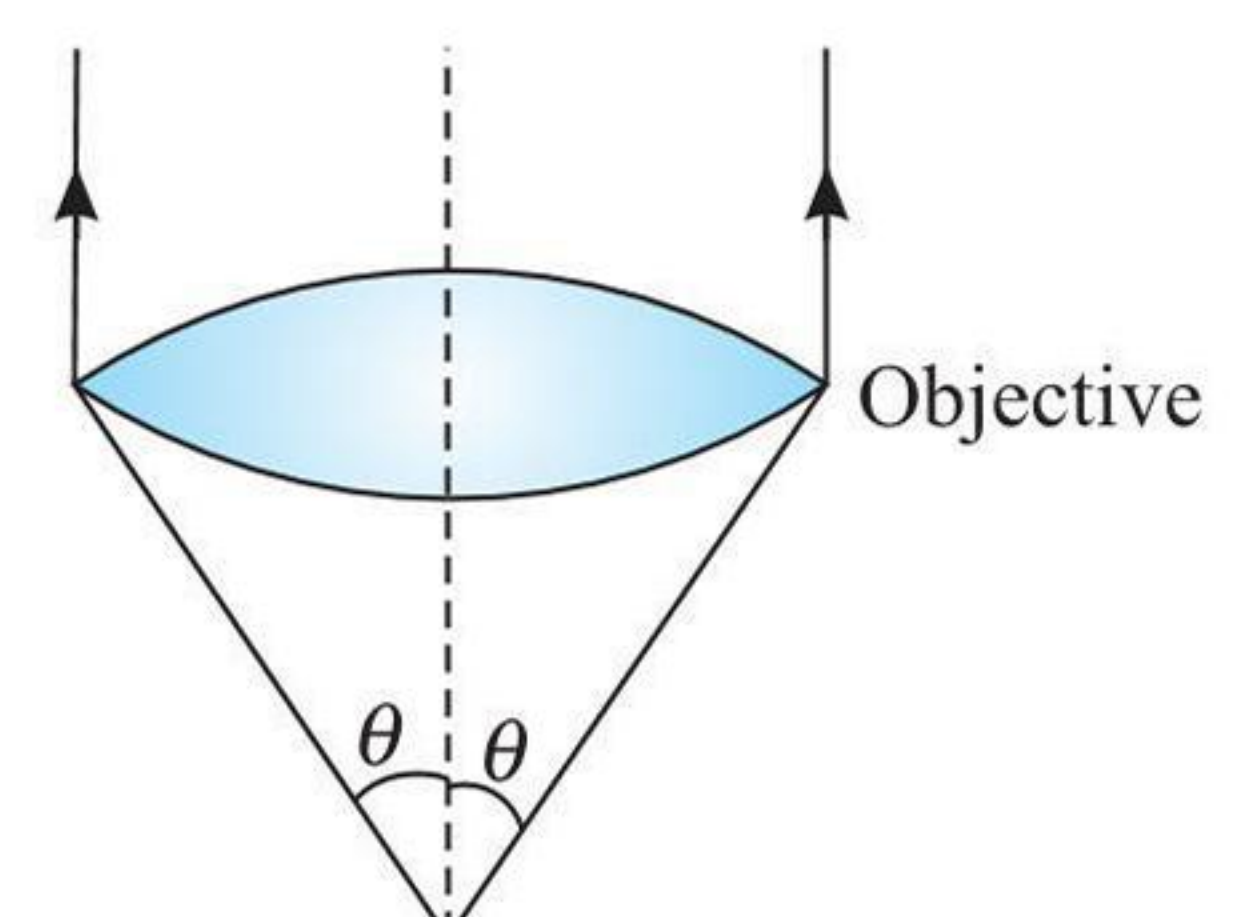
Consider that two objects A and B lying very close to each other are observed through a microscope. The objects are placed just beyond the focus of the objective (O) of the microscope. Let λ be the wavelength of the light used to observe the objects. The objective forms the diffraction images A' and B' of the two objects as shown in figure.



Let d be the least distance between the two objects so that their diffraction images A' and B' are just resolved. It can be proved that

$$d = \frac{1.22 \lambda}{2 \sin \theta} \quad \dots(i)$$

θ = half the angle of cone of light from each point object or the angle subtended by each point object on the radius of the objective (see figure).



The factor 1.22 appears due to circular aperture of the objective. If the space between the point objects and the objective lens is filled with medium (usually oil) of refractive index μ , then the above relation modifies to

$$d = \frac{1.22 \lambda}{2\mu \sin \theta} \quad \dots(ii)$$

The least distance given by Eq. (ii) is called the **limit of resolution** of the microscope.

The resolving power of a microscope is defined as the reciprocal of the least separation between two close objects, so that they appear just separated, when seen through the microscope.

Therefore, resolving power of microscope, $R = \frac{1}{d} = \frac{2\mu \sin \theta}{1.22 \lambda}$

From the above expression, the following conclusions can be drawn:

1. The resolving power of a microscope increases with increase in the value of the refractive index of the medium between its objective and the object. For this reason, **oil immersion objective microscopes** are used to achieve high resolving power.
2. The resolving power of a microscope increases with decrease in the value of the wavelength of the light used to illuminate the object. Since the wavelength of ultraviolet light is less than that of the visible light, the microscopes employing ultraviolet light for illuminating the objects are used to achieve high resolving power. Such microscopes are called **ultra microscopes**.

RESOLVING POWER OF THE EYE

Eye is an efficient natural optical instrument. The diameter (a) of the iris is about 2 mm and for green light ($\lambda = 500 \text{ nm} = 5 \times 10^{-5} \text{ cm}$), The limit of angular resolution of the eye,

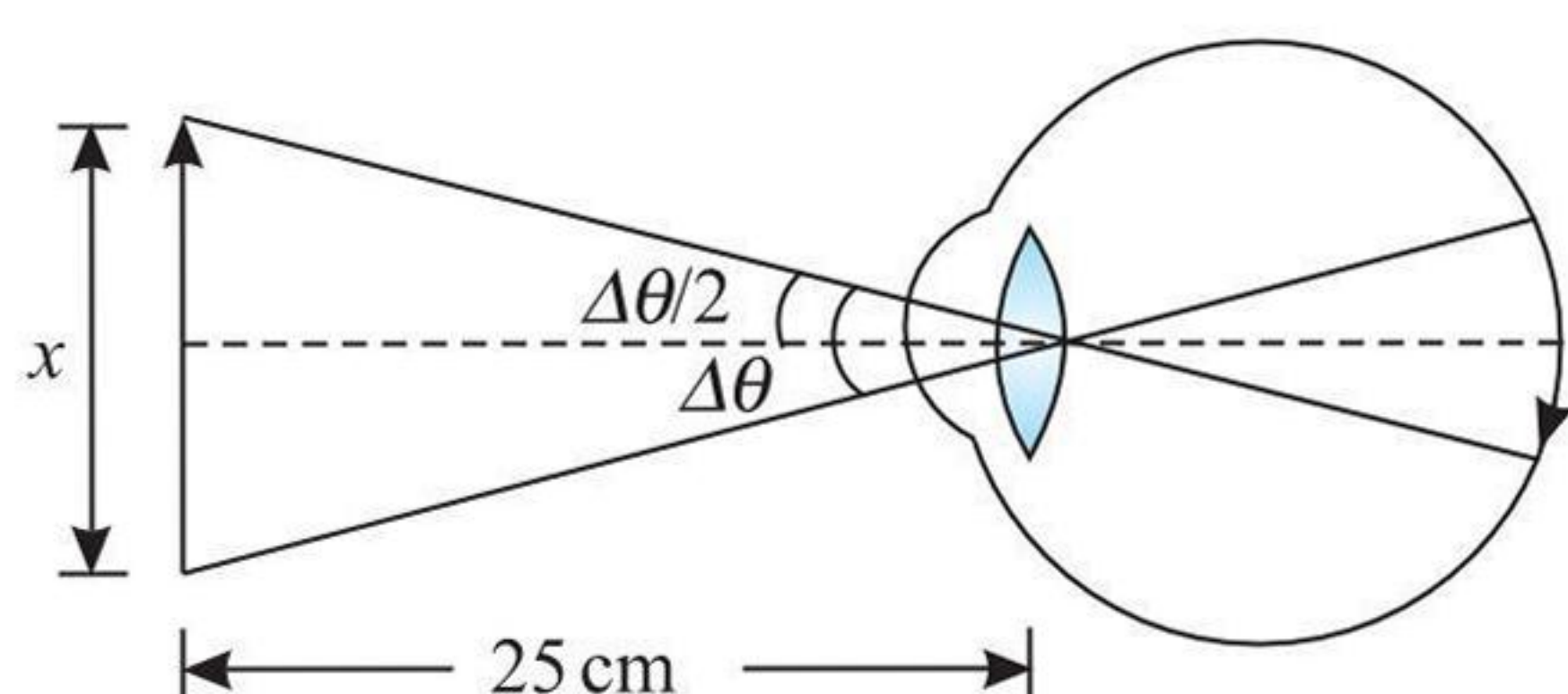
$$\Delta\theta = \frac{1.22 \lambda}{a} = \frac{(1.22)(5 \times 10^{-5} \text{ cm})}{2.00 \times 10^{-3} \text{ m}} = 3.05 \times 10^{-4} \text{ rad}$$

An average eye is not capable of resolving objects with an angular separation of less than $3 \times 10^{-4} \text{ rad}$. The reason for this is the size of the cones, which are closely packed at the fovea. The diameter of each cone is $2 \times 10^{-6} \text{ m}$. For two points to be clearly resolved, their images formed at the retina must be such that the two excited cones are separated by at least one cone.

Smallest Size an Eye can Resolve

The limit of angular resolution of the eye, $3 \times 10^{-4} \text{ rad}$, leads to the following smallest size that a normal eye can resolve. Suppose two point objects separated $x \text{ cm}$ are placed at distance of distinct vision (25 cm) from the eye as shown in figure. Since $\Delta\theta$ is small,

$$\tan\left(\frac{\Delta\theta}{2}\right) \approx \frac{\Delta\theta}{2} = \frac{x/2}{25 \text{ cm}}$$



$$\text{or } \Delta\theta = \frac{x}{25 \text{ cm}}$$

$$\text{or } x = (25 \text{ cm}) \Delta\theta = (25 \text{ cm})(3 \times 10^{-4} \text{ rad}) = 7.5 \times 10^{-3} \text{ cm} = 0.075 \text{ mm} \quad (\text{taking } \Delta\theta = 3 \times 10^{-4} \text{ rad})$$

ILLUSTRATION 4.10

Calculate the smallest angular separation resolved by the human eye. Given: Aperture $a = 2.5 \text{ mm}$ and effective $\lambda = 5500 \text{ \AA}$. If a scale with mm markings is viewed by the unaided eye, deduce the largest distance to which the markings will be visible.

Sol. Here, $\lambda = 5500 \text{ \AA} = 5500 \times 10^{-10} \text{ m}$,

$$a = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$$

$$\Delta\theta = 1.22 \left(\frac{\lambda}{a} \right) = 1.22 \left(\frac{5500 \times 10^{-10} \text{ m}}{2.5 \times 10^{-3} \text{ m}} \right) = 2.7 \times 10^{-4} \text{ rad} \quad \dots(i)$$

Angle subtended by 1mm markings from a distance $x \text{ m}$ i.e.,

$$\theta = \frac{1 \times 10^{-3} \text{ m}}{x \text{ m}} = \frac{1}{x} \times 10^{-3} \quad \dots(ii)$$

$$\text{From Eqs. (i) and (ii), } \frac{1}{x} \times 10^{-3} = 2.7 \times 10^{-4}$$

$$\text{or } x = \frac{10^{-3}}{2.7 \times 10^{-4}} = 3.7 \text{ m}$$

Thus, a normal eye can resolve objects that are separated by $\sim 0.1 \text{ mm}$ and placed at a distance of distinct vision. This is approximately equal to thickness of a human hair.

CONCEPT APPLICATION EXERCISE 4.2

1. Two parallel pillars are 11 km away from an observer. Find the minimum distance between the pillars so that they can be seen separately.
2. A telescope is used to resolve two stars separated by $4.6 \times 10^{-6} \text{ rad}$. If the wavelength of light used is $5460 \text{ \AA}$, what should be the aperture of the objective of the telescope?
3. A microscope has objective of aperture 8 mm and focal length 2.5 cm. Estimate its resolving power. Given $\lambda = 5500 \text{ \AA}$.
4. Calculate the least value of the angular separation of two stars which can be resolved by a telescope of 200 cm aperture. If the aperture of the human eye be 2 mm and if the focal length of the eyepiece be 1 inch, what must be the minimum focal of the objective if resolving power of the telescope is to be utilized. Take $\lambda = 5500 \text{ \AA}$.
5. Diameter of human eye lens is 2 mm. What will be the minimum distance between two points to resolve them, which are situated at a distance of 50 m from eye? The wavelength of light is $5,000 \text{ \AA}$.

ANSWERS

- | | | |
|--------------------------------------|-------------|---------------------------------|
| 1. 3.2 m | 2. 0.1488 m | 3. $2 \times 10^{-6} \text{ m}$ |
| 4. $3.35 \times 10^{-7} \text{ rad}$ | 5. 1.525 cm | |

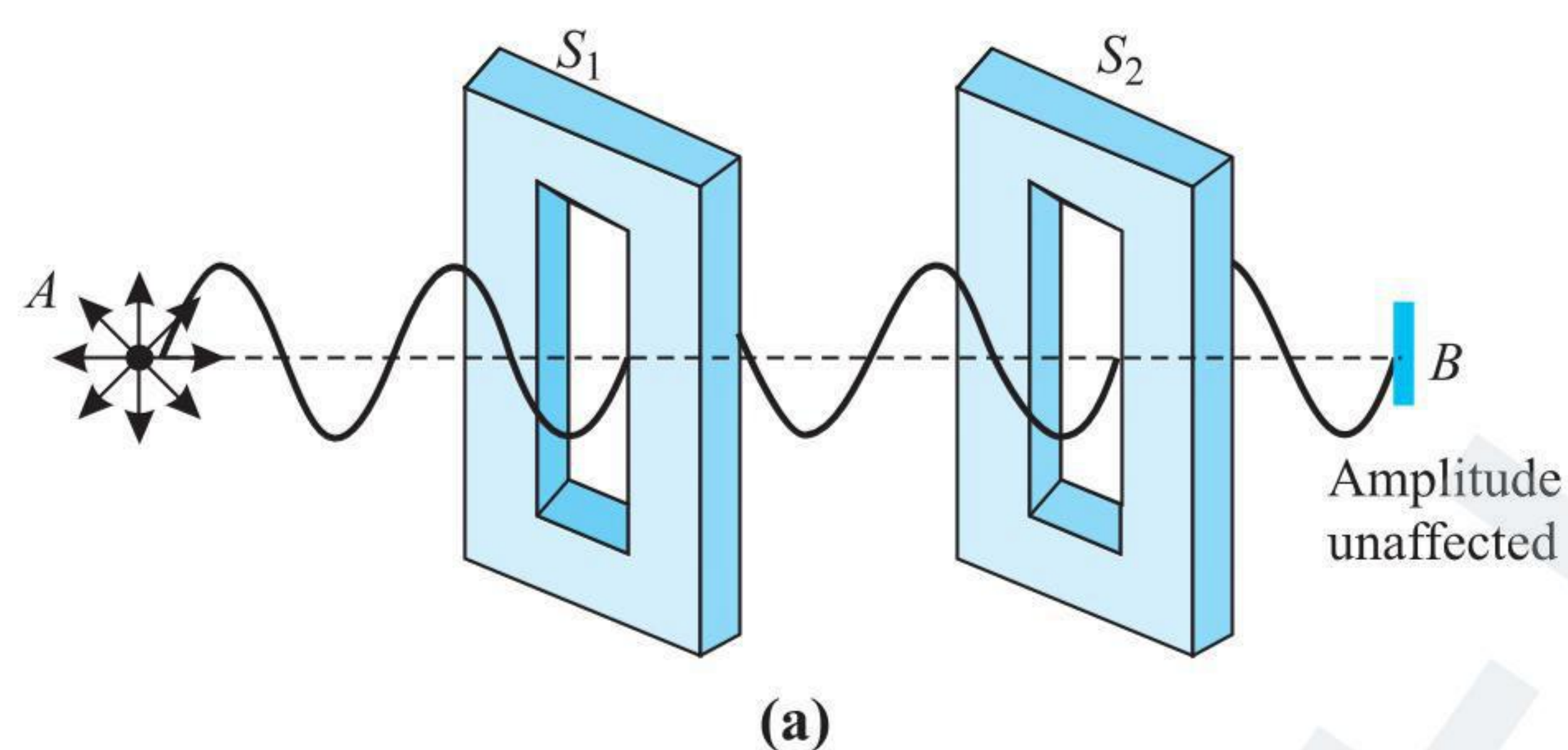
POLARIZATION

Through the phenomena of interference and diffraction, we established without doubt the wave nature of light. As both these phenomena of interference and diffraction are exhibited by either type of wave, they do not tell us the exact nature of light. It was only the discovery of another phenomenon, known as ‘polarization’ which conclusively proved the transverse nature of light waves.

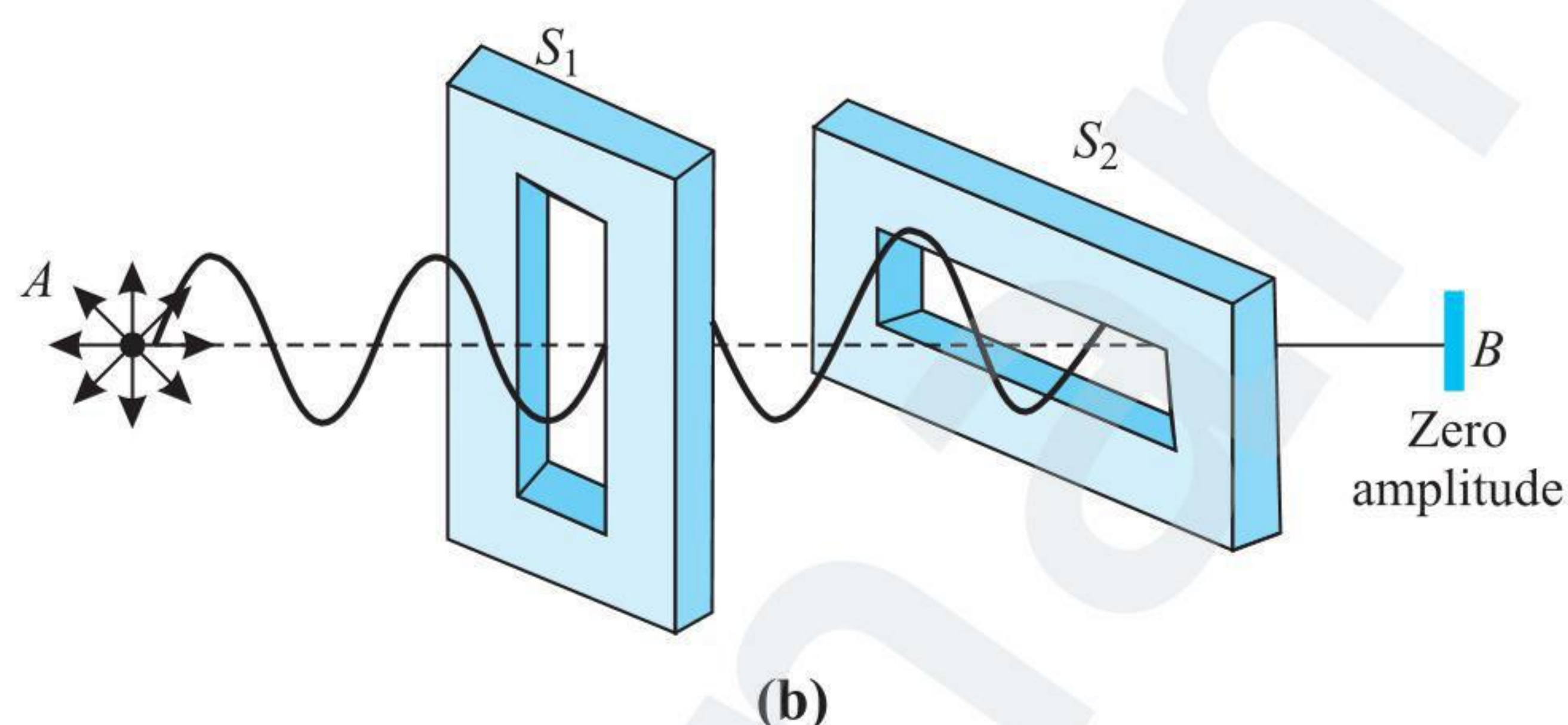
A wave exhibiting different properties in different directions is said to be **polarized** and the phenomenon is referred to as the **polarization**.

To understand the polarization of the waves, the following **mechanical analogy** is to be considered.

Two vertical slits S_1 and S_2 are placed parallel to each other and a string is passed through them. The end B of the string is fixed and by continuously jerking the string at the end A in all directions perpendicular to AB , transverse waves are set up in it in all possible perpendicular planes. Then, transverse wave, whose vibrations are parallel to the length of the slit, will only pass through the slit S_1 . If the slit S_2 is parallel to S_1 , then amplitude of vibrations will remain unaffected [see Fig. (a)].



On the other hand, when the slit S_2 is at right angle to the slit S_1 , no vibrations will pass through it i.e. amplitude of vibrations will become zero [see Fig. (b)].



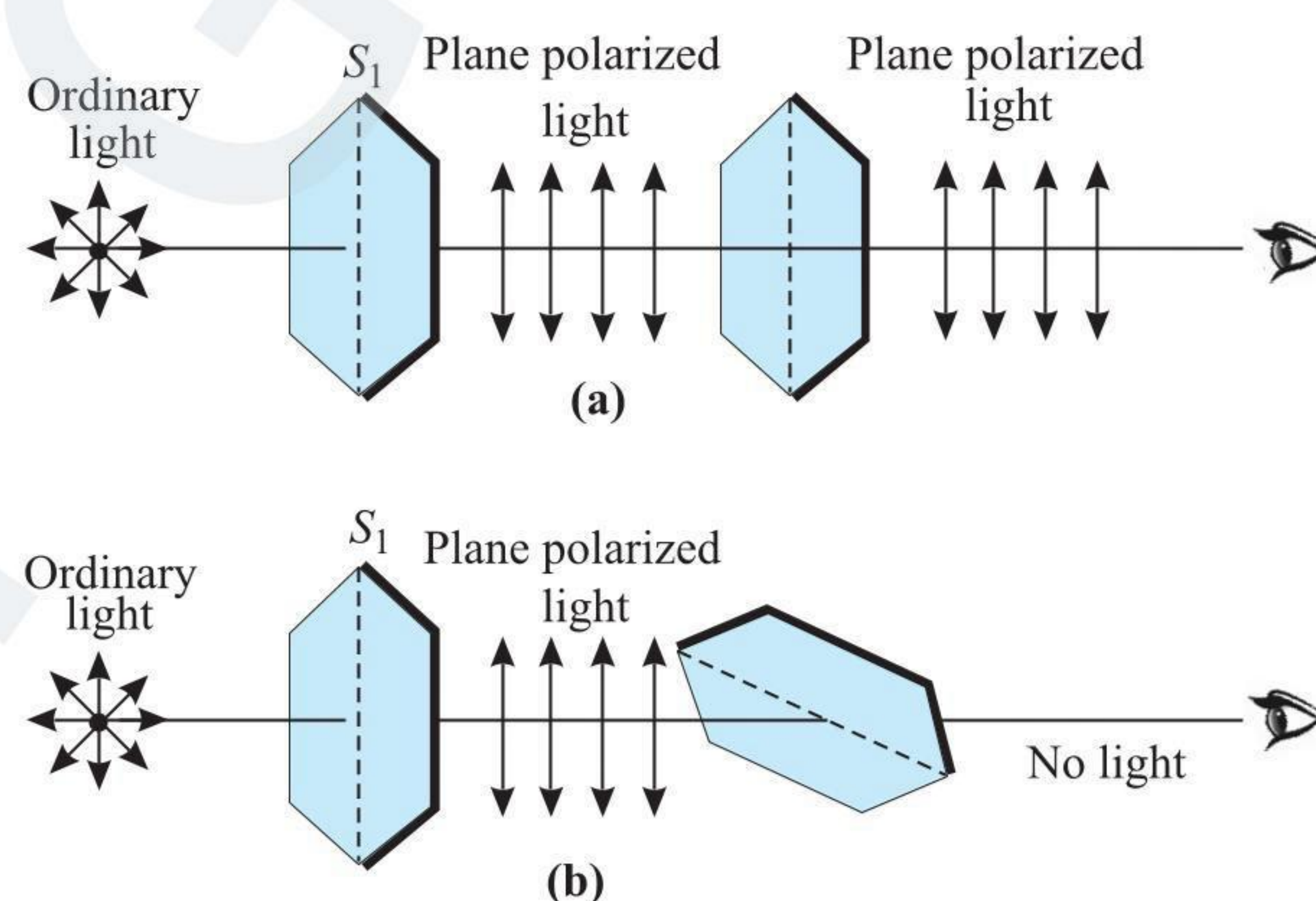
Thus, in the wave that emerges from S_1 , all the vibrations are restricted to the vertical plane and as such the wave is said to be **plane-polarized**. The first slit is called the **polarizer** as it produces polarized vibrations. Since the slit S_2 is also vertical, this plane polarized wave will be able to pass through it and remain plane polarized. But if S_2 is gradually rotated about the string as the axis, the amplitude of vibrations passing through it goes on decreasing till it becomes zero when S_2 is at right angles to S_1 . Thus, no wave emerges from S_2 as shown in Fig. (b). This confirms the polarized state of vibrations passing through S_1 , and as S_2 detects this state, it is called the **analyzer**.

If the string is replaced by a spring in which longitudinal waves can be generated, these will pass through both the slits unaffected, irrespective of the two slits being parallel or perpendicular to each other.

From the above mechanical experiment, it follows that if on rotating the slit S_2 , the amplitude of vibration in the string decreases and becomes zero, when the two slits are at right angles, then the wave motion is transverse in nature. On the other hand, if the amplitude of vibration remains unaffected, the wave motion is a longitudinal one.

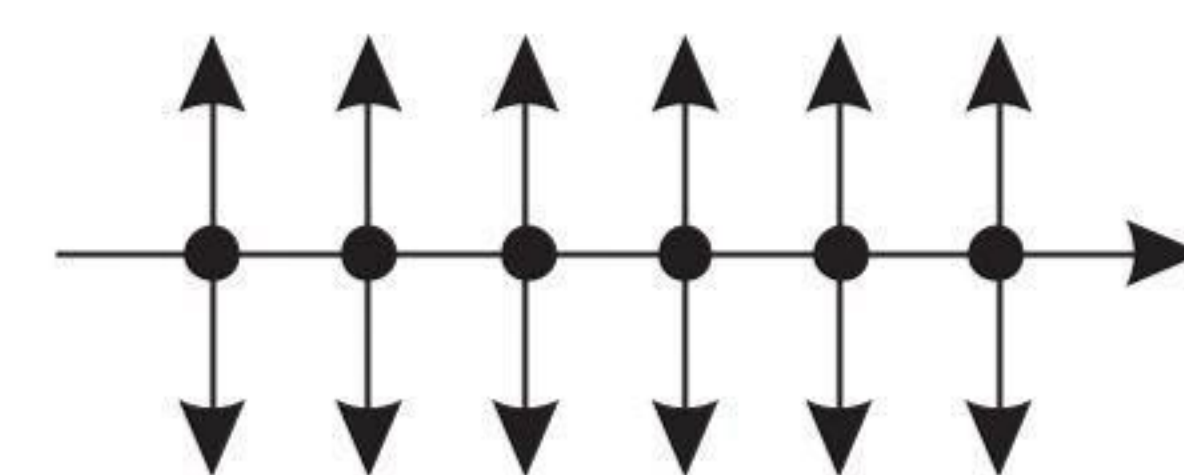
Thus, polarization is a characteristic of transverse waves only. The **polarization of light waves** can be demonstrated by an optical experiment similar to the mechanical experiment described earlier.

A Nicol prism or tourmaline crystal cut parallel to its crystallographic axis behaves in a similar manner for light waves as the slit does in case of the wave motion produced in the string. Suppose that the ordinary light is incident normally on a pair of such crystals S_1 and S_2 . It is observed that the intensity of the emergent light is maximum, when the axis of the two crystals are parallel [Fig. (a)] and minimum, when the axis of the two crystals become at right angles to each other [Fig. (b)]. In view of the mechanical experiment discussed earlier, it follows that light is a transverse wave motion.



Light is electromagnetic in nature and an electromagnetic wave consists of varying electric and magnetic fields, such that the two fields are mutually perpendicular to each other and to the direction of propagation of waves. The optical phenomena i.e. the phenomena concerning light are described by the **electric field vector** in electromagnetic waves. Therefore, the transverse nature of light may be attributed to the vibrations of electric field vector in a direction perpendicular to the direction of propagation of light.

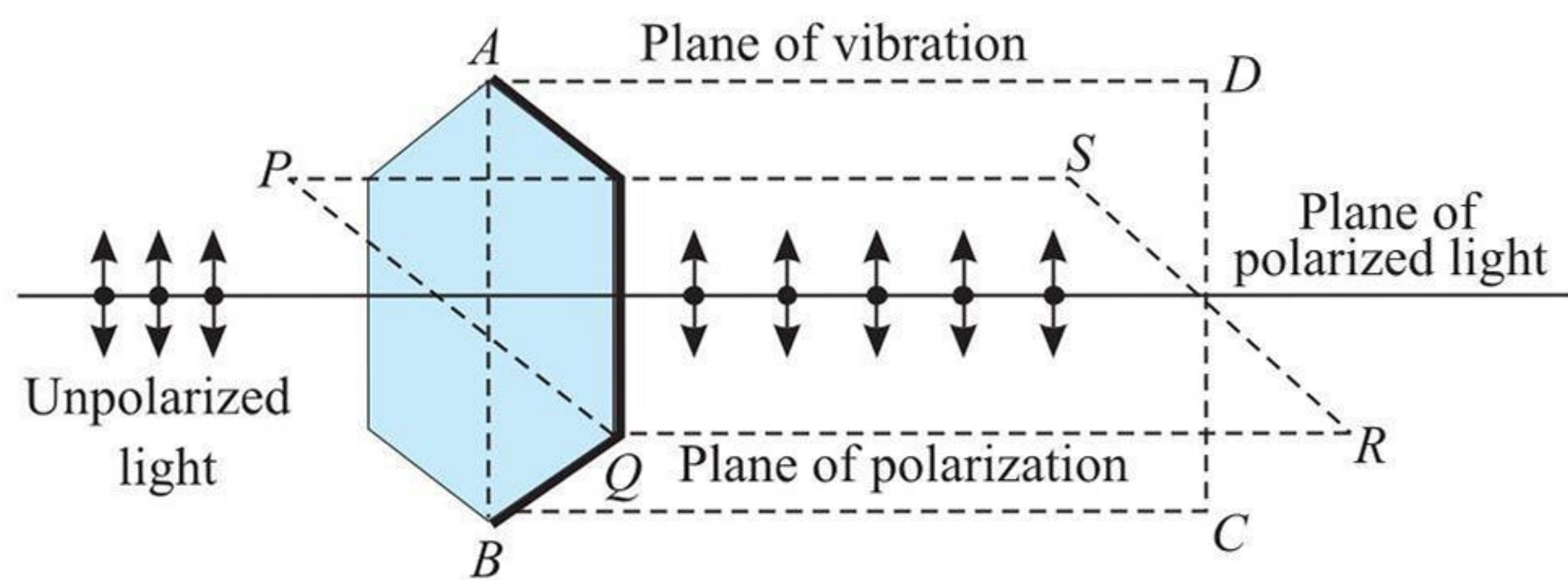
In an ordinary or unpolarized ray of light, the vibrations of electric field vector are symmetrically distributed in all the directions but perpendicular to the direction of propagation of the light. Such a ray of light is called a ray of **ordinary or unpolarized light**. It is schematically represented as shown in figure. The arrows represent vibrations in the plane of the paper, while the dots represent vibrations in a direction perpendicular to the plane of the paper.



SOME USEFUL TERMS AND DEFINITIONS

Polarization of light: It is the phenomenon, due to which the vibrations of light are restricted in a particular plane, is called the polarization of light.

When ordinary light is passed through a tourmaline crystal, it gets polarized, i.e., its vibrations originally present in all directions perpendicular to the direction of propagation are now confined to only one of these directions which are parallel to its crystallographic axis AB . This plane contains the direction of vibration and the direction of propagation of light.



Plane of vibration: The plane within which the vibrations of the polarized light are confined is known as the **plane of vibration**. It is represented by the plane ($ABCD$) in figure.

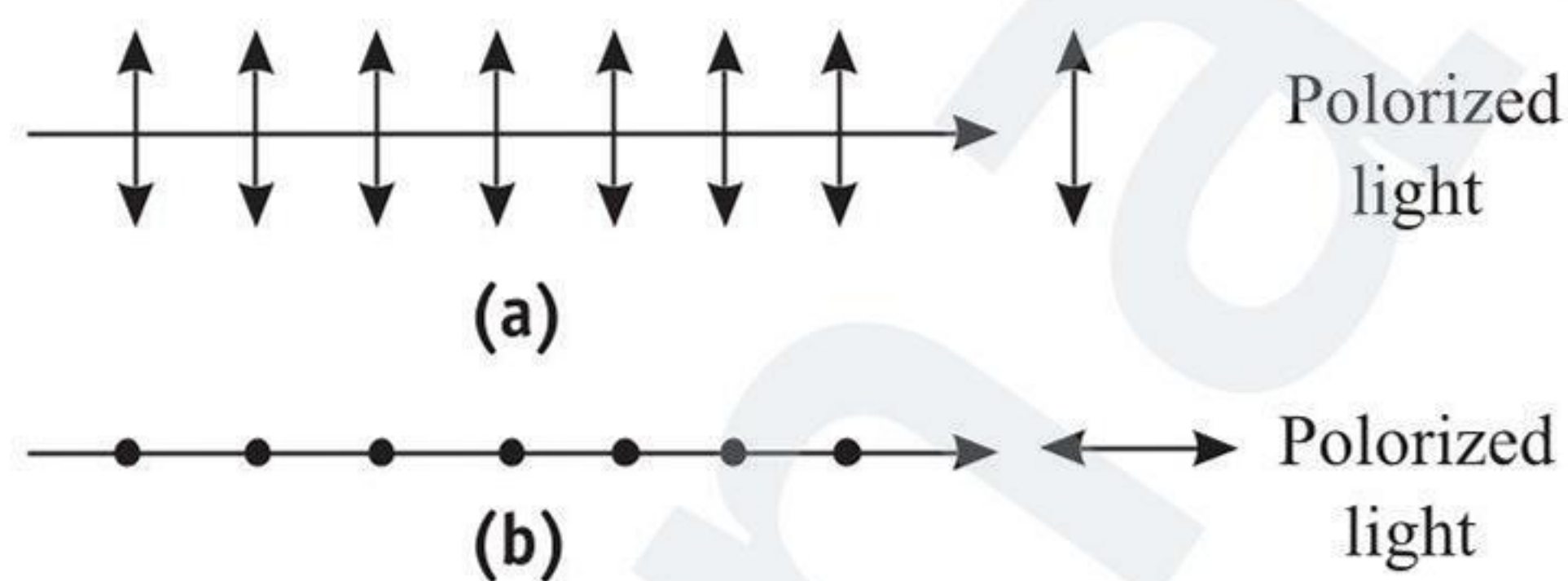
Plane of polarization: It is the plane at right angles to the plane of vibration and passing through the direction of propagation of light is known as the **plane of polarization**. The plane ($PQRS$) perpendicular to the plane of vibrations is called the plane of polarization.

Plane polarized light: It may be defined as the light, in which the vibrations of the light (vibrations of the electric vector) are restricted to a particular plane.

According to the electromagnetic theory of light, the electric vector acts as light vector. Therefore, in a plane polarized light, the electric vector vibrates along a fixed line in a plane perpendicular to the direction of propagation.

In Fig. (a), the plane polarized component whose vibrations are confined to the plane of paper but perpendicular to the direction of propagation, has been represented. The plane of polarization is therefore perpendicular to the plane of the paper.

Figure (b), on the other hand, is the representation of the other plane polarized component whose plane of polarization is in the plane of the paper and the vibrations which are perpendicular to the plane of the paper are represented by dots on the direction of propagation.

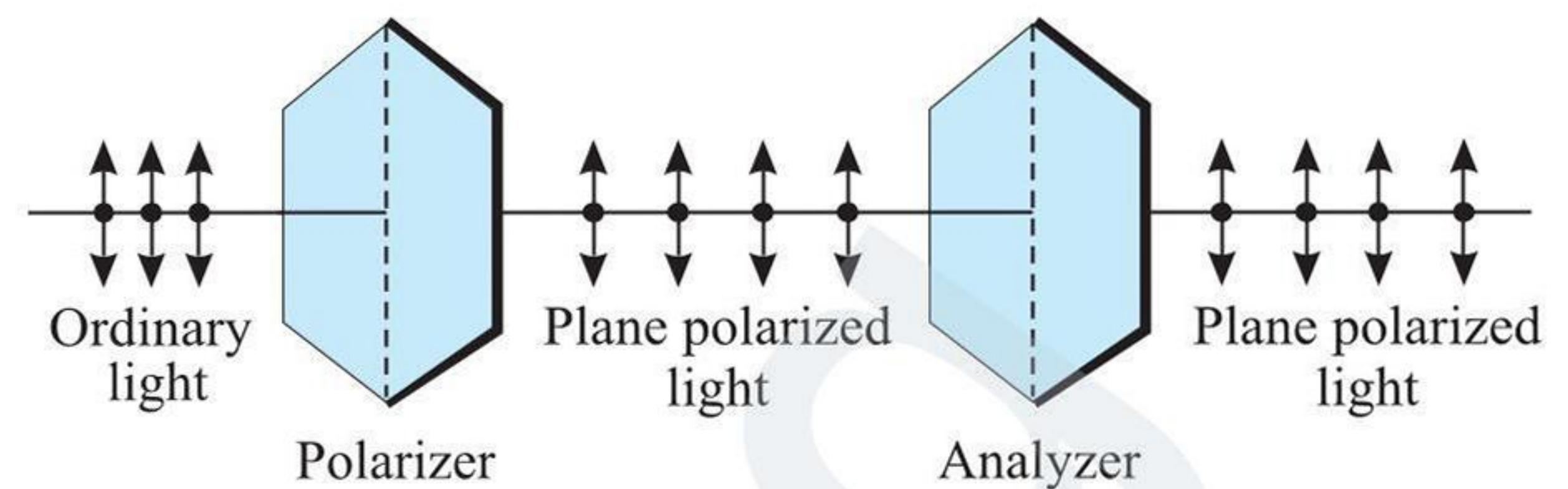


DETECTING PLANE POLARIZED LIGHT

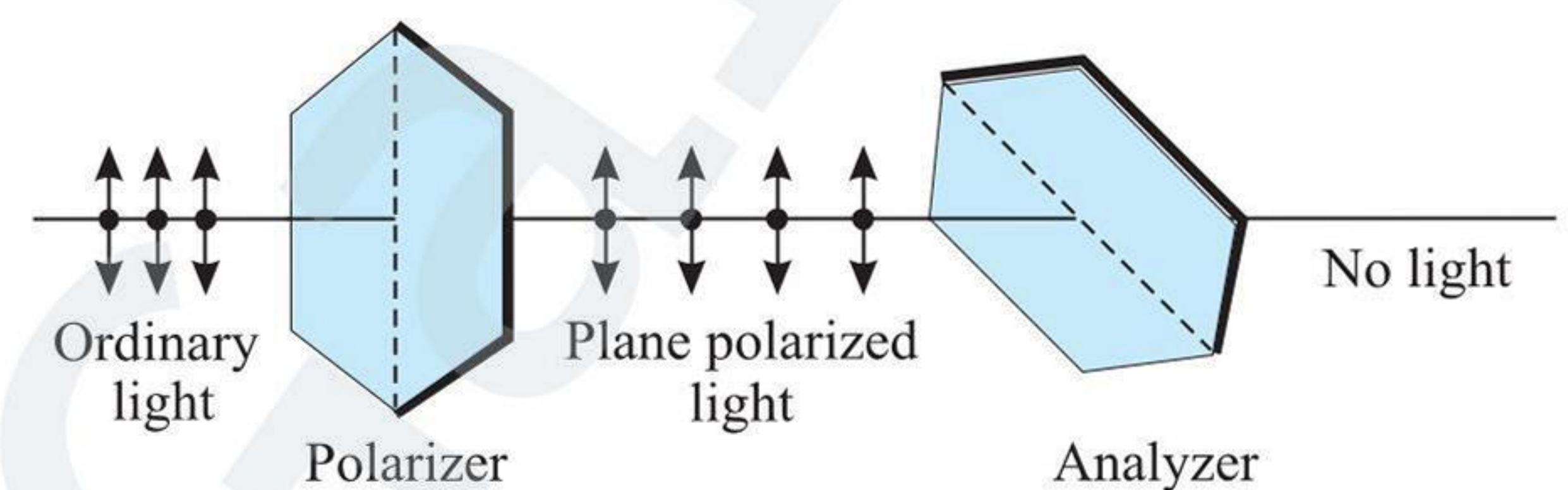
We cannot make distinction between the unpolarized light and the plane polarized light through the naked eye or the polarizer alone. Another such crystal required to analyze the nature of light is called **analyzer**.

A tourmaline crystal or a Nicol prism used to produce plane polarized light is called **polarizer**. If the polarizer is rotated in the path of the ordinary light, the intensity of the light transmitted from the polarizer remains unchanged. It is because, in each orientation of the polarizer, the plane polarized light is obtained, which has vibrations in a direction parallel to the axis of the crystal in that orientation.

If we rotate the analyzer in the path of the light transmitted from the polarizer, so that the axes of the polarizer and the analyzer are parallel to each other, then the intensity of light is found to remain unaffected (or maximum) as shown in Fig. (a).



But when the axis of the polarizer and the analyzer are perpendicular to each other, the intensity of light becomes minimum. In the positions shown in Fig.(b), the polarizer and the analyzer are said to be in **crossed positions**.

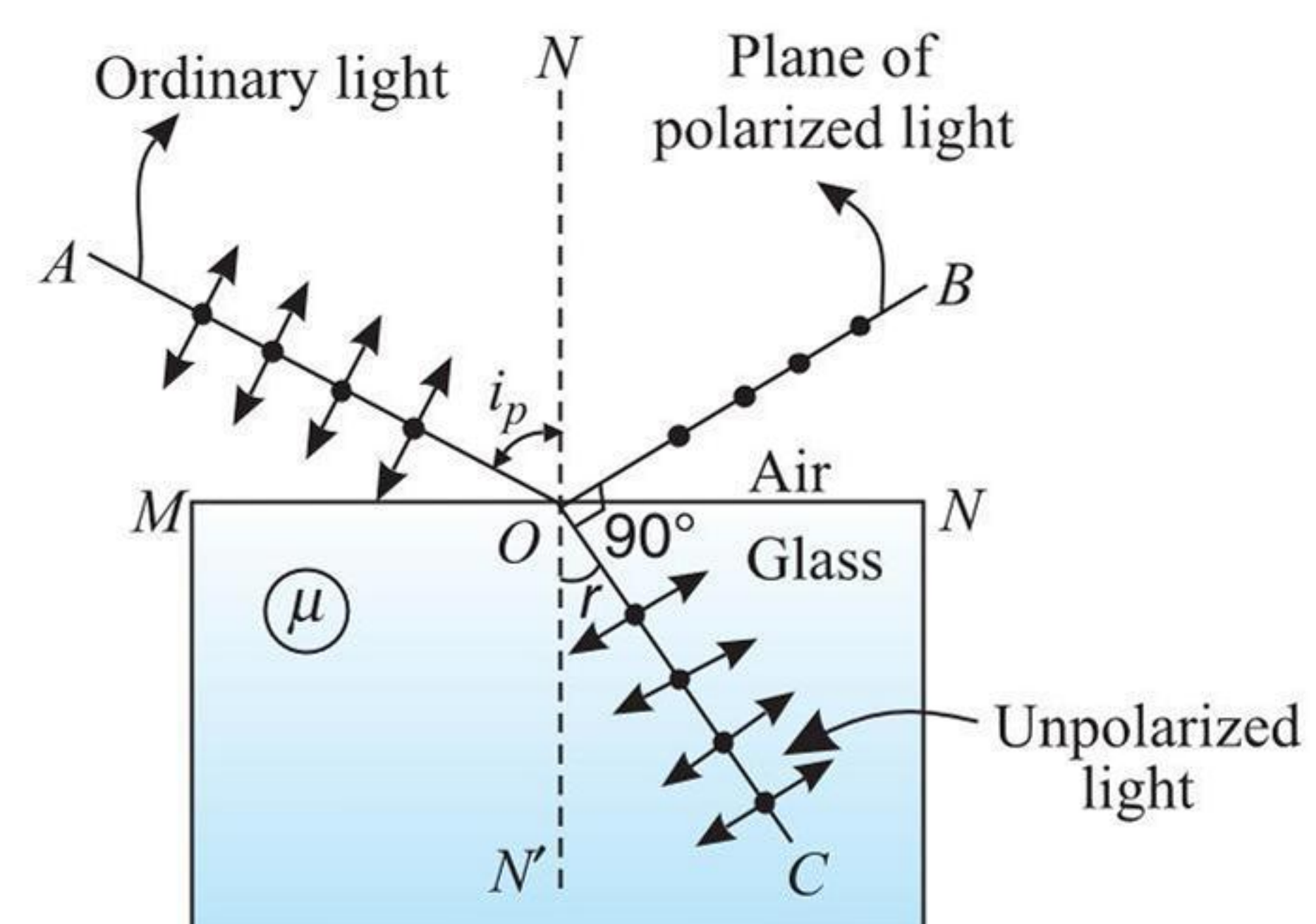


POLARIZATION BY REFLECTION

When unpolarized light is reflected from a plane boundary between two transparent media, such as air and glass or air and water, the reflected light is partially polarized. The degree of polarization depends on the angle of incidence and the ratio of wave speeds in the two media.

One of the simplest method to produce plane polarized light is by reflection. When unpolarized light is reflected from a plane boundary between two transparent media, such as air and glass or air and water, the reflected light is partially polarized. The degree of polarization increases as the angle of incidence is increased. At a particular value of angle of incidence, the reflected beam becomes completely polarized. This angle of incidence is called the **polarizing angle** for that medium.

Let MN represent **the reflecting surface** separating two media, say air and glass (see figure). Consider a ray AO incident on the surface MN at an angle i_p with the normal NON' . OB and OC represent the corresponding reflected and refracted rays. If the reflected beam is examined, it is found to be polarized, but not completely.



If the angle of incidence be gradually changed and the reflected beam examined in every case, it will be observed that for one particular value of the angle of incidence, this beam is completely polarized. This angle is called the **polarizing angle** (i_p) for that

medium. The plane polarized light has vibrations perpendicular to the plane of paper. The light travelling along the path OC (refracted light) is unpolarized light.

BREWSTER'S LAW

The explanation for obtaining the reflected beam as plane polarized at the polarizing angle was given by David Brewster. According to him,

‘When a ray of light is incident at polarizing angle, the reflected ray is at right angle to the refracted ray.’

Thus, when $i = i_p$, $r = (90 - i_p)$.

$$\text{As } \mu = \frac{\sin i}{\sin r}, \mu = \frac{\sin i_p}{\sin(90 - i_p)} = \frac{\sin i_p}{\cos i_p}$$

$$\text{or } \mu = \tan i_p \quad \dots(i)$$

This relation is known as the **Brewster's law**, it states that:

When light is incident at polarizing angle at the interface of a refracting medium, the refractive index of the medium is equal to the tangent of the polarizing angle.

The polarizing angle is sometimes called the **Brewster's angle**.

The value of the polarizing angle for glass is about 57.5° and that for water is about 37° .

ILLUSTRATION 4.11

The critical angle between a given transparent medium and air is denoted by C . A ray of light in air medium enters this transparent medium at an angle of incidence equal to polarizing angle i_p . Deduce a relation for the angle of refraction in terms of C .

Sol. According to Brewster's law, when a ray of light is incident at polarizing angle,

$$\mu = \tan i_p \quad \dots(i)$$

If C is critical angle for the transparent medium and air, then

$$\mu = \frac{1}{\sin C} \quad \dots(ii)$$

From the Eqs. (i) and (ii), we have

$$\tan i_p = \frac{1}{\sin C} \quad \dots(iii)$$

If r is angle of refraction, then $i_p = 90^\circ - r$

$$\text{or } \tan i_p = \tan(90^\circ - r) = \cot r = \frac{1}{\tan r} \quad \dots(iv)$$

From the Eqs. (iii) and (iv), we have $\tan r = \sin C$

$$\text{or } r = \tan^{-1}(\sin C)$$

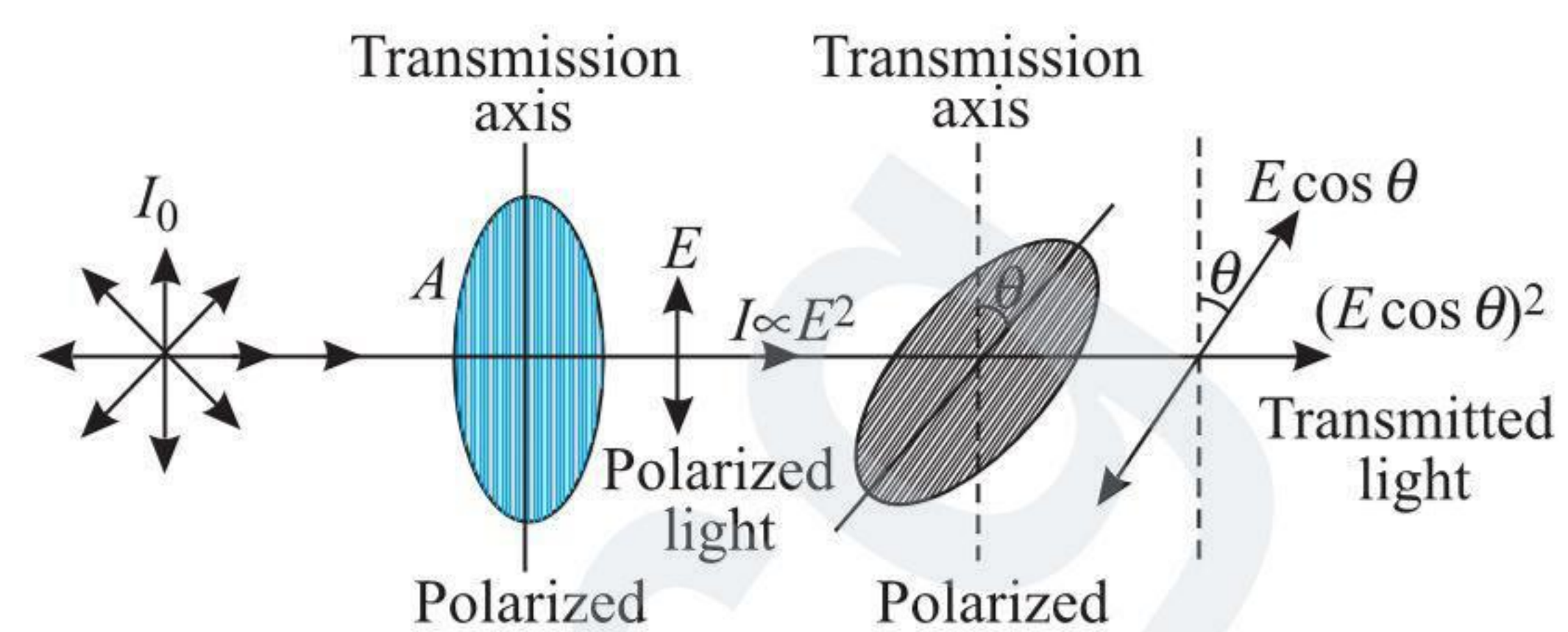
MALUS'S LAW

The dependence of intensity of transmitted light on the angle between the analyzer and the polarizer was investigated by E.L. Malus.

It states that when a completely plane polarized light beam is incident on an analyzer, the intensity of the emergent light varies as the square of the cosine of the angle between the planes of transmission of the analyzer and the polarizer.

In Fig. (a), an unpolarized light beam incident on the polarizer where the transmission axis is oriented vertically. The light that is passing through this sheet is polarized vertically as shown, where the transmitted electric field vector is E . A second

polarizing sheet, called the analyzer, intercepts this beam with its transmission axis at an angle θ to the axis of the polarizer. The component of E perpendicular to the axis of the analyzer is completely absorbed, and the component of E parallel to the axis of the analyzer is transmitted.



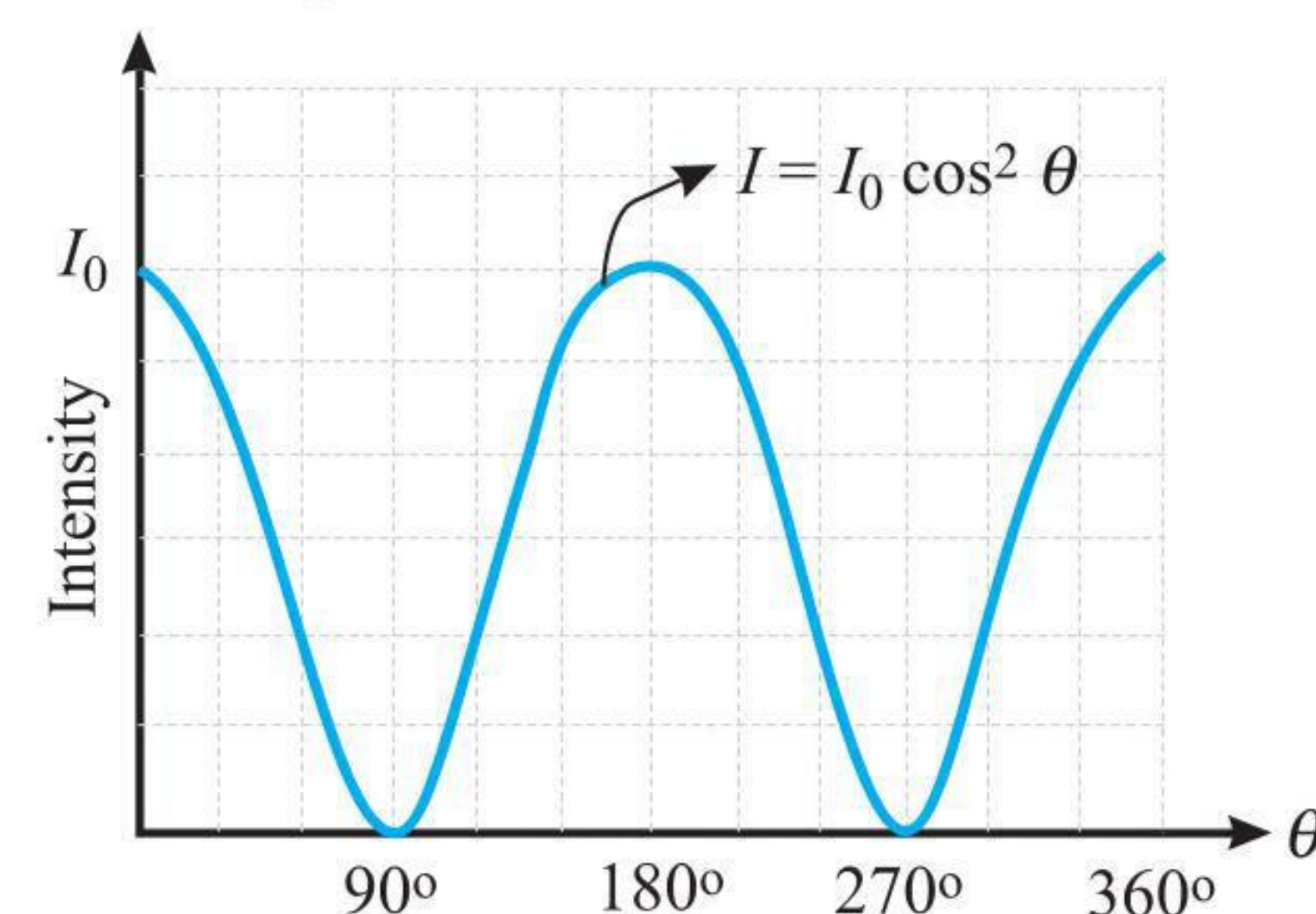
As the component of E parallel to the axis of the analyzer is $E \cos \theta$ and as the transmitted intensity varies as the square of the transmitted amplitude, we conclude that the transmitted intensity (I_θ) varies as

$$I_\theta = I \cos^2 \theta \quad \dots(i)$$

$$[\text{as } I \propto E^2 \text{ and } I_\theta \propto (E \cos \theta)^2]$$

where I is the intensity of polarized wave incident on the analyzer. This expression is known as **Malus's law** and is applicable to any two polarizing materials whose transmission axes are at an angle θ to each other.

Intensity curve: As the angle ' θ ' between the transmission directions of polarizer and analyzer is varied, the intensity ' I ' of the light transmitted by the analyzer varies as a function of $\cos^2 \theta$, as shown in figure.



Graph of intensity I through analyzer versus angle θ between polarizer and analyzer.

Important Points:

- When $\theta = 0^\circ$ or 180° , $\cos \theta = \pm 1$, so that $I = I_0$
So when the transmission directions of polarizer and analyzer are parallel or antiparallel to each other, the maximum intensity of plane polarized light is transmitted by the analyzer and is equal to the intensity emerging from the polarizer.
- When $\theta = 90^\circ$, $\cos \theta = 0$, so that $I = 0$
So when the transmission directions of polarizer and analyzer are perpendicular to each other, the intensity of light transmitted through the analyzer is zero.
- When a beam of unpolarized light is incident on the polarizer,

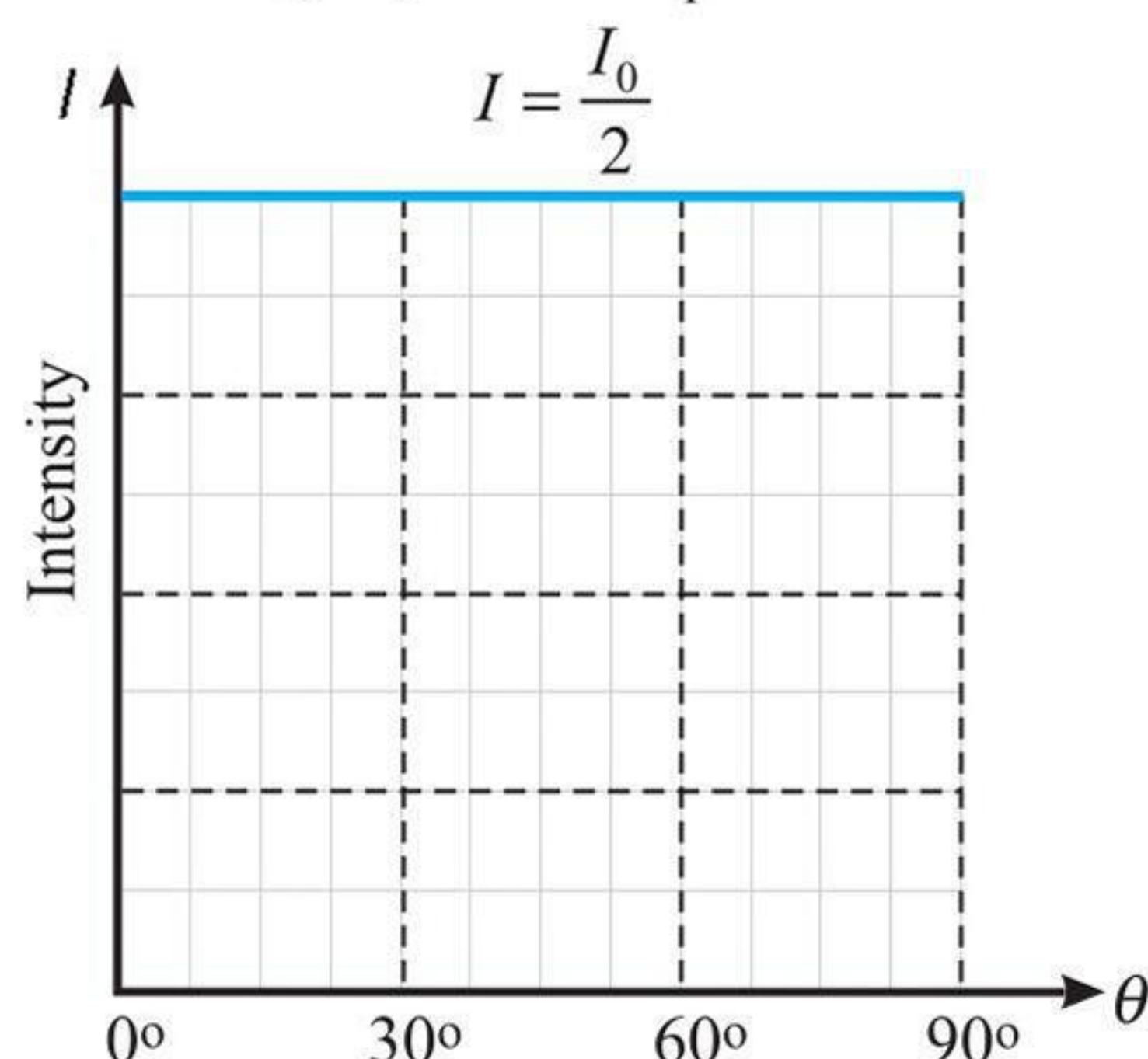
$$I = I_0 \overline{\cos^2 \theta} = I_0 \times \frac{1}{2} (1 + \overline{\cos 2\theta})$$

$$= \frac{1}{2} I_0 (1 + \overline{\cos 2\theta}) = \frac{1}{2} I_0 (1 + 0) = \frac{1}{2} I_0$$

ILLUSTRATION 4.12

Light from an ordinary source (say, a sodium lamp) is passed through a polarized sheet P_1 . The transmitted light is then made to pass through a second polaroid sheet P_2 which can be rotated so that the angle (θ) between the two polaroid sheets varies from 0° to 90° . Show graphically the variation of the intensity of light, transmitted by P_1 and P_2 , as a function of the angle θ . Take the incident beam intensity as I_0 .

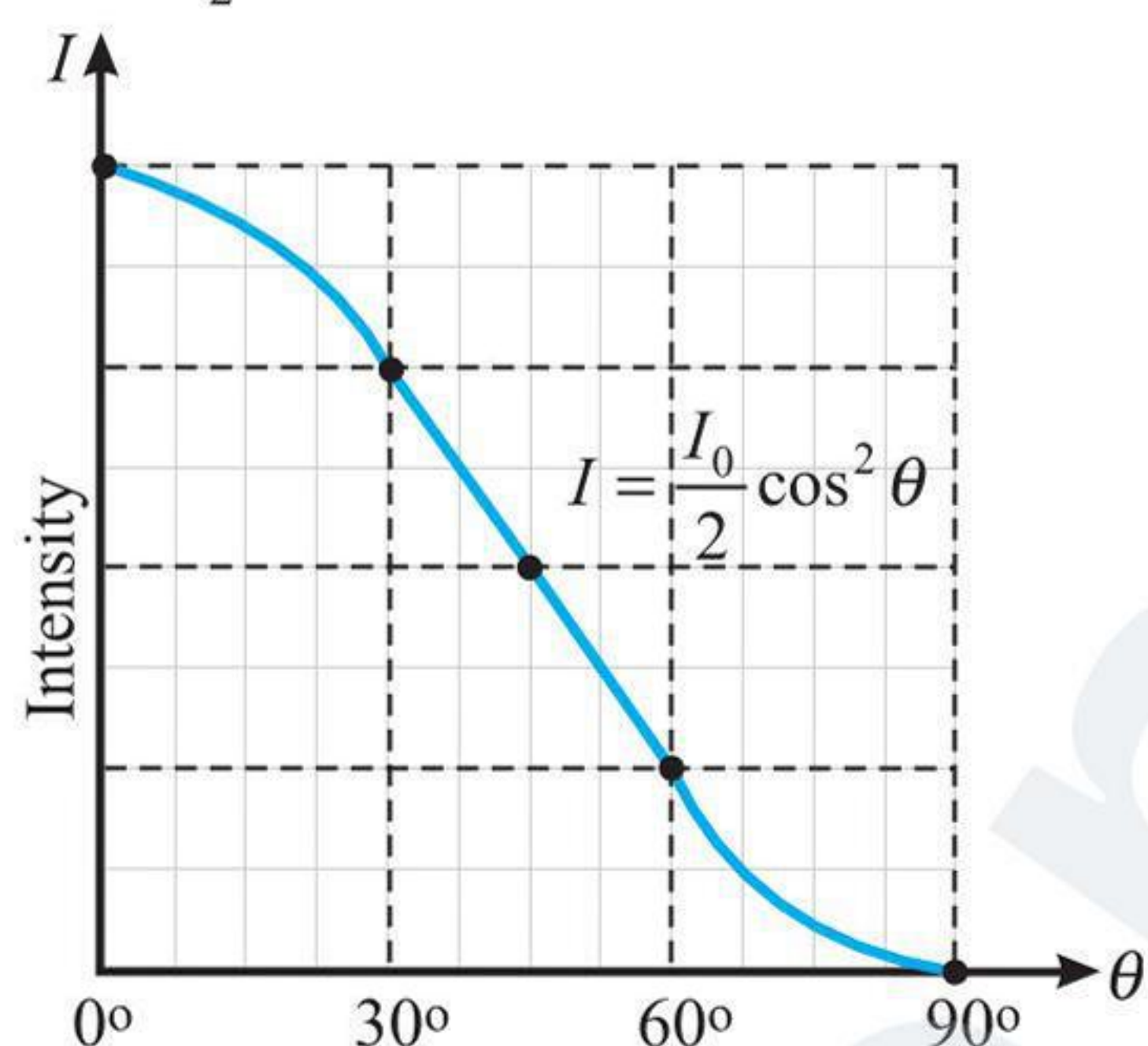
Sol. The intensity of light passing through P_1 remains constant equal to $I_0/2$. So I vs. θ graph for P_1 is as shown below.



The intensity of light transmitted by analyzer P_2 varies as function of $\cos^2 \theta$ as per relation:

$$I = \frac{I_0}{2} \cos^2 \theta$$

So I vs. θ graph for P_2 is as shown below.

**ILLUSTRATION 4.13**

A polaroid examines two adjacent plane-polarized light beams A and B whose planes of polarization are mutually at right angles. In one position of the polaroid, the beam B shows zero intensity. From this position a rotation of 30° shows the two beams of equal intensities. Find the intensity ratio I_A/I_B of the two beams.

Sol. The planes of polarization of light beams A and B are mutually at right angles. Initially, the beam B shows zero intensity. Therefore, $\theta = 90^\circ$ for beam B and $\theta = 0^\circ$ for beam A . When the polaroid is rotated through 30° , we have $\theta = 60^\circ$, for beam A and $\theta = 30^\circ$ for beam B . In this position, Intensity of emerging beam A = Intensity of emerging beam B

$$\therefore I_A \cos^2 30^\circ = I_B \cos^2 60^\circ$$

$$\text{or } \frac{I_A}{I_B} = \frac{\cos^2 60^\circ}{\cos^2 30^\circ} = \frac{(1/2)^2}{(\sqrt{3}/2)^2} = \frac{1}{3} = 1:3$$

ILLUSTRATION 4.14

Unpolarized light falls on two polarizing sheets placed one on top of the other. What must be the angle between the characteristic directions of the sheets if the intensity of the transmitted light is: (a) one third the maximum intensity of the unpolarized light of the incident beam, (b) one third the maximum intensity of the incident beam? Assume that the sheet is ideal, i.e., it reduces the intensity of unpolarized light by exactly 50%.

Sol.

(a) If I_0 is the intensity of unpolarised light, the intensity of the polarized beam transmitted by the first sheet, i.e.,

$$I = \frac{1}{2} I_0 \quad \dots(i)$$

If θ is the angle between the characteristic directions of the two sheets, intensity of polarized light transmitted by second sheet, i.e.,

$$I_\theta = I \cos^2 \theta \quad \dots(ii)$$

$$\text{For } I_\theta = \frac{1}{3} I, \cos^2 \theta = \frac{1}{3}$$

$$\text{or } \theta = \cos^{-1} \left[\pm (1/\sqrt{3}) \right] = \cos^{-1} (\pm 0.577) = \pm 55^\circ$$

(b) From Eqs. (i) and (ii), $I_\theta = \frac{1}{2} I_0 \cos^2 \theta$

$$\text{For } I_\theta = \frac{1}{3} I_0, \frac{1}{3} = \frac{1}{2} \cos^2 \theta$$

$$\text{or } \theta = \cos^{-1} \left(\pm \sqrt{\frac{2}{3}} \right) = \cos^{-1} (\pm 0.8165) = \pm 35^\circ$$

ILLUSTRATION 4.15

Two polaroids are crossed to each other. Now, one of them is rotated through 60° . What percentage of incident unpolarized light will pass through the system?

Sol. Let I_0 be the intensity of the incident unpolarized light on the system of the two polaroids. As the incident light is unpolarized, the intensity of light on being transmitted from the first polaroid will become

$$I = \overline{I_0 \cos^2 \theta} = \frac{I_0}{2} \quad \left(\because \overline{\cos^2 \theta} = \frac{1}{2} \right)$$

If θ is the angle between the transmission planes of the two polaroids, then the intensity of light on passing through the second polaroid is given by

$$I' = I \cos^2 \theta = \frac{I_0}{2} \cos^2 \theta$$

Since the two polaroids were initially crossed, the angle between the transmission planes of the two polaroids on rotating through 60° will become $\theta = 90^\circ - 60^\circ = 30^\circ$.

$$\therefore I' = \frac{I_0}{2} \times \cos^2 30^\circ = \frac{I_0}{2} \times \left(\frac{\sqrt{3}}{2} \right)^2 = 0.375 I_0$$

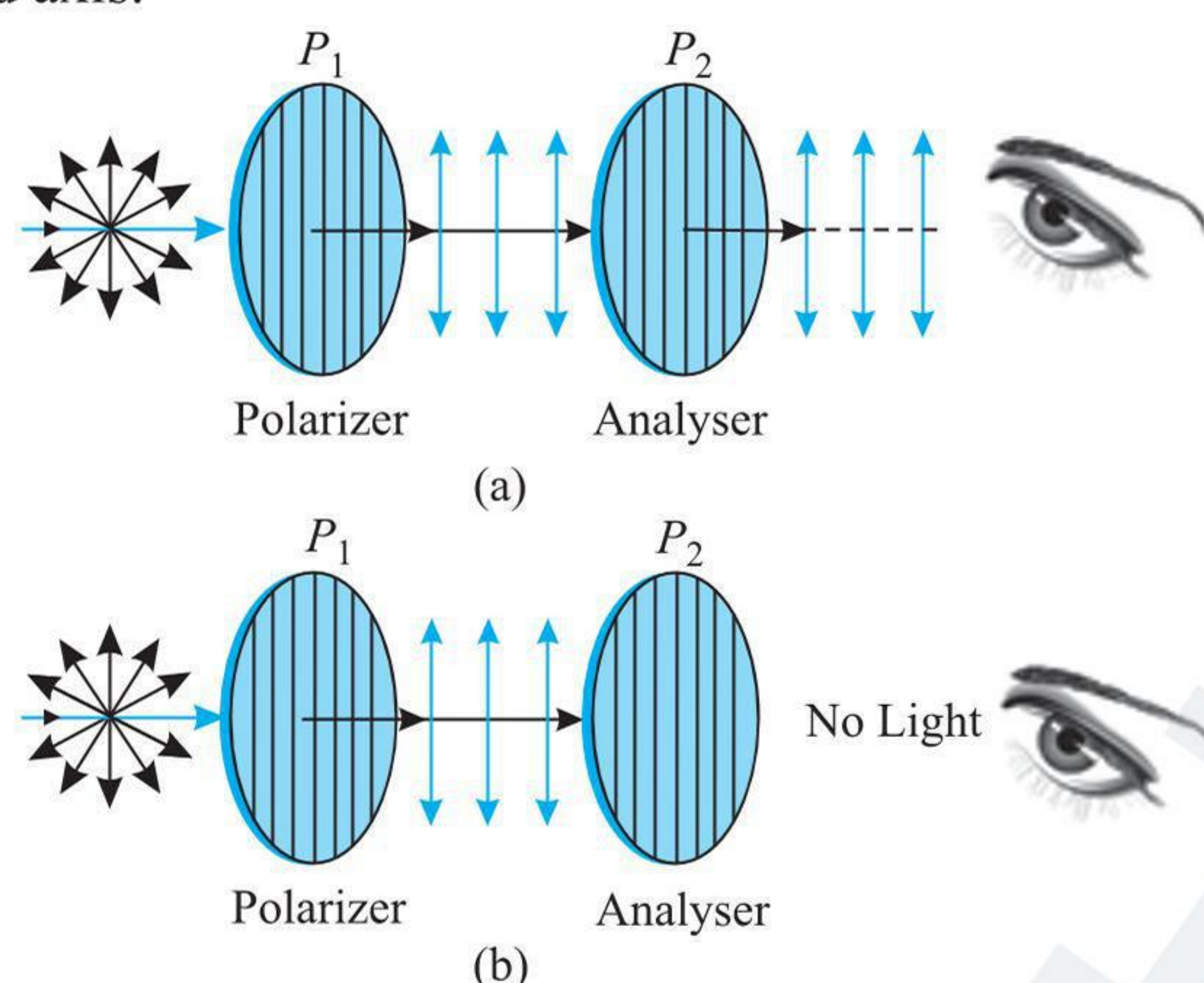
Hence, the percentage of incident light that passes through the system,

$$\frac{I'}{I} \times 100 = 0.375 \times 100 = 37.5\%$$

POLAROIDS

A Nicol prism or a tourmaline crystal cannot be used to produce plane polarized beam of light of large cross-section. It is because of the reason that very large crystals of calcite and tourmaline are not available in practice. Polaroids are thin commercial sheets which make use of the property of selective absorption to produce an intense beam of plane polarized light.

When a paste of quinine iodosulphate made in nitrocellulose is squeezed out through a fine slit, the needle-shaped crystals of quinine iodosulphate align themselves parallel to their optic axis. These crystals are highly dichroic. They absorb one of the doubly refracted beams completely. The thin polarizing sheet so obtained is enclosed between two thin glass plates for mechanical support and we get a polaroid. Each polaroid has a characteristic direction called polaroid axis (shown by parallel lines). A polaroid transmits only those vibrations which are parallel to its polaroid axis.



As shown in the figure, when a beam of unpolarized light falls on a polaroid P_1 it transmits only those vibrations which are parallel to its polaroid axis. It absorbs the vibrations in the perpendicular direction. Thus, the transmitted light is plane-polarized. This can be examined by using a second polaroid P_2 .

However, when the polaroid P_2 is rotated through 90° , so that the transmission planes of the two polaroids become perpendicular to each other, then no light comes out. The two polaroids are said to be *crossed*.

Important Points:

- A polaroid sheet can also be obtained from a sheet of polyvinyl alcohol. When such a sheet is subjected to a large strain, the molecules get oriented in the direction of the applied strain. When the sheet is impregnated with iodine, the material becomes *dichroic*. The polaroid sheet so obtained is called **H-polaroid**.
- If the stretched sheet of polyvinyl alcohol is heated in the presence of a dehydrating agent such as HCl, it becomes strongly dichroic as well as very stable. Such a polaroid sheet is called **K-polaroid**.

DETECTION OF PLANE POLARIZED LIGHT BY POLAROIDS

Detection of unpolarized and polarized lights: The given light is passed through a polaroid and the polaroid is rotated about the

direction of incident light. The intensity of the emergent light is observed.

- If on rotating the polaroid through one complete rotation, there is no change in the intensity of emergent light, then the given light is unpolarized.
- If the intensity of emergent light shows alternate rise and fall and becomes twice maximum and twice zero in one complete rotation of the polaroid, then the given light is plane-polarized or **linearly polarized**.
- If the intensity of emergent light becomes twice maximum and twice minimum (and not zero) in one complete rotation of the polaroid, then the given light is **partially polarized**.

USES OF POLAROIDS

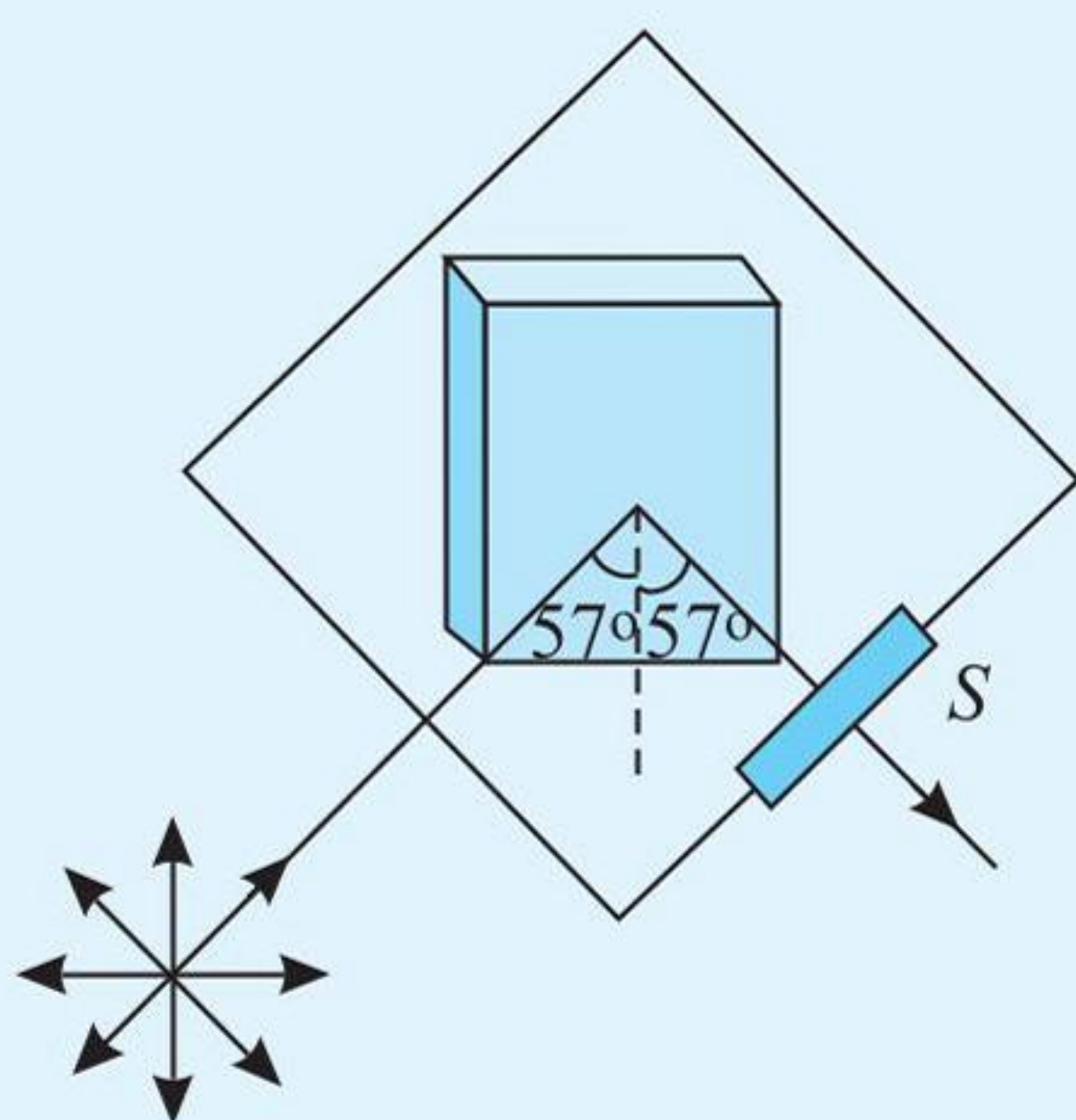
The polaroids are widely used and a few important uses are as listed below:

- In sunglasses and camera filters:** Sunglasses and camera filters are made of polarizing sheets to reduce glare of light produced by reflection from shiny surface such as water surface.
- In wind screens:** The wind screens and car head lights of motor cars are fitted with polaroid films with their axes inclined at 45° to the horizontal. When two cars approach each other from opposite directions, the transmission planes of their wind screens will be perpendicular to each other, so the glare of their head light is completely eliminated. Each driver sees the road by the light sent by his own car.
- In windowpanes of aeroplanes:** The polaroids are used in windowpanes of a train and especially of an aeroplane. They help to control the light entering through the windows. One of the polaroids is fixed while the other can be rotated to control the amount of light coming in.
- In 3-D movies:** The pictures taken by a stereoscopic camera, when seen with the help of polaroid spectacles, create three-dimensional effect. The two 3-D glasses are really polarizing glasses with their directions of polarization perpendicular to each other. So one eye sees one image and other sees a slightly different image.

CONCEPT APPLICATION EXERCISE 4.3

- A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ . If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is

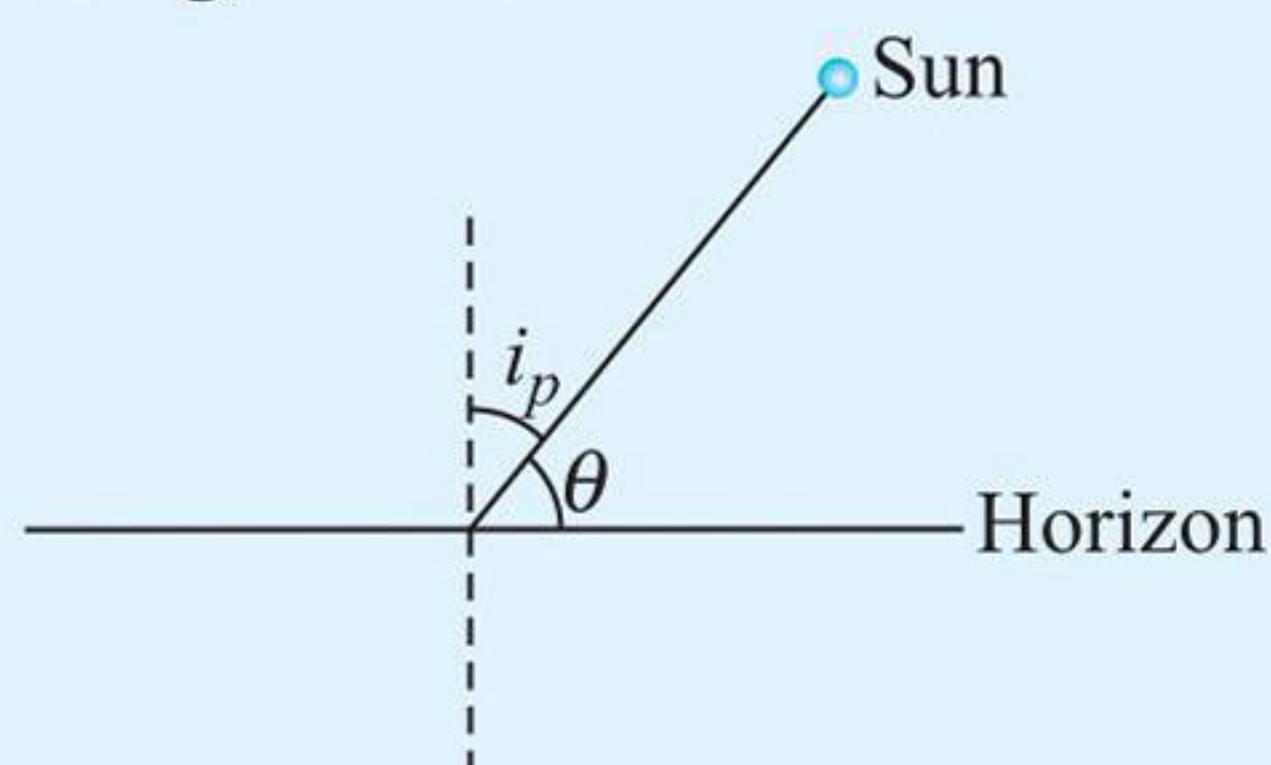
(a) $90 + \phi$	(b) $\sin^{-1}(\mu \cos \phi)$
(c) 90°	(d) $90^\circ - \sin^{-1}(\sin \phi / \mu)$
- Figure represents a glass plate placed vertically on a horizontal table with a beam of unpolarized light falling on its surface at the polarizing angle of 57° with the normal. The electric vector in the reflected light on screen S will vibrate with respect to the plane of incidence in a



- (a) Vertical plane (b) Horizontal plane
 (c) Plane making an angle of 45° with the vertical
 (d) Plane making an angle of 57° with the horizontal

3. When sunlight is incident on water at an angle of 53° , the reflected light is found to be completely plane-polarized. Determine: (a) angle of refraction of light and (b) refractive index of water.

4. Sunlight is reflected from the smooth surface of a pond. What is the Sun's altitude when the polarization of the reflected light is greatest?



$$\left[\text{Refractive index of water } \mu_{\text{water}} = \frac{4}{3} \right]$$

5. Two polaroids are used to study polarization. One of them (the polarizer) is kept fixed and the other (the analyzer) is initially kept with its axis parallel to the polarizer axis. The analyzer is then rotated through angles of 45° , 90° and 180° in turn. How would the intensity of light coming out of the analyzer be affected for these angles of rotation, as compared to the initial intensity and why?
6. Two polarizing sheets have their polarizing directions parallel so that the intensity of the transmitted light is maximum. Through what angle must the either sheet be turned if the intensity is to drop by one-half?

ANSWERS

1. (c) 2. (a) 3. (a) 37° (b) 1.327
 4. 37° 6. $\pm 45^\circ$ or $\pm 135^\circ$

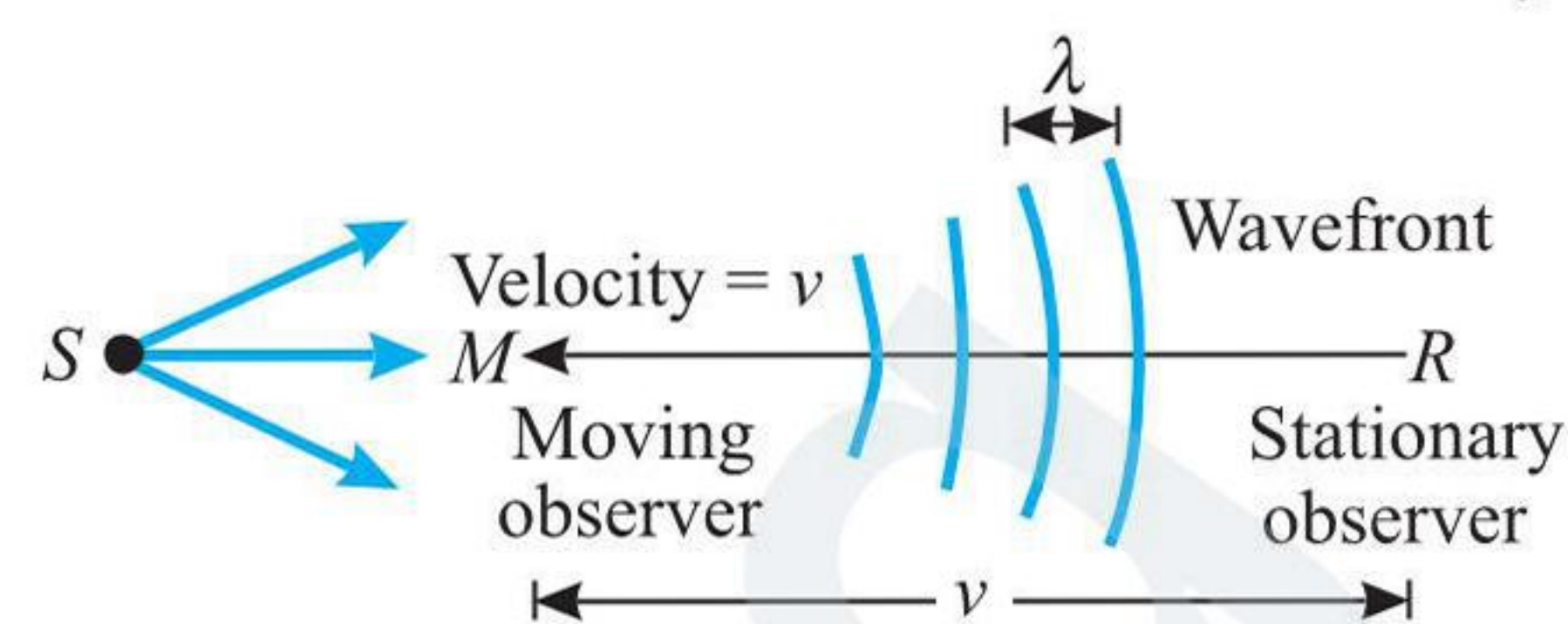
DOPPLER EFFECT FOR LIGHT

You have already learnt Doppler effect for sound waves. When a source of sound travels towards an observer, the apparent frequency is higher than the frequency actually emitted by the source. When the source moves away, the apparent frequency is lower than the actual frequency. Doppler effect is a basic property of all waves and so occurs in case of light also.

Whenever there is a relative motion between source of light and observer, the frequency of light received by the observer is different from the frequency actually emitted by the source. This phenomenon of the apparent change in the frequency of light is called **Doppler effect** for light.

EXPRESSION FOR THE APPARENT FREQUENCY OF LIGHT

Suppose a source of light emits waves of frequency f and wavelength λ . If c is the speed of light, then $\lambda = \frac{c}{f}$



Suppose an observer moves towards the source with velocity v . In one second, the source and observer come closer by a distance v .

$\therefore$ Apparent frequency = Number of light waves emitted per second by the source + Number of light waves contained in distance v

$$\text{or } f' = f + \frac{v}{\lambda} = f + \frac{v}{c/f} = f + v \cdot \frac{f}{c}$$

$$\text{or } f' = f \left(1 + \frac{v}{c} \right) \quad \dots(i)$$

Clearly, $f' > f$ i.e., the apparent frequency increases when source and observer approach each other.

When source and observer move away from each other, the apparent frequency can be obtained by replacing v by $-v$ in the above equation. Then

$$f' = f \left(1 - \frac{v}{c} \right) \quad \dots(ii)$$

Clearly, $f' < f$ i.e., the apparent frequency decreases when source and observer move away from each other.

Blue shift and red shift: Equations (i) and (ii) can be combined together as

$$f' = f \left(1 \pm \frac{v}{c} \right)$$

$$\text{or } f' - f = \pm \frac{v}{c} \cdot f$$

The frequency change $\Delta f = f' - f$ is called Doppler shift.

Putting $f = \frac{c}{\lambda}$ and $f' = \frac{c}{\lambda'}$, we get $\frac{c}{\lambda'} - \frac{c}{\lambda} = \pm \frac{v}{c} \cdot \frac{c}{\lambda}$

$$\text{or } \frac{\lambda - \lambda'}{\lambda'} = \pm \frac{v}{c}$$

$$\text{But } \frac{\lambda - \lambda'}{\lambda'} \simeq \frac{\lambda - \lambda'}{\lambda}$$

$$\therefore \frac{\lambda - \lambda'}{\lambda} = \pm \frac{v}{c}$$

$$\text{or } \lambda - \lambda' = \pm \frac{v}{c} \lambda$$

- (i) When source and observer approach each other, positive sign is taken. Then $\lambda - \lambda'$ is positive or $\lambda' < \lambda$, i.e., the wavelengths in the middle part of the visible spectrum shift towards the blue region. This is called **blue shift**.

- (ii) When source and observer move away from each other, negative sign is taken. Then $\lambda - \lambda'$ is negative or $\lambda' > \lambda$, i.e., the wavelengths in the middle part of the visible spectrum shift towards the red region. This is called **red shift**.

Applications of Doppler Effect:

1. Light received from stars and galaxies shows a red shift which indicates that the universe is expanding.
2. By measuring Doppler shift in the e.m. wave reflected from an automobile, the speed of the automobile can be determined.
3. Doppler shift of light received from Saturn rings shows that the rings consist of a number of discontinuous satellites.
4. By measuring Doppler shift in the light received from eastern and western edges of the Sun, the speed of rotation of the Sun has been determined to be 2 km s^{-1} from east to west relative to the earth.

CONCEPT APPLICATION EXERCISE 4.4

1. If the shift of wavelength of light emitted by a star is towards violet, then this shows that star is
 - (a) stationary
 - (b) moving towards earth
 - (c) moving away from earth
 - (d) information is incomplete

2. A star emits light of wavelength $5500 \text{ \AA}$. It appears blue to an observer on the earth, it means
 - (a) star is going away from the earth
 - (b) star is stationary
 - (c) star is coming towards earth
 - (d) none of the above
3. A star is moving towards the earth with a speed of $9.0 \times 10^6 \text{ ms}^{-1}$. If the wavelength of a particular spectral line emitted by it is $6000 \text{ \AA}$, then find the apparent wavelength.
4. A spaceship is moving with a velocity of 10^8 m/s towards a star that emits radiations of wavelength $6 \times 10^{11} \text{ m}$. Calculate the wavelength of the radiation as received by the crew in the ship.
5. What speed should a galaxy move with respect to us that the sodium line at 589.0 nm is observed at 589.6 nm ?

ANSWERS

- | | | |
|-----------------------------------|----------------------------------|-----------------------|
| 1. (b) | 2. (c) | 3. $5820 \text{ \AA}$ |
| 4. $4.5 \times 10^{-7} \text{ m}$ | 5. $3.1 \times 10^5 \text{ m/s}$ | |

Exercises

Single Correct Answer Type

- In case of linearly polarized light, the magnitude of the electric field vector
 - does not change with time
 - varies periodically with time
 - increases and decreases linearly with time
 - is parallel to the direction of propagation
- A diffraction is obtained by using a beam of red light. What will happen if the red light is replaced by the blue light?
 - Bands will narrower and crowded together
 - Bands become broader and move further apart
 - No change will take place
 - Bands disappear
- When an unpolarized light of intensity I_0 is incident on a polarizing sheet, the intensity of the light which does not get transmitted is
 - $\frac{1}{2}I_0$
 - $\frac{1}{4}I_0$
 - zero
 - I_0
- In a diffraction pattern by a wire, on increasing diameter of wire, fringe width
 - decreases
 - increases
 - remains unchanged
 - increasing or decreasing will depend on wavelength
- What will be the angular width of central maxima in Fraunhofer diffraction when light of wavelength $6000\text{ \AA}$ is used and slit width is $12 \times 10^{-5}\text{ cm}$?
 - 2 rad
 - 3 rad
 - 1 rad
 - 8 rad
- If we observe the single slit Fraunhofer diffraction with wavelength λ and slit width d , the width of the central maxima is 2θ . On decreasing the slit width for the same λ
 - θ increases
 - θ remains unchanged
 - θ decreases
 - θ increases or decreases depending on the intensity of light
- In order to see diffraction, the thickness of the film is
 - 100 $\text{\AA}$
 - 10,000 $\text{\AA}$
 - 1 mm
 - 1 cm
- A spectral line $\lambda = 5000\text{ \AA}$ in the light coming from a distant star is observed as a $5200\text{ \AA}$. What will be recession velocity of the star?
 - $1.15 \times 10^7\text{ cm/s}$
 - $1.15 \times 10^7\text{ m/s}$
 - $1.15 \times 10^7\text{ km/s}$
 - 1.15 km/s
- The apparent wavelength of the light from a star moving away from the earth is 0.01% more than its real wavelength. Then the velocity of star is
 - 60 km/s
 - 15 km/s
 - 150 km/s
 - 30 km/s
- A star emitting light of wavelength $5896\text{ \AA}$ is moving away from the earth with a speed of 3600 km/s . The wavelength of light observed on earth will
 - decrease by $5825.25\text{ \AA}$
 - increase by $5966.75\text{ \AA}$
 - decrease by $70.75\text{ \AA}$
 - increase by $70.75\text{ \AA}$

($c = 3 \times 10^8\text{ m/s}$ is the speed of light)
- A star moves away from earth at speed $0.8c$ while emitting light of frequency $6 \times 10^{14}\text{ Hz}$. What frequency will be observed on the earth (in units of 10^{14} Hz) ($c = \text{speed of light}$)?
 - 0.24
 - 1.2
 - 30
 - 3.3
- A light source approaches the observer with velocity $0.8c$. The doppler shift for the light of wavelength $5500\text{ \AA}$ is
 - 4400 $\text{\AA}$
 - 1833 $\text{\AA}$
 - 3167 $\text{\AA}$
 - 7333 $\text{\AA}$
- In the spectrum of light of a luminous heavenly body the wavelength of a spectral line is measured to be $4747\text{ \AA}$ while actual wavelength of the line is $4700\text{ \AA}$. The relative velocity of the heavenly body with respect to earth will be (velocity of light is $3 \times 10^8\text{ m/s}$)
 - $3 \times 10^5\text{ m/s}$ moving towards the earth
 - $3 \times 10^5\text{ m/s}$ moving away from the earth
 - $3 \times 10^6\text{ m/s}$ moving towards the earth
 - $3 \times 10^6\text{ m/s}$ moving away from the earth
- $6563\text{ \AA}$ line emitted by hydrogen atom in a star is found to be red shifted by $5\text{ \AA}$. The speed with which the star is receding from the earth is
 - $17.29 \times 10^9\text{ m/s}$
 - $4.29 \times 10^7\text{ m/s}$
 - $3.39 \times 10^5\text{ m/s}$
 - $2.29 \times 10^5\text{ m/s}$
- A slit of width a is illuminated by white light. For red light ($\lambda = 6500\text{ \AA}$), the first minima is obtained at $\theta = 30^\circ$. Then the value of a will be
 - 3250 $\text{\AA}$
 - $6.5 \times 10^{-4}\text{ mm}$
 - 1.24 microns
 - $2.6 \times 10^{-4}\text{ cm}$
- The light of wavelength $6328\text{ \AA}$ is incident on a slit of width 0.2 mm perpendicularly, the angular width of central maxima will be
 - 0.36°
 - 0.18°
 - 0.72°
 - 0.90°
- A slit of size 0.15 cm is placed at 2.1 m from a screen. On illuminated it by a light of wavelength $5 \times 10^{-5}\text{ cm}$. The width of central maxima will be
 - 70 mm
 - 0.14 mm
 - 1.4 mm
 - 0.14 cm

18. What will be the angle of diffraction for the first minimum due to Fraunhofer diffraction with sources of light of wave length 550 nm and slit of width 0.55 mm?
 (1) 0.001 rad (2) 0.01 rad
 (3) 1 rad (4) 0.1 rad
19. Angular width (β) of central maximum of a diffraction pattern on a single slit does not depend upon
 (1) distance between slit and source
 (2) wavelength of light used
 (3) width of the slit
 (4) frequency of light used
20. A single slit of width 0.20 mm is illuminated with light of wavelength 500 nm. The observing screen is placed 80 cm from the slit. The width of the central bright fringe will be
 (1) 1 mm (2) 2 mm
 (3) 4 mm (4) 5 mm
21. The observed wavelength of light coming from a distant galaxy is found to be increased by 0.5% as compared with that coming from a terrestrial source. The galaxy is
 (1) stationary with respect to the earth
 (2) approaching the earth with velocity of light
 (3) receding from the earth with the velocity of light
 (4) receding from the earth with a velocity equal to 1.5×10^6 m/s
22. A rocket is moving away from the earth at a speed of 6×10^7 m/s. The rocket has blue light in it. What will be the wavelength of light recorded by an observer on the earth? (Wavelength of blue light = 4600 Å)
 (1) 4600 Å (2) 5520 Å
 (3) 3680 Å (4) 3920 Å
23. Yellow light is used in single slit diffraction experiment with slit width 0.6 mm. If yellow light is replaced by X-rays then the pattern will reveal
 (1) that the central maxima is narrower
 (2) no diffraction pattern
 (3) more number of fringes
 (4) less number of fringes
24. A plane wavefront ($\lambda = 6 \times 10^{-7}$ m) falls on a slit 0.4 mm wide. A convex lens of focal length 0.8 m placed behind the slit focusses the light on a screen. What is the linear diameter of second maximum?
 (1) 6 mm (2) 12 mm
 (3) 3 mm (4) 9 mm
25. Red light is generally used to observe diffraction pattern from single slit. If blue light is used instead of red light, then diffraction pattern
 (1) will be more clear (2) will contract
 (3) will expand (4) will not be visualized
26. Light of wavelength 589.3 nm is incident normally on the slit of width 0.1 mm. What will be the angular width of the central diffraction maximum at a distance of 1 m from the slit?
 (1) 0.68° (2) 1.02°
 (3) 0.34° (4) None of these
27. A beam of light of wavelength 600 nm from a distant source falls on a single slit 1 mm wide and the resulting diffraction pattern is observed on a screen 2 m away. The distance between the first dark fringes on either side of the central bright fringe is
 (1) 1.2 mm (2) 1.2 cm
 (3) 2.4 cm (4) 2.4 mm
28. Direction of the first secondary maximum in the Fraunhofer diffraction pattern at a single slit is given by (a is the width of the slit)
 (1) $a \sin \theta = \frac{\lambda}{2}$ (2) $a \cos \theta = \frac{3\lambda}{2}$
 (3) $a \sin \theta = \lambda$ (4) $a \sin \theta = \frac{3\lambda}{2}$
29. A parallel monochromatic beam of light is incident normally on a narrow slit. A diffraction pattern is formed on a screen placed perpendicular to the direction of incident beam. At the first maximum of the diffraction pattern the phase difference between the rays coming from the edges of the slit is
 (1) 0 (2) $\frac{\pi}{2}$
 (3) π (4) 2π
30. A light wave is incident normally over a slit of width 24×10^{-5} cm. The angular position of second dark fringe from the central maxima is 30° . What is the wavelength of light?
 (1) 6000 Å (2) 5000 Å
 (3) 3000 Å (4) 1500 Å
31. A parallel beam of monochromatic light of wavelength 5000 Å is incident normally on a single narrow slit of width 0.001 mm. The light is focused by a convex lens on a screen placed on the focal plane. The first minimum will be formed for the angle of diffraction equal to
 (1) 0° (2) 15°
 (3) 30° (4) 60°
32. In the far field diffraction pattern of a single slit under polychromatic illumination, the first minimum with the wavelength λ_1 is found to be coincident with the third maximum at λ_2 . So
 (1) $3\lambda_1 = 0.3\lambda_2$ (2) $3\lambda_1 = \lambda_2$
 (3) $\lambda_1 = 3.5\lambda_2$ (4) $0.3\lambda_1 = 3\lambda_2$
33. Light of wavelength $\lambda = 5000$ Å falls normally on a narrow slit. A screen placed at a distance of 1 m from the slit and perpendicular to the direction of light. The first minima of the diffraction pattern is situated at 5 mm from the centre of central maximum. The width of the slit is
 (1) 0.1 mm (2) 1.0 mm
 (3) 0.5 mm (4) 0.2 mm
34. A single slit of width a is illuminated by violet light of wavelength 400 nm and the width of the diffraction pattern is measured as y . When half of the slit width is covered and illuminated by yellow light of wavelength 600 nm, the width of the diffraction pattern is
 (1) the pattern vanishes and the width is zero
 (2) $y/3$
 (3) $3y$
 (4) none of these
35. The angle of polarization for any medium is 60° , what will be critical angle for this?

- (1) $\sin^{-1} \sqrt{3}$ (2) $\tan^{-1} \sqrt{3}$
 (3) $\cos^{-1} \sqrt{3}$ (4) $\sin^{-1} \frac{1}{\sqrt{3}}$
36. A polaroid is placed at 45° to an incoming light of intensity I_0 . Now the intensity of light passing through polaroid after polarization would be?
 (1) I_0 (2) $I_0 / 2$
 (3) $I_0 / 4$ (4) Zero
37. Two Nicols are oriented with their principal planes making an angle of 60° . The percentage of incident unpolarized light which passes through the system is
 (1) 50% (2) 100%
 (3) 12.5% (4) 37.5%
38. Unpolarized light falls on two polarizing sheets placed one on top of the other. What must be the angle between the characteristic directions of the sheets if the intensity of the final transmitted light is one-third the maximum intensity of the first transmitted beam?
 (1) 75° (2) 55°
 (3) 35° (4) 15°
39. When unpolarized light beam is incident from air onto glass ($n = 1.5$) at the polarizing angle
 (1) Reflected beam is polarized 100 percent
 (2) Reflected and refracted beams are partially polarized
 (3) The reason for (1) is that almost all the light is reflected
 (4) All of the above
40. When the angle of incidence on a material is 60° , the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1})
 (1) 3×10^8 (2) $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$
 (3) $\sqrt{3} \times 10^8$ (4) 0.5×10^8
41. Two polaroids are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid is placed between these polaroids then the intensity of light emerging from the last polaroid will be
 (1) $\left(\frac{I_0}{8}\right) \sin^2 2\theta$ (2) $\left(\frac{I_0}{4}\right) \sin^2 2\theta$
 (3) $\left(\frac{I_0}{2}\right) \cos^4 \theta$ (4) $I_0 \cos^4 \theta$
42. Angular width of central maxima in the Fraunhofer diffraction pattern of a slit is measured. The slit is illuminated by light of wavelength $6000 \text{ \AA}$. When the slit is illuminated by light of another wavelength, the angular width decreases by 30%. The wavelength of this light will be
 (1) $6000 \text{ \AA}$ (2) $4200 \text{ \AA}$
 (3) $3000 \text{ \AA}$ (4) $1800 \text{ \AA}$
43. A beam of natural light falls on a system of 6 polaroids, which are arranged in succession such that each polaroid is turned through 30° with respect to the preceding one. The percentage of incident intensity that passes through the system will be
 (1) 100% (2) 50%
 (3) 30% (4) 12%
44. If I_0 is the intensity of the principal maximum in the single slit diffraction pattern, then what will be its intensity when the slit width is doubled?
 (1) I_0 (2) $\frac{I_0}{2}$
 (3) $2I_0$ (4) $4I_0$
45. A beam of electron is used in an YDSE. The slit width is d . When the velocity of electron is increased, then
 (1) no interference is observed
 (2) fringe width increases
 (3) fringe width decreases
 (4) fringe width remains same
46. The time period of rotation of the Sun is 25 days and its radius is $7 \times 10^8 \text{ m}$. The Doppler shift for the light of wavelength $6000 \text{ \AA}$ emitted from the surface of the Sun will be
 (1) $0.04 \text{ \AA}$ (2) $0.40 \text{ \AA}$
 (3) $4.00 \text{ \AA}$ (4) $40.0 \text{ \AA}$
47. In hydrogen spectrum the wavelength of H_α line is 656 nm whereas in the spectrum of a distant galaxy, H_α line wavelength is 706 nm . Estimated speed of the galaxy with respect to earth is
 (1) $2 \times 10^8 \text{ m/s}$ (2) $2 \times 10^7 \text{ m/s}$
 (3) $2 \times 10^6 \text{ m/s}$ (4) $2 \times 10^5 \text{ m/s}$
48. A rocket is going towards moon with a speed v . The astronaut in the rocket sends signals of frequency ν towards the moon and receives them back on reflection from the moon. What will be the frequency of the signal received by the astronaut? (Take $v \ll c$)
 (1) $\frac{c}{c-v} \nu$ (2) $\frac{c}{c-2v} \nu$
 (3) $\frac{2v}{c} \nu$ (4) $\frac{2c}{v} \nu$
49. The periodic time of rotation of a certain star is 22 days and its radius is $7 \times 10^8 \text{ metre}$. If the wavelength of light emitted by its surface be $4320 \text{ \AA}$, the Doppler shift will be (1 day = 86400 sec)
 (1) $0.033 \text{ \AA}$ (2) $0.33 \text{ \AA}$
 (3) $3.3 \text{ \AA}$ (4) $33 \text{ \AA}$
50. In a single slit diffraction of light of wavelength λ by a slit of width d , the size of the central maximum on a screen at a distance b is
 (1) $2b\lambda + d$ (2) $\frac{2b\lambda}{d}$
 (3) $\frac{2b\lambda}{d} + d$ (4) $\frac{2b\lambda}{d} - d$
51. In a single slit diffraction experiment first minimum for red light (660 nm) coincides with first maximum of some other wavelength λ' . The value of λ' is
 (1) $4400 \text{ \AA}$ (2) $6600 \text{ \AA}$
 (3) $2000 \text{ \AA}$ (4) $3500 \text{ \AA}$
52. The ratio of intensities of consecutive maxima in the diffraction pattern due to a single slit is
 (1) 1:4:9 (2) 1:2:3
 (3) $1: \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$ (4) $1: \frac{1}{\pi^2} : \frac{9}{\pi^2}$

53. The ratio of the intensity at the centre of a bright fringe to the intensity at a point one-quarter of the distance between two fringe from the centre is
(1) 2
(3) 4

(2) 1/2
(4) 16
54. A beam of plane polarized light falls normally on a polarizer of cross-sectional area $3 \times 10^{-4} \text{ m}^2$. Flux of energy of incident ray in 10^{-3} W . The polarizer rotates with an angular frequency of 31.4 rad/s . The energy of light passing through the polarizer per revolution will be

(1) 10^{-4} joule
(3) 10^{-2} joule

(2) 10^{-3} joule
(4) 10^{-1} joule
55. Unpolarized light of intensity 32 Wm^{-2} passes through three polarizers such that transmission axes of the first and second polarizer makes an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be

(1) 32 Wm^{-2}
(3) 8 Wm^{-2}

(2) 3 Wm^{-2}
(4) 4 Wm^{-2}

Answers Key

EXERCISES

Single Correct Answer Type

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (2) | 2. (1) | 3. (1) | 4. (1) | 5. (3) | 21. (4) | 22. (2) | 23. (2) | 24. (1) | 25. (2) |
| 6. (1) | 7. (2) | 8. (2) | 9. (4) | 10. (4) | 26. (1) | 27. (4) | 28. (4) | 29. (4) | 30. (1) |
| 11. (2) | 12. (3) | 13. (4) | 14. (4) | 15. (3) | 31. (3) | 32. (3) | 33. (1) | 34. (3) | 35. (4) |
| 16. (1) | 17. (3) | 18. (1) | 19. (1) | 20. (3) | 36. (2) | 37. (3) | 38. (2) | 39. (1) | 40. (3) |
| | | | | | 41. (1) | 42. (2) | 43. (4) | 44. (4) | 45. (2) |
| | | | | | 46. (1) | 47. (2) | 48. (2) | 49. (1) | 50. (3) |
| | | | | | 51. (1) | 52. (3) | 53. (1) | 54. (1) | 55. (2) |

Chapter 4

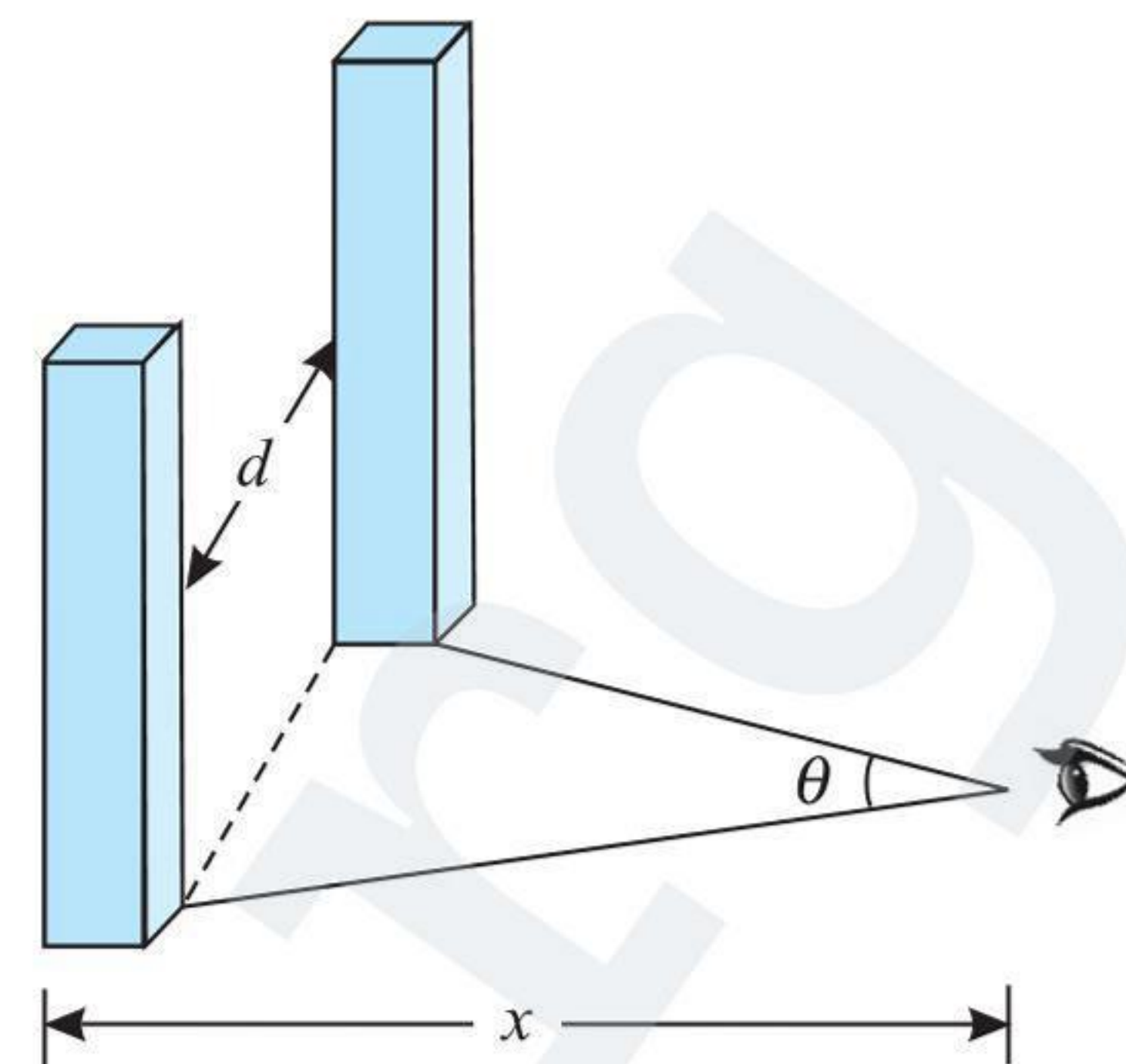
Concept Application Exercises

Exercise 4.1

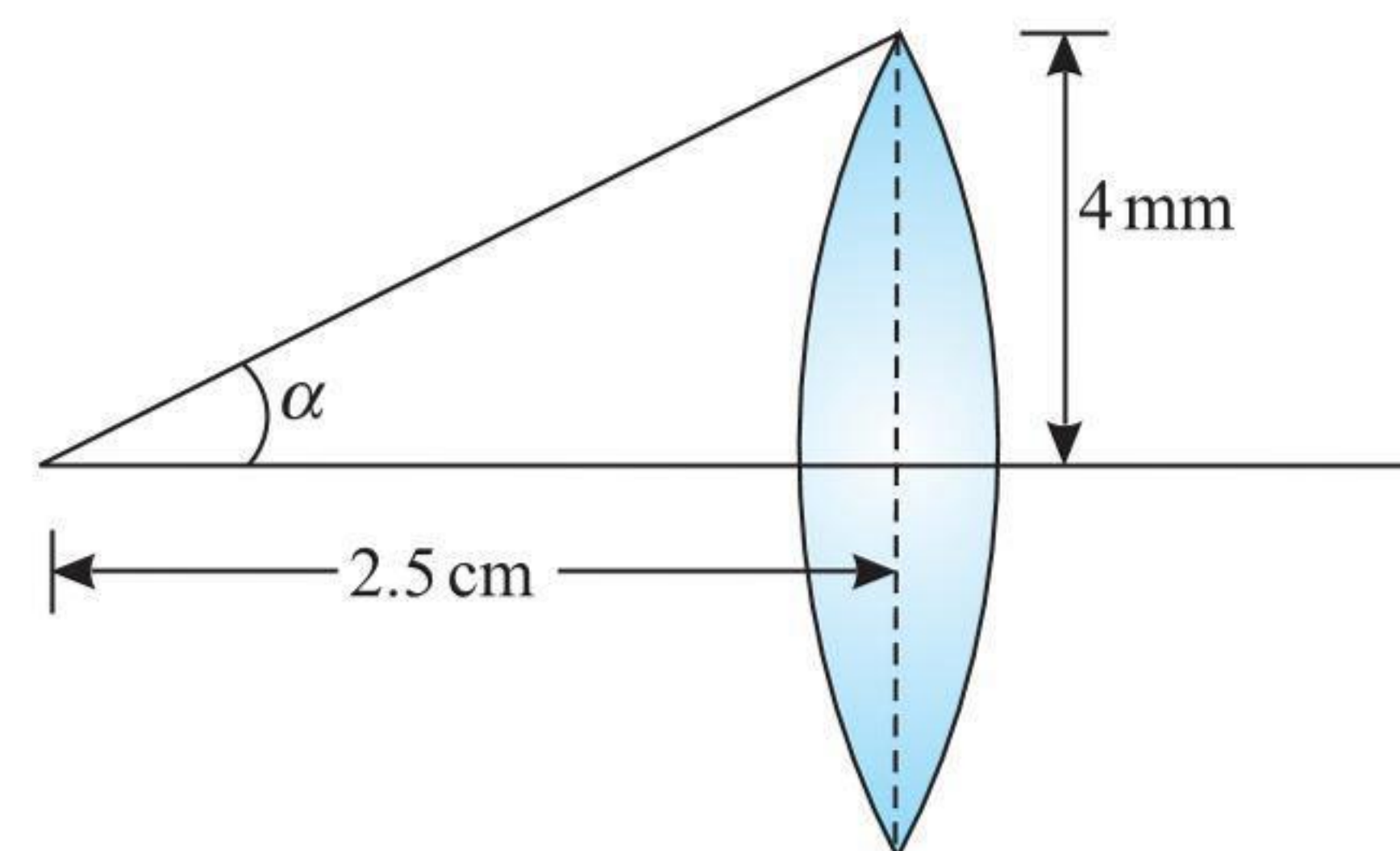
- Here, $a = 0.15 \text{ cm} = 0.15 \times 10^{-2} \text{ m}$,
 $\lambda = 5 \times 10^{-5} \text{ cm} = 5 \times 10^{-7} \text{ m}$, $D = 2.1 \text{ m}$
 Now, $\beta = \frac{2D\lambda}{a} = \frac{2 \times 2.1 \times 5 \times 10^{-7}}{0.15 \times 10^{-2}} = 1.4 \times 10^{-3} \text{ m} = 1.4 \text{ mm}$
- Here, $\lambda = 5900 \times 10^{-10} \text{ m}$
 $a = 11.8 \times 10^{-5} \text{ cm} = 11.8 \times 10^{-7} \text{ m}$
 The angular position of first minimum is given by
 $\sin \theta_1 = \frac{\lambda}{a} = \frac{5900 \times 10^{-10}}{11.8 \times 10^{-7}} = 0.5$
 or $\theta_1 = 30^\circ$
 The angular width of central maximum, $2\theta_1 = 2 \times 30^\circ = 60^\circ$
- Here, wavelength of microwaves, $\lambda = \frac{c}{\nu} = \frac{3 \times 10^{10}}{12 \times 10^9} = 2.5 \text{ cm}$
 Width of the slit, $a = 10 \text{ cm}$
 The direction of the first minimum on either side of the central maximum is given by $\theta = \frac{\lambda}{a}$.
 In other words, the angular spread of the central maximum is given by
 $2\theta = \frac{2\lambda}{a} = \frac{2 \times 2.5}{10} = 0.5 \text{ rad}$
 $= (0.5 \text{ rad}) \left(\frac{180^\circ}{\pi \text{ rad}} \right) = 28.7^\circ$
- For third minimum, $\theta = \frac{3\lambda}{d}$
 $\theta = \frac{3(4.00 \times 10^{-7} \text{ m})}{5.0 \times 10^{-5} \text{ m}} = 0.024 \text{ rad} = 0.024 \left(\frac{180^\circ}{\pi} \right) = 1.4^\circ$
 Width of central maximum $= \frac{2\lambda D}{d}$
 $= \frac{2(4.0 \times 10^{-7} \text{ m})(1.0 \text{ m})}{5.0 \times 10^{-5} \text{ m}} = 1.6 \times 10^{-2} \text{ m} = 1.6 \text{ cm}$
- Angular spread of central maximum on either side of incident light is given by, $\sin \theta = \frac{\lambda}{a}$
 $\therefore$ Slit width, $a = \frac{\lambda}{\sin \theta} = \frac{5000 \times 10^{-10}}{\sin 30^\circ} = 10^{-6} \text{ m}$
 For $\theta = 90^\circ$, we have $a = \frac{\lambda}{\sin 90^\circ} = \frac{5000 \times 10^{-10}}{1} = 5 \times 10^{-7} \text{ m}$.
- Here $d = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$, $\lambda = 500 \text{ nm} = 500 \times 10^{-9} \text{ m}$
 The distance up to which a beam of light can travel without significant broadening is called Fresnel distance and its value is given by
 $D_F = \frac{d^2}{\lambda} = \frac{(3 \times 10^{-3})^2}{500 \times 10^{-9}} = 18 \text{ m}$

Exercise 4.2

- As limit of resolution of eye is $\left(\frac{1}{60} \right)^\circ$, the pillars will be seen distinctly if $\theta > \left(\frac{1}{60} \right)^\circ$ i.e., $\frac{d}{x} > \left(\frac{1}{60} \right) \times \frac{\pi}{180}$



- $$\Rightarrow d > \frac{\pi \times x}{60 \times 180}$$
- $$\Rightarrow d > \frac{3.14 \times 11 \times 10^3}{60 \times 180} \Rightarrow d > 3.2 \text{ m}$$
- Here $d\theta = 4.6 \times 10^{-6} \text{ rad}$,
 $\lambda = 5460 \text{ \AA} = 5460 \times 10^{-10} \text{ m}$
 As $d\theta = \frac{1.22 \lambda}{D}$
 $\therefore$ Aperture of the telescope,
 $D = \frac{1.22 \lambda}{d\theta} = \frac{1.22 \times 5460 \times 10^{-10}}{4.6 \times 10^{-6}} = 0.1488 \text{ m}$
 - We assume that the object is placed a little beyond its focal distance, say 2.5 cm.



- Refer figure, $\tan \alpha = \frac{4 \text{ mm}}{2.5 \text{ cm}} = \frac{0.4}{2.5} = 0.16$
 Since $\sin \alpha$ is small, $\sin \alpha \approx \tan \alpha = 0.16$
 Thus, $\Delta x = \frac{1.22 \lambda}{2\mu \sin \alpha} = \frac{1.22(5500 \times 10^{-10} \text{ m})}{2 \times 1 \times 0.16} \approx 2 \times 10^{-6} \text{ m}$
- $\Delta \theta = 1.22 \left(\frac{\lambda}{a} \right) = 1.22 \left(\frac{5500 \times 10^{-8} \text{ cm}}{200 \text{ cm}} \right) = 3.35 \times 10^{-7} \text{ rad}$
 - Here, diameter of the human eye, $D = 2 \text{ mm} = 2 \times 10^{-3} \text{ m}$
 distance of the two points from the eye, $y = 50 \text{ m}$
 and wavelength of light used $= 5000 \text{ \AA} = 5 \times 10^{-7} \text{ m}$
 The angular diameter of the two points at the eye, $d\theta = \frac{d}{y}$
 Now, resolving power the eye, $\frac{1}{d\theta} = \frac{D}{1.22 \lambda}$
 or $\frac{y}{d} = \frac{D}{1.22 \lambda}$
 or $d = \frac{1.22 \lambda}{D} \times y = \frac{1.22 \times 5 \times 10^{-7}}{2 \times 10^{-3}} \times 50$
 $= 1.525 \times 10^{-2} \text{ m} = 1.525 \text{ cm}$

Exercise 4.3

- (c) At polarizing angle, the reflected and refracted rays are mutually perpendicular to each other.
- (a) When unpolarized light is made incident at polarizing angle, the reflected light is plane polarized in a direction perpendicular to the plane of incidence.
Therefore $\vec{E}$ in reflected light will vibrate in vertical plane with respect to plane of incidence.

3. (a) At $i_p + r_p = 90^\circ$

$$\therefore r_p = 90^\circ - i_p = 90^\circ - 53^\circ = 37^\circ$$

(b) $\mu = \tan i_p = \tan 53^\circ = \frac{4}{3}$

4. Refer to figure in question. Incident light at the Brewster's angle (i_p) has the greatest polarization, where

$$i_p = \tan^{-1}(\mu) = \tan^{-1} \frac{4}{3} = 53^\circ$$

If θ is the altitude angle for greatest polarization,

$$\theta = 90^\circ - i_p = 90^\circ - 53^\circ = 37^\circ$$

5. Let I_0 be the intensity when the axis of analyzer is parallel to the axis of polarizer. By the law of Malus, $I = I_0 \cos^2 \theta$.

(i) When $\theta = 45^\circ$, $I = I_0 \cos^2 45^\circ = \frac{I_0}{2}$.

(ii) When $\theta = 90^\circ$, $I = I_0 \cos^2 90^\circ = 0$.

(iii) When $\theta = 180^\circ$, $I = I_0 \cos^2 180^\circ = I_0$.

6. Here $I = \frac{I_0}{2}$

Using Malus law, $I = I_0 \cos^2 \theta$

$$\therefore \frac{I_0}{2} = I_0 \cos^2 \theta$$

or $\cos \theta = \pm \frac{1}{\sqrt{2}}$

Hence $\theta = \pm 45^\circ, \pm 135^\circ$

The same effect occurs no matter which sheet is rotated or in which direction it is rotated.

Exercise 4.4

- (b) Shifting towards ultraviolet region shows that apparent wavelength decreased. Therefore, the source is moving towards the earth.
- (c) Blue radiations have the wavelength around 4600 Å. It shows that apparent wavelength is smaller than the real wavelength. It means that the star is proceeding towards earth.

3. $\Delta\lambda = \frac{v}{c} \cdot \lambda = \frac{9 \times 10^6}{3 \times 10^8} \times 6000 \text{ Å} = 180 \text{ Å}$

Apparent wavelength, $\lambda' = \lambda - \Delta\lambda = 6000 - 180 = 5820 \text{ Å}$.

4. $v' = v \left(\frac{c+v}{c} \right)$, $\frac{v'}{v} = 1 + \frac{v}{c}$

or $\frac{\lambda}{\lambda'} = 1 + \frac{v}{c}$

or $\lambda' = \frac{\lambda}{1 + v/c} = \frac{6 \times 10^{-7}}{(4/3)}$

or $\lambda' = 4.5 \times 10^{-7} \text{ m}$

5. $\frac{\Delta\lambda}{\lambda} = \frac{v_s}{c}$

or $v_s = \left(\frac{\Delta\lambda}{\lambda} \right) c = \frac{(589.6 - 589.0) \text{ nm}}{589.0 \text{ nm}} \times (3 \times 10^8 \text{ m/s})$
 $= 3.1 \times 10^5 \text{ m/s}$

Exercises

Single Correct Answer Type

1. (2) The magnitude of electric field vector varies periodically with time because it is in the form of electromagnetic wave.

2. (1) Band width $\propto \lambda$,
 $\therefore \lambda_{\text{blue}} < \lambda_{\text{red}}$, hence for blue light the diffraction bands become narrower and crowded together.

3. (1) Intensity of the polarized light coming out of polarizing sheet

$$\text{will be } I = \int_0^{2\pi} I_0 \cos^2 \theta d\theta$$

On solving, we get $I = \frac{I_0}{2}$

4. (1) $\beta = \frac{\lambda D}{d}$, where D = distance of screen from wire, d = diameter of wire

5. (3) Angular width $= \frac{2\lambda}{d} = \frac{2 \times 6000 \times 10^{-10}}{12 \times 10^{-5} \times 10^{-2}} = 1 \text{ rad}$

6. (1) $2\theta = \frac{2\lambda}{d}$ (where d = slit width)

As d decreases, θ increases.

7. (2) Thickness of the film must be of the order of wavelength of light falling on film (i.e. visible light)

8. (2) $\Delta\lambda = 5200 - 5000 = 200 \text{ Å}$

Now $\frac{\Delta\lambda}{\lambda'} = \frac{v}{c}$

$$\Rightarrow v = \frac{c\Delta\lambda}{\lambda'} = \frac{3 \times 10^8 \times 200}{5000} = 1.2 \times 10^7 \text{ m/s} \approx 1.15 \times 10^7 \text{ m/s}$$

9. (4) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c}$

$$\Rightarrow v = \frac{c}{\lambda} \Delta\lambda = \frac{c}{\lambda} (\lambda' - \lambda) = c \times \frac{0.01}{100}$$

$$= 3 \times 10^4 \text{ m/s} = 30 \text{ km/s}$$

10. (4) $\Delta\lambda = \frac{v}{c} \lambda = \frac{3600 \times 10^3}{3 \times 10^8} \times 5896 = 70.75 \text{ Å}$

So the increased wavelength of light is observed.

11. (2) Observed frequency $v' = v \left(1 - \frac{v}{c} \right)$

$$\Rightarrow v' = 6 \times 10^{14} \left(1 - \frac{0.8c}{c} \right) = 1.2 \times 10^{14} \text{ Hz}$$

12. (3) According to Doppler's principle, $\lambda' = \lambda \sqrt{\frac{1-v/c}{1+v/c}}$
 For $v = c$,

$$\lambda' = 5500 \sqrt{\frac{1-0.8}{1+0.8}} = 1833.3$$

$$\therefore \text{Shift} = 5500 - 1833.3 = 3167 \text{ Å}$$

13. (4) $\Delta\lambda = \frac{v_s}{c} \lambda$

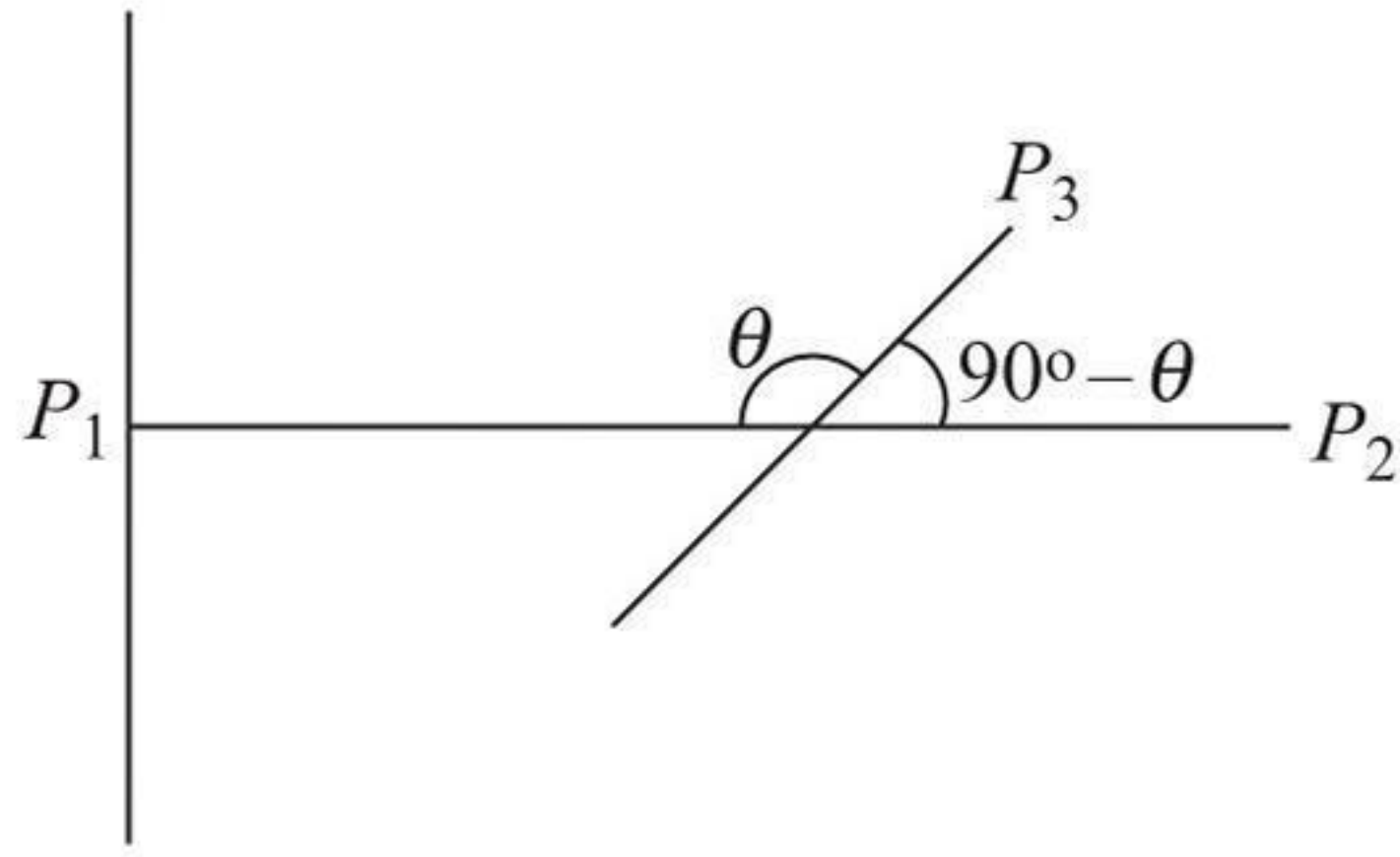
$$\Rightarrow v_s = \frac{\Delta\lambda \cdot c}{\lambda} = \frac{47 \times 3 \times 10^8}{4700} = 3 \times 10^6 \text{ m/s}$$

away from earth

14. (4) $\frac{\Delta\lambda}{\lambda} = \frac{v}{c} \Rightarrow v = \frac{\Delta\lambda}{\lambda} \cdot c = \frac{5}{6563} \times (3 \times 10^8) = 2.29 \times 10^5 \text{ m/s}$

15. (3) For first minima, $\theta = \frac{\lambda}{a}$
 or $a = \frac{\lambda}{\theta}$
 $\therefore a = \frac{6500 \times 10^{-8} \times 6}{\pi}$ (As $30^\circ = \frac{\pi}{6}$ radian)
 $= 1.24 \times 10^{-4} \text{ cm} = 1.24 \text{ micron}$
16. (1) The angular half width of the central maxima is given by
 $\sin \theta = \frac{\lambda}{a}$
 $\Rightarrow \theta = \frac{6328 \times 10^{-10}}{0.2 \times 10^{-3}} \text{ rad}$
 $= \frac{6328 \times 10^{-10} \times 180}{0.2 \times 10^{-3} \times \pi} \text{ degree} = 0.18^\circ$
 Total width of central maxima $= 2\theta = 0.36^\circ$
17. (3) Width of central maxima $= \frac{2\lambda D}{d}$
 $= \frac{2 \times 2.1 \times 5 \times 10^{-7}}{0.15 \times 10^{-2}} = 1.4 \times 10^{-3} \text{ m} = 1.4 \text{ mm}$
18. (1) Using $d \sin \theta = n\lambda$, for $n=1$
 $\sin \theta = \frac{\lambda}{d} = \frac{550 \times 10^{-9}}{0.55 \times 10^{-3}} = 10^{-3} = 0.001 \text{ rad}$
19. (1) For single slit diffraction pattern, $d \sin \theta = \lambda$ (d = slit width)
 Angular width $= 2\theta = 2 \sin^{-1} \left(\frac{\lambda}{d} \right)$
 It is independent of D i.e. distance between screen and slit
20. (3) Width of central bright fringe
 $= \frac{2\lambda D}{d} = \frac{2 \times 500 \times 10^{-9} \times 80 \times 10^{-2}}{0.20 \times 10^{-3}} = 4 \times 10^{-3} \text{ m} = 4 \text{ mm}$
21. (4) $\frac{\Delta \lambda}{\lambda} = \frac{v}{c}$, Now $\Delta \lambda = \frac{0.5}{100} \lambda \Rightarrow \frac{\Delta \lambda}{\lambda} = \frac{0.5}{100}$
 $\therefore v = \frac{0.5}{100} \times c = \frac{0.5}{100} \times 3 \times 10^8 = 1.5 \times 10^6 \text{ m/s}$
 Increase in λ indicates that the star is receding.
22. (2) $\frac{\Delta \lambda}{\lambda} = \frac{v}{c} = \frac{6 \times 10^7}{3 \times 10^8} = 0.2$
 $\Delta \lambda = \lambda' - \lambda = 0.2\lambda \Rightarrow \lambda' = 1.2\lambda = 1.2 \times 4600 = 5520 \text{ Å}$
23. (2) Diffraction is obtained when the slit width is of the order of wavelength of e.m. waves (or light). Here wavelength of X-rays (1–100 Å) is very-very lesser than slit width (0.6 mm). Therefore, no diffraction pattern will be observed.
24. (1) For secondary maxima, $d \sin \theta = \frac{5\lambda}{2} \Rightarrow d\theta = d \cdot \frac{x}{D(\approx f)} = \frac{5\lambda}{2}$
 $\Rightarrow 2x = \frac{5\lambda f}{d} = \frac{5 \times 0.8 \times 10^{-7}}{4 \times 10^{-4}} = 6 \times 10^{-3} \text{ m} = 6 \text{ mm}$
25. (2) $\lambda_{\text{Blue}} < \lambda_{\text{Red}}$
 Therefore fringe pattern will contract because fringe width $\propto \lambda$
26. (1) Angular width of central maxima $= \frac{2\lambda}{d}$
 $= \frac{2 \times 589.3 \times 10^{-9}}{0.1 \times 10^{-3}} \text{ rad} = 0.0117 \times \frac{180^\circ}{\pi} = 0.68^\circ$
27. (4) Distance between the first dark fringes on either side of central maxima = width of central maxima
 $= \frac{2\lambda D}{d} = \frac{2 \times 600 \times 10^{-9} \times 2}{1 \times 10^{-3}} = 2.4 \text{ mm}$
28. (4) For n^{th} secondary maxima path difference
 $d \sin \theta = (2n+1) \frac{\lambda}{2} \Rightarrow a \sin \theta = \frac{3\lambda}{2}$
29. (4) The phase difference (ϕ) between the wavelets from the top edge and the bottom edge of the slit is $\phi = \frac{2\pi}{\lambda} (d \sin \theta)$ where d is the slit width. The first minima of the diffraction pattern occurs at $\sin \theta = \frac{\lambda}{d}$ so $\phi = \frac{2\pi}{\lambda} \left(d \times \frac{\lambda}{d} \right) = 2\pi$
30. (1) For second dark fringe $d \sin \theta = 2\lambda$
 $\Rightarrow 24 \times 10^{-5} \times 10^{-2} \times \sin 30^\circ = 2\lambda$
 $\Rightarrow \lambda = 6 \times 10^{-7} \text{ m} = 6000 \text{ Å}$
31. (3) For the first minima $d \sin \theta = \lambda$
 $\Rightarrow \sin \theta = \frac{\lambda}{d} \Rightarrow \theta = \sin^{-1} \left(\frac{5000 \times 10^{-10}}{0.001 \times 10^{-3}} \right) = 30^\circ$
32. (3) Position of first minima = position of third maxima
 i.e., $\frac{1 \times \lambda_1 D}{d} = \frac{(2 \times 3 + 1) \lambda_2 D}{d} \Rightarrow \lambda_1 = 3.5 \lambda_2$
33. (1) Position of n^{th} minima, $x_n = \frac{n\lambda D}{d}$
 $\Rightarrow 5 \times 10^{-3} = \frac{1 \times 5000 \times 10^{-10} \times 1}{d}$
 $\Rightarrow d = 10^{-4} \text{ m} = 0.1 \text{ mm}$
34. (3) In single slit experiment,
 Width of central maxima (y) $= \frac{2\lambda D}{d}$
 $\Rightarrow \frac{y'}{y} = \frac{\lambda'}{\lambda} \times \frac{d}{d/2} = \frac{600}{d/2} \times \frac{d}{400} \Rightarrow y' = 3y$
35. (4) By using $\mu = \tan \theta_p \Rightarrow \mu = \tan 60^\circ = \sqrt{3}$,
 also $C = \sin^{-1} \left(\frac{1}{\mu} \right) \Rightarrow C = \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$
36. (2) $I = I_0 \cos^2 \theta = I_0 \cos^2 45^\circ = \frac{I_0}{2}$
37. (3) Intensity of polarized light from first polarizer $= \frac{100}{2} = 50$
 $I = 50 \cos^2 60^\circ = \frac{50}{4} = 12.5$
38. (2) $I' = \frac{I}{2} \cos^2 \theta = \frac{I}{6}$
 or $\cos \theta = \frac{1}{\sqrt{3}}$
 $\therefore \theta = 55^\circ$
39. (1) According to Brewster's law, when a beam of ordinary light (i.e., unpolarized) is reflected from a transparent medium (like glass), the reflected light is completely plane polarized at certain angle of incidence called the angle of polarization.
40. (3) From Brewster's law, $\mu = \tan i_p \Rightarrow \frac{c}{v} = \tan 60^\circ = \sqrt{3}$
 $\Rightarrow v = \frac{c}{\sqrt{3}} = \frac{3 \times 10^8}{\sqrt{3}} = \sqrt{3} \times 10^8 \text{ m s}^{-1}$

41. (1) No light is emitted from the second polaroid, so P_1 and P_2 are perpendicular to each other.



Let the initial intensity of light is I_0 . So, intensity of light after transmission from first polaroid = $\frac{I_0}{2}$.

Intensity of light emitted from P_3 , $I_1 = \frac{I_0}{2} \cos^2 \theta$

Intensity of light transmitted from last polaroid i.e. from $P_2 = I_1 \cos^2 (90^\circ - \theta) = \frac{I_0}{2} \cos^2 \theta \sin^2 \theta = \frac{I_0}{8} (2 \sin \theta \cos \theta)^2 = \frac{I_0}{8} \sin^2 2\theta$

42. (2) Angular width $\beta = \frac{2\lambda}{d} \Rightarrow \beta \propto \lambda$

$$\Rightarrow \frac{\beta_1}{\beta_2} = \frac{\lambda_1}{\lambda_2} \Rightarrow \frac{\beta}{\frac{70}{100}\beta} = \frac{6000}{\lambda_2} \Rightarrow \lambda_2 = 4200 \text{ \AA}$$

43. (4) If I is the final intensity and I_0 is the initial intensity then

$$I = \frac{I_0}{2} (\cos^2 30^\circ)^5$$

$$\text{or } \frac{I}{I_0} = \frac{1}{2} \times \left(\frac{\sqrt{3}}{2} \right)^{10} = 0.12$$

44. (4) When the slit width is doubled, the half angular width of the central maximum, which is λ/a , reduces to half. The area of the central diffraction band becomes $1/4$. So the intensity of the central maximum becomes four times.

45. (2) Momentum of the electron will increase. So the wavelength ($\lambda = h/p$) of electrons will decrease and fringe width decreases as $\beta \propto \lambda$.

46. (1) $\Delta\lambda = \lambda \frac{v}{c}$ and $v = r\omega$

$$v = 7 \times 10^8 \times \frac{2\pi}{25 \times 24 \times 3600}, c = 3 \times 10^8 \text{ m/s}$$

$$\therefore \Delta\lambda = 0.04 \text{ \AA}$$

47. (2) $v = \frac{c\Delta\lambda}{\lambda} = \frac{3 \times 10^8 \times (706 - 656)}{656} = \frac{1500}{656} \times 10^7 = 2 \times 10^7 \text{ m/s}$

48. (2) In this case, we can assume as if both the source and the observer are moving towards each other with speed v .

$$\begin{aligned} \text{Hence } v' &= \frac{c - u_o}{c - u_s} v = \frac{c - (-v)}{c - v} v = \frac{c + v}{c - v} v \\ &= \frac{(c + v)(c - v)}{(c - v)^2} v = \frac{c^2 - v^2}{c^2 + v^2 - 2vc} v \end{aligned}$$

$$\text{Since } v \ll c, \text{ therefore } v' = \frac{c^2 v}{c^2 - 2vc} = \frac{c}{c - 2v} v$$

49. (1) $\Delta\lambda = \lambda \frac{v}{c}$ where $v = r\omega = r \times \left(\frac{2\pi}{T} \right)$

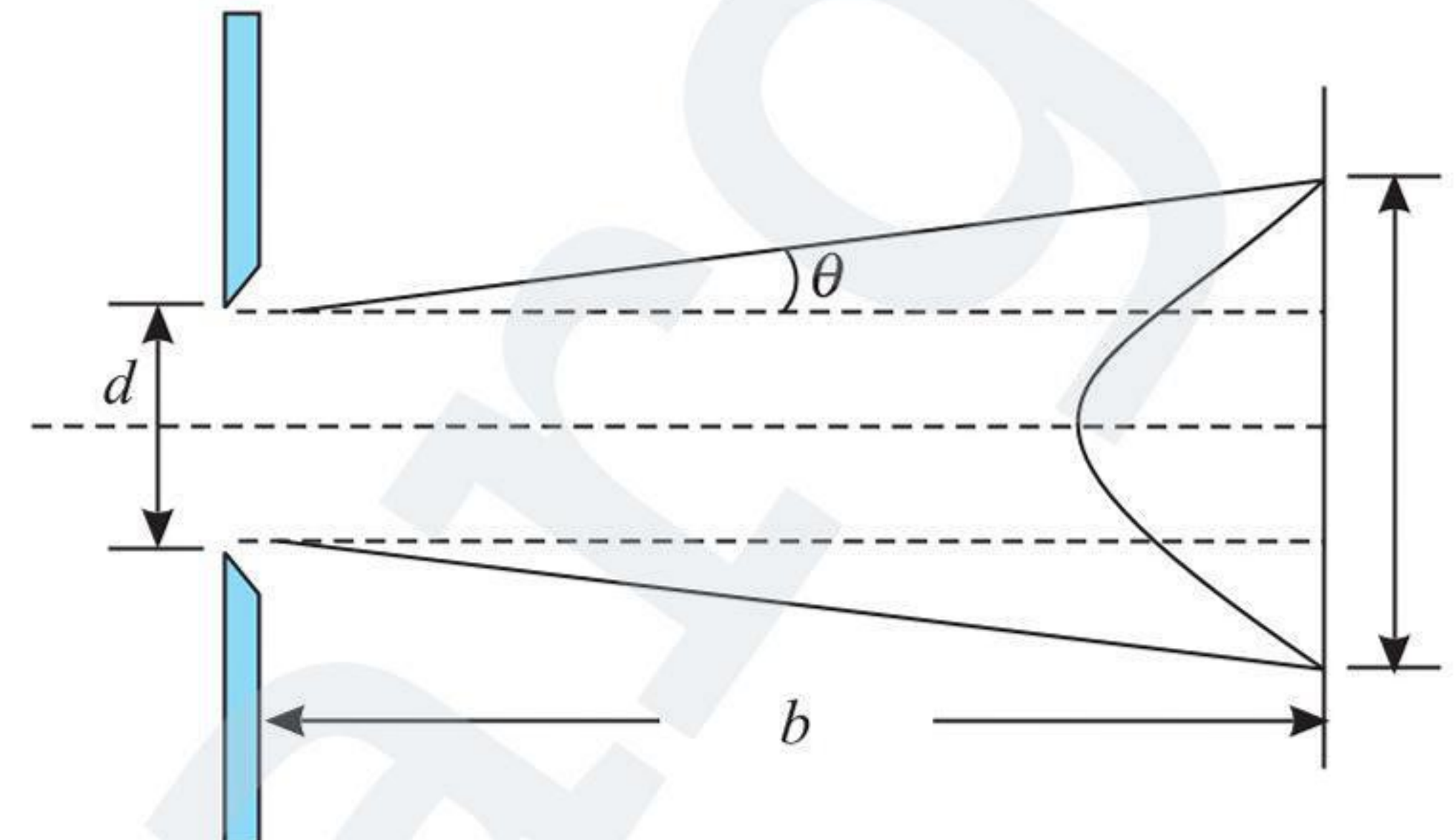
$$\therefore \Delta\lambda = \frac{4320 \times 7 \times 10^8 \times 2 \times 3.14}{3 \times 10^8 \times 22 \times 86400} = 0.033 \text{ \AA}$$

50. (3) The direction in which the first minima occurs is θ (say).

$$\text{Then } d \sin \theta = \lambda$$

$$\text{or } d\theta = \lambda$$

$$\text{or, } \theta = \frac{\lambda}{d} \quad (\because \theta = \sin \theta \text{ when } \theta \text{ small})$$



$$\text{Width of central maximum} = 2b\theta + d = \frac{2b\lambda}{d} + d$$

51. (1) In a single slit diffraction experiment, position of minima is given by $d \sin \theta = n\lambda$

So for first minima of red, $\sin \theta = 1 \times \left(\frac{\lambda_R}{d} \right)$ and as first maxima is midway between first and second minima, for wavelength λ' , its position will be

$$d \sin \theta' = \frac{\lambda' + 2\lambda'}{2} \Rightarrow \sin \theta' = \frac{3\lambda'}{2d}$$

According to given condition $\sin \theta = \sin \theta'$

$$\Rightarrow \lambda' = \frac{2}{3} \lambda_R$$

$$\text{So } \lambda' = \frac{2}{3} \times 6600 = 4400 \text{ nm} = 4400 \text{ \AA}$$

52. (3) $I = I_0 \left[\frac{\sin \alpha}{\alpha} \right]^2$, where $\alpha = \frac{\phi}{2}$

$$\text{For } n^{\text{th}} \text{ secondary maxima, } d \sin \theta = \left(\frac{2n+1}{2} \right) \lambda$$

$$\Rightarrow \alpha = \frac{\phi}{2} = \frac{\pi}{\lambda} [d \sin \theta] = \left(\frac{2n+1}{2} \right) \pi$$

$$\Rightarrow I = I_0 \left[\frac{\sin \left(\frac{2n+1}{2} \pi \right)}{\left(\frac{2n+1}{2} \pi \right)} \right]^2 = \frac{I_0}{\left\{ \frac{(2n+1)}{2} \pi \right\}^2}$$

$$\text{So } I_0 : I_1 : I_2 = I_0 : \frac{4}{9\pi^2} I_0 : \frac{4}{25\pi^2} I_0 = 1 : \frac{4}{9\pi^2} : \frac{4}{25\pi^2}$$

53. (1) $I = 4I_0 \cos^2 \frac{\phi}{2}$

$$\text{At central position } I_1 = 4I_0 \quad \dots(i)$$

Since the phase difference between two successive fringes is $2x$, the phase difference between two points separated by a distance equal to one quarter of the distance between the two,

$$\text{successive fringes is equal to } \delta = (2\pi) \left(\frac{1}{4} \right) = \frac{\pi}{2} \text{ radian}$$

$$\Rightarrow I_2 = 4I_0 \cos^2 \left(\frac{\pi}{2} \right) = 2I_0 \quad \dots(ii)$$

Using Eqs. (i) and (ii), $\frac{I_1}{I_2} = \frac{4I_0}{2I_0} = 2$

54. (1) Using Malus law, $I = I_0 \cos^2 \theta$

As here polarizer is rotating i.e. all the values of θ are possible.

$$I_{av} = \frac{1}{2\pi} \int_0^{2\pi} I d\theta = \frac{1}{2\pi} \int_0^{2\pi} I_0 \cos^2 \theta d\theta$$

On integration we get $I_{av} = \frac{I_0}{2}$

where $I_0 = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{p}{A} = \frac{10^{-3}}{3 \times 10^{-4}} = \frac{10}{3} \text{ watt m}^{-2}$

$$\therefore I_{av} = \frac{1}{2} \times \frac{10}{3} = \frac{5}{3} \text{ watt}$$

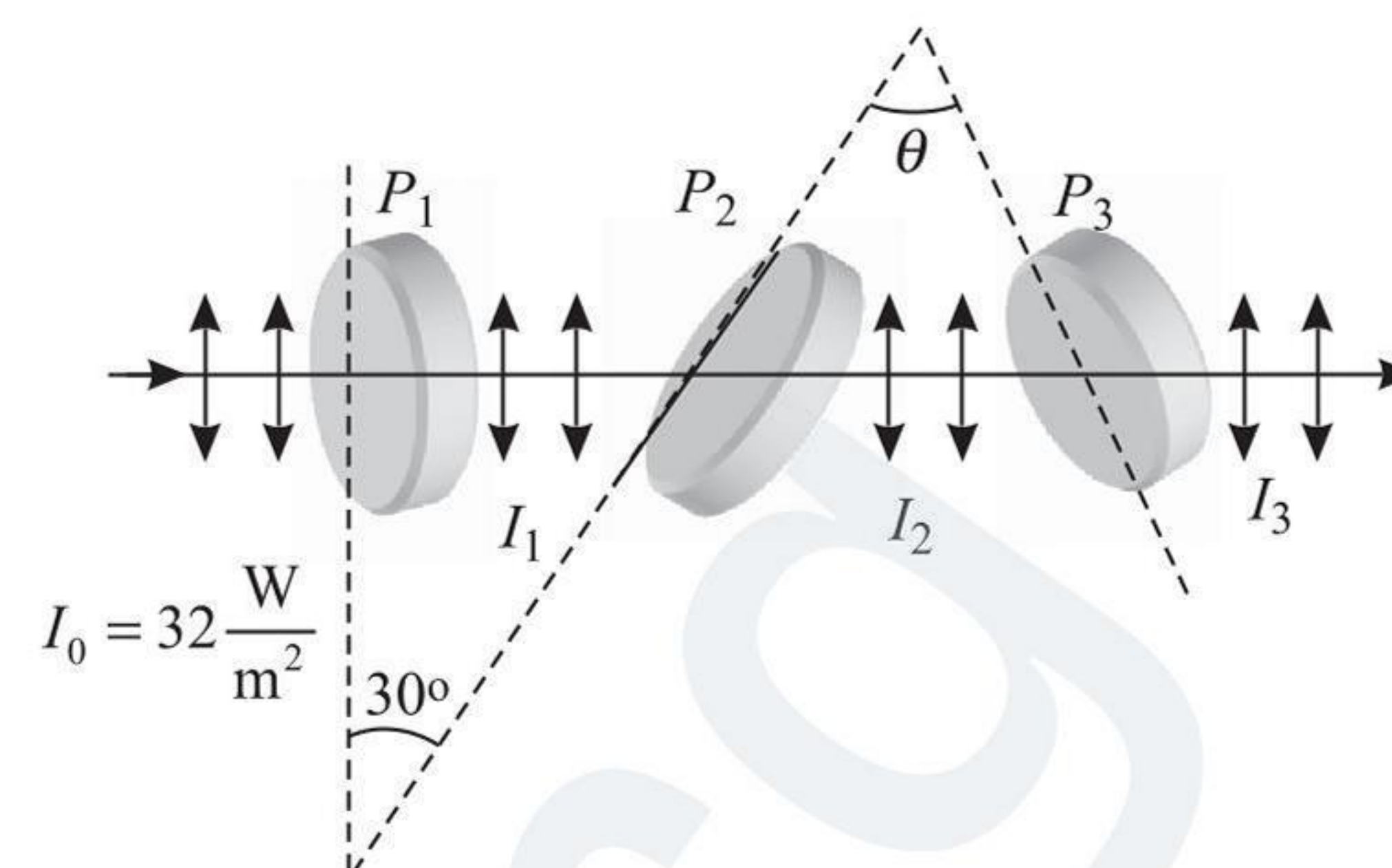
and time period $T = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{31.4} = \frac{1}{5} \text{ s}$

$\therefore$ Energy of light passing through the polarizer per revolution

$$= I_{av} \times \text{Area} \times T = \frac{5}{3} \times 3 \times 10^{-4} \times \frac{1}{5} = 10^{-4} \text{ J}$$

55. (2) Angle between P_1 and $P_2 = 30^\circ$ (given)

Angle between P_2 and $P_3 = \theta = 90^\circ - 30^\circ = 60^\circ$



The intensity of light transmitted by P_1 is $I_1 = \frac{I_0}{2} = \frac{32}{2} = 16 \text{ Wm}^{-2}$

According to Malus law the intensity of light transmitted by P_2 is

$$I_2 = I_1 \cos^2 30^\circ = 16 \left(\frac{\sqrt{3}}{2} \right)^2 = 12 \text{ W m}^{-2}$$

Similarly intensity of light transmitted by P_3 is

$$I_3 = I_2 \cos^2 \theta = 12 \cos^2 60^\circ = 12 \left(\frac{1}{2} \right)^2 = 3 \text{ W m}^{-2}$$